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Channel Knowledge Map Prediction with Deep Learning for 6G UAV-Enabled Networks

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Resum

Aquest TFG estudia la predicció de mapes de coneixement de canal per a xarxes 6G amb UAVs. A partir del conjunt de dades CKM (66 ciutats reals, 16,180 mostres, mapes urbans de 513 × 513 píxels amb màscares LoS/NLoS i altura d'antena contínua entre 12 i 478 m) es prediuen pèrdua de trajecte, dispersió temporal / delay spread i dispersió angular / angular spread sota un protocol estricte de validació per ciutats completes. Després de provar U-Nets, cGANs, experts i priors físics, el treball convergeix en un model híbrid: priors interpretables calibrats per pèrdua de trajecte i spreads, més un model final amb cap GMM residual conjunt. En els resultats finals, aquest cap s'avalua mitjançant la predicció determinista de la mitjana de la mescla, no com una densitat predictiva calibrada.

El resultat final no és un prior pur ni una xarxa neuronal pura: el model final usa deep learning com a corrector multitasca condicionat per la geometria urbana, l'altura del UAV, les màscares LoS/NLoS i els mapes prior. En el split final de test de 14 ciutats no vistes, els priors congelats assoleixen 1.94 dB, 28.10 ns i 13.74 graus RMSE, mentre que el model final amb residuals ancorats als priors els millora fins a 1.65 dB, 26.56 ns i 11.39 graus. Això fa que els priors siguin una baseline forta per si mateixa, no només un pas intermedi: concentren la calibració per règims de densitat urbana, altura d'edificis, visibilitat i altura d'antena, i deixen al model neuronal només el residual local. El resultat és una estratègia de divideix i venceràs més que una nova llei universal de propagació.

Resumen

Este TFG estudia la predicción de mapas de conocimiento de canal para redes 6G con UAVs. A partir del conjunto de datos CKM (66 ciudades reales, 16,180 muestras, mapas urbanos de 513 × 513 píxeles con máscaras LoS/NLoS y altura de antena continua entre 12 y 478 m) se predicen pérdida de trayecto, dispersión temporal / delay spread y dispersión angular / angular spread bajo un protocolo estricto de validación por ciudades completas. Tras probar U-Nets, cGANs, expertos y priors físicos, el trabajo converge en un modelo híbrido: priors interpretables calibrados para pérdida de trayecto y spreads, más un modelo final con cabeza GMM residual conjunta. En los resultados finales, esa cabeza se evalúa mediante la predicción determinista de la media de la mezcla, no como una densidad predictiva calibrada.

El resultado final no es un prior puro ni una red neuronal pura: el modelo final usa deep learning como corrector multitarea condicionado por la geometría urbana, la altura del UAV, las máscaras LoS/NLoS y los mapas prior. En el split final de test de 14 ciudades no vistas, los priors congelados alcanzan 1.94 dB, 28.10 ns y 13.74 grados RMSE, mientras que el modelo final con residuales anclados a los priors los mejora hasta 1.65 dB, 26.56 ns y 11.39 grados. Esto convierte a los priors en una baseline fuerte por sí misma, no solo en un paso intermedio: concentran la calibración por regímenes de densidad urbana, altura de edificios, visibilidad y altura de antena, y dejan al modelo neuronal solo el residual local. El resultado es una estrategia de divide y vencerás más que una nueva ley universal de propagación.

Summary

This thesis studies channel knowledge map prediction for 6G UAV-enabled networks. Using the CKM dataset (66 real cities, 16,180 samples, 513×513 urban maps with LoS/NLoS masks and a continuous UAV antenna height between 12 and 478 m), the project predicts path loss, delay spread, and angular spread under a strict city-holdout evaluation protocol, where entire cities are reserved for validation and test. After testing U-Nets, cGANs, expert models, and physical priors, the project converges on a hybrid design: calibrated interpretable priors for path loss and spreads, plus a joint residual GMM-head model. In the final reported results, that head is evaluated through its deterministic mixture-mean prediction, not as a calibrated predictive density.

The final system is neither a pure prior nor a pure neural network: the final model adds deep learning as a multi-task residual corrector conditioned on urban geometry, UAV height, LoS/NLoS masks, and prior maps. On the final 14-city unseen test split, the frozen priors achieve 1.94 dB, 28.10 ns, and 13.74-degree RMSE, while the final prior-anchored residual GMM-head model improves them to 1.65 dB, 26.56 ns, and 11.39-degree RMSE. This makes the priors a strong baseline in their own right, not just an intermediate step: they concentrate the calibration over urban-density, building-height, visibility, and antenna-height regimes, leaving the neural model to learn only the local residual. The result is a divide-and-conquer strategy rather than a new universal propagation law.

For all those who failed throughout their journey.

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Abbreviations

2D two-dimensional

3D three-dimensional

3GPP Third Generation Partnership Project

6G sixth-generation mobile network

A2G air-to-ground

ABS aerial base station

AS angular spread

BT bachelor thesis

cGAN conditional generative adversarial network

CKM channel knowledge map

CNN convolutional neural network

COST-231 European Cooperation in Science and Technology 231

CPU central processing unit

CUDA Compute Unified Device Architecture

DirectML Direct Machine Learning

DL deep learning

DNN deep neural network

DPM Dominant Path Model

DS delay spread

DS-RPP distribution shift radio path loss prediction

EMA exponential moving average

ETSETB Escola Tecnica Superior d'Enginyeria de Telecomunicacio de Barcelona

FiLM feature-wise linear modulation

FLOPs floating-point operations

FM-RME foundation model-empowered radio map estimation

FR3 Frequency Range 3

FSPL free-space path loss

GAN generative adversarial network

GMM Gaussian mixture model

GPU graphics processing unit

GRETST Grau en Enginyeria de Tecnologies i Serveis de Telecomunicacio

GT ground truth

HDF5 Hierarchical Data Format version 5

HF high-frequency

HPC high-performance computing

ICASSP IEEE International Conference on Acoustics, Speech and Signal Processing

IEEE Institute of Electrical and Electronics Engineers

IRT Intelligent Ray Tracing

ITU International Telecommunication Union

JSON JavaScript Object Notation

KL Kullback-Leibler divergence

LoS line of sight

MAE mean absolute error

MATLAB Matrix Laboratory

MDN mixture density network

ML machine learning

MLP multi-layer perceptron

mmWave millimetre wave

MSE mean squared error

NLL negative log-likelihood

NLoS non-line of sight

NMSE normalized mean squared error

NPZ NumPy zipped archive

OLS ordinary least squares

PARS Programa Academic de Recorregut Successiu

PCT Parallel Computing Toolbox

PDE partial differential equation

PINN physics-informed neural network

PL path loss

PMHHNet path-loss map network with high-frequency and height awareness

PMNet path-loss map prediction network

PNG Portable Network Graphics

PSNR peak signal-to-noise ratio

RGB red-green-blue

RM3D RadioMap3DSeer

RMa rural macro

RMSE root mean squared error

RQ research question

RQ1 research question 1

RQ2 research question 2

RQ3 research question 3

RSSI received signal strength indicator

RT ray tracing

RTX NVIDIA RTX graphics-processing platform

Rx receiver

SDG Sustainable Development Goal

SOA state of the art

SSIM structural similarity index measure

SWA stochastic weight averaging

TR Technical Report

TTA test-time augmentation

Tx transmitter

U-Net U-shaped convolutional encoder-decoder network

U2G UAV-to-ground

UAV unmanned aerial vehicle

UMa urban macro

UMi urban micro

USC University of Southern California

VQGAN vector-quantized generative adversarial network

VRAM video random-access memory

WiCo-PG Wireless Channel Foundation Model for Pathloss Map Generation

WINNER Wireless World Initiative New Radio

YAML YAML Ain't Markup Language

CHAPTER 1

Introduction

Reliable radio coverage prediction is a central requirement for future 6G systems in which UAVs may act as aerial base stations, relays, or temporary coverage nodes. A classical ray-tracing simulator can produce detailed channel knowledge maps, but this is too expensive to repeat every time the transmitter height, the city morphology, or the deployment assumptions change. In the 6G literature, a channel knowledge map is a site-specific, location-tagged repository of channel information that can support environment-aware planning and reduce online channel-acquisition overhead [51, 52]. This thesis studies whether a learned surrogate can predict dense channel maps directly from urban geometry while preserving enough physical structure to generalize to cities not seen during training.

The project is formulated as dense image-to-image regression on 513×513 pixel city maps. The input is primarily a building-height topology map, a geometrically derived LoS/NLoS mask, and the UAV antenna height. The outputs studied through the project are path loss, delay spread, and angular spread. The final line of work combines interpretable ML-calibrated priors with deep residual models, so the neural network learns corrections around a strong physical anchor instead of rediscovering the whole propagation law from scratch.

1.1 Work goals

The initial technical targets were defined in physical units and were kept as the main success criteria throughout the project. The proposal also included a runtime objective, because a surrogate is only useful if it is clearly faster than generating the same maps through ray tracing:

- path-loss RMSE below 5 dB;
- delay-spread RMSE below 50 ns;
- angular-spread RMSE below 20°;
- separate reporting for global, LoS, and NLoS regions;
- a 10 to 100× reduction in channel-map generation time compared with ray tracing.

These numerical targets were set in the project proposal and kept unchanged throughout the thesis. They are meant to match the order of magnitude at which a ray-tracing surrogate starts being useful for planning and coverage decisions. A few dB of path-loss RMSE is compatible with practical link-budget margins, and tens of ns of delay spread and tens of degrees of angular spread stay well inside typical multipath spread values reported by 3GPP TR 38.901 for UMa/UMi scenarios. Chapter 4 checks the runtime target explicitly by comparing the measured final-generator time with the measured MATLAB ray-tracing time on the same Barcelona geometry and hardware.

The most important methodological objective was not only to obtain a low average RMSE, but to do so under a strict city-holdout protocol. In this setting, complete cities are reserved for validation and testing. This is a harder and more deployment-relevant condition than random pixel or random tile splits, because the model must extrapolate to urban layouts it has never seen.

1.2 Research questions, answers and contributions

The thesis is organised around three research questions, which also define the three main scientific claims of the final pipeline:

- **RQ1.** Can a single deep-learning surrogate predict per-pixel path loss, delay spread, and angular spread on 513×513 channel-knowledge maps, under strict city-holdout evaluation, with errors compatible with the thesis targets?
- **RQ2.** Which part of the prediction should be left to a calibrated analytic prior and which part should be left to the neural network, so that the hybrid model is both accurate and interpretable?
- **RQ3.** How does the continuous UAV transmitter height (12 m to 478 m) affect the prediction error, and is the same architecture able to cover the full height range without per-height retraining?

In the final version of the project these questions can be answered directly. RQ1 is answered positively for the fixed CKM simulation contract: the selected prior-anchored residual model predicts all three dense targets on unseen cities below the original error targets. RQ2 is answered by a division of labour: stable radial and visibility-dependent structure is handled by frozen calibrated priors, while the neural network learns bounded residual corrections. RQ3 is answered by the height-bin analysis: the same architecture covers the full UAV height range, but low-altitude and NLoS-heavy regimes remain the hardest cases and therefore require explicit reporting rather than a single global RMSE.

The main contributions corresponding to these questions are:

- A reproducible pipeline for dense 513×513 CKM prediction on the FR3 UAV dataset, with ground-only masking, city-holdout splits, and sinusoidal FiLM conditioning on continuous UAV altitude.
- An ML-calibrated path-loss prior based on an enhanced two-ray LoS model and a regime-wise OLS NLoS head, achieving 1.75 dB LoS and 3.40 dB NLoS RMSE under city-holdout evaluation.
- An ML-calibrated spread prior combining log-domain ridge regression with geometry and morphology features, reaching 28.10 ns delay-spread RMSE and 13.74° angular-spread RMSE on the same split.
- A prior-anchored joint neural residual GMM-head model that shares a single backbone across the three targets, conditioned on the frozen path-loss and spread priors.
- A distribution-first diagnosis of the previous PMHHNet family that identifies NLoS mode collapse and motivates the distribution-first GMM architecture. The distribution analysis is released as supporting documentation alongside the code.

1.3 Thesis outline

The state-of-the-art chapter surveys cGAN and U-Net image-to-image baselines, the PMNet and RMTransformer families on the ICASSP 2023 challenge, physics-hybrid and PINN approaches, AIRMap, Gao et al., PathFinder, and distribution-aware and heteroscedastic methods; it closes with the fairness axes along which the final model is positioned. The methodology chapter describes the CKM dataset, the ground-mask protocol, the shared PMHHNet architecture with sinusoidal height conditioning, and the four prior-detail subsections for the path-loss prior, spread priors, and residual GMM. The results chapter reports the quantitative outcomes: the LoS waterfall, per-expert and per-height breakdowns, spread-prior results, and the final test evaluation on unseen cities. The sustainability and conclusions chapters close the thesis, followed by the appendices with the representative validation panels and the final CKM generator documentation.

1.4 Requirements and specifications

The project uses the CKM HDF5 dataset `CKM_Dataset_270326.h5`. The canonical dataset contains 66 real-city scenarios and 16180 samples. Each sample is a 513×513 pixel raster with the following fields:

Table 1.1: *Main CKM dataset fields used in the thesis.*

Field	Shape	Type	Role
`topology_map`	513^2	float32	Building height in metres. Pixels with buildings are not valid receiver locations.
`los_mask`	513^2	uint8	Binary LoS/NLoS visibility mask.
`path_loss`	513^2	uint8	Ray-traced path loss in dB, quantized at approximately 1 dB.
`delay_spread`	513^2	uint16	Delay-spread target, stored as integers and evaluated in ns after decoding.
`angular_spread`	513^2	uint8	Angular-spread target, stored as integers and evaluated in degrees after decoding.
`uav_height`	scalar	float32	UAV transmitter height, used for conditioning and physical priors.

All losses and metrics are computed only on ground pixels. Building pixels are kept in the input topology because they explain propagation, but they are excluded from error accumulation because the receiver is assumed to be on the ground. The fixed receiver height used in the analytic priors is 1.5 m. The transmitter is the UAV antenna, whose height varies continuously across the dataset.

1.5 Methods and procedures

The work started from a U-Net and pix2pix-style cGAN baseline [14]. This was a natural first approach because the task is dense map prediction, but the adversarial component did not solve the physical errors that mattered most. Later experiments changed decoder upsampling, added multi-scale supervision, separated metrics and regimes, introduced physical prior channels, and finally moved to topology-partitioned experts.

The largest methodological change came after the models repeatedly failed on NLoS tails. Histogram analysis showed that the path-loss distribution in NLoS regions is not well described by a single Gaussian-like target. This motivated the distribution-first architecture: a sample-level GMM head followed by a map head that assigns pixels to distribution components. The same idea was then applied to delay and angular spreads, where spike-plus-tail behaviour is even more visible.

In the final stage, deep learning changed role again. The path-loss branch provided a strong interpretable prior from FSPL, enhanced two-ray LoS propagation, and calibrated NLoS corrections. The spread-prior module provided log-domain ridge-calibrated priors for delay and angular spread. The final model then combined these frozen priors with a shared multi-task residual GMM-head network.

1.6 Work plan

The original work plan assumed a mostly linear path: literature review, dataset preparation, baseline U-Net/cGAN training, model improvement, evaluation, and writing. The actual project became more experimental because each stage exposed a different limitation. Figure 1.1 therefore shows the plan as the major technical phases rather than as all 80 individual experiments.

The main deviation from the initial plan was the discovery that a standard image-to-image CNN could look plausible in LoS-dominated regions while still failing badly in NLoS regions. This forced several changes: stricter ground-only masking, regime-level diagnostics, topology experts, physical priors, and finally the distribution-first and prior-anchored designs. The result is a project whose final method is more hybrid and more interpretable than the initial proposal anticipated.

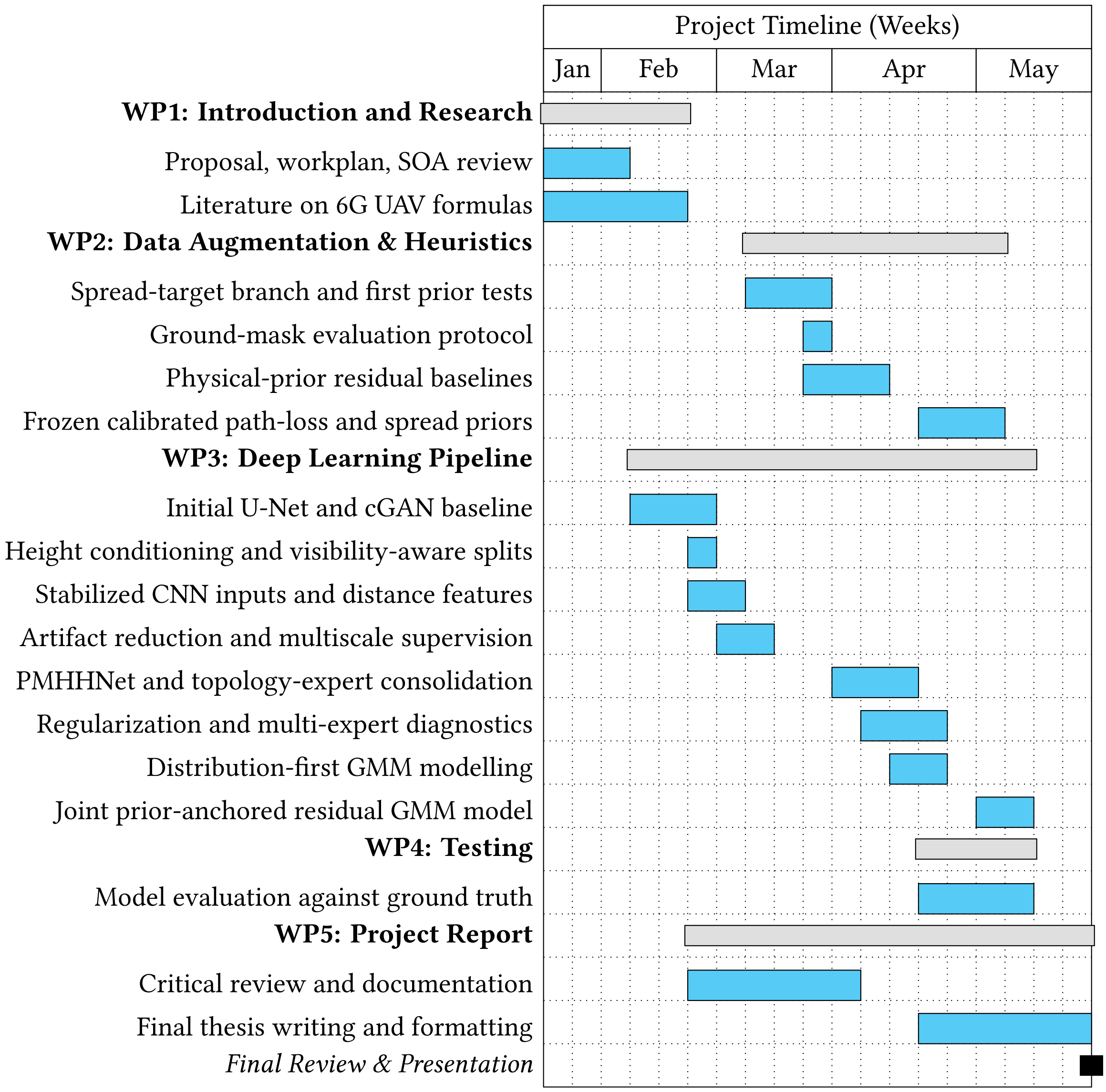

Figure 1.1: *Gantt diagram of the project, grouped by the main architectural phases rather than by individual experiments.*

CHAPTER 2

State of the Art and Technical Background

Radio-map prediction sits at the intersection of classical wireless propagation modelling, statistical channel analysis, and dense computer-vision regression. This chapter surveys the relevant state of the art across four axes that jointly motivate the design choices in the final residual GMM-head model: (i) classical and empirical propagation models, (ii) deep learning for radio-map estimation, (iii) physics-informed and hybrid neural approaches, and (iv) distributional and multi-target channel modelling. A final section frames fair-comparison criteria and positions the final model within the literature.

2.1 Channel knowledge maps as the thesis context

The term channel knowledge map (CKM) refers to a site-specific database of location-tagged channel knowledge, such as path loss, channel gain, angles, or delay information, used to make future radio systems environment-aware [51, 52]. This framing is more precise than a generic radio map: the map is not only an image-like coverage estimate, but a queryable prior that can reduce channel-state acquisition overhead, guide beamforming, support planning, and provide initial channel knowledge before online measurements are available.

Recent CKM literature also clarifies why the thesis uses a large offline ray-traced dataset rather than sparse field samples. Xu and Zeng [46] analyse how CKM construction error depends on spatial sample density and on the number of neighbouring samples used for online prediction, showing the explicit cost/accuracy trade-off of measurement-driven CKM construction. For a thesis whose goal is to learn dense 513×513 targets under city holdout, this motivates the simulation-first route: it provides full supervision everywhere, while the final model is still evaluated as a surrogate for a ray-traced CKM rather than as a replacement for field calibration.

The data ecosystem has also expanded since the first path-loss challenge. RadioMap3DSeer [49] and the ICASSP 2023 challenge [50] provide the closest open benchmark family for path-loss image prediction, while the directive-antenna dataset of Jaensch et al. [17] and CKMImageNet [45] show the broader move toward open, location-specific datasets with richer channel and environment representations. The CKM dataset used in this thesis sits in the same data-driven tradition, but differs in its continuous UAV height conditioning, real-city holdout split, and simultaneous path-loss, delay-spread, and angular-spread targets.

2.2 Classical propagation models and channel statistics

2.2.1 Free-space and two-ray propagation

The most fundamental path-loss term is the *free-space path loss* (FSPL), which depends only on carrier frequency f and Tx-Rx distance d:

$$\mathrm{PL}_{\mathrm{FSPL}} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right). \tag{2.1}$$

FSPL is the single-direct-ray reference baseline in unobstructed conditions, not a lower bound on the final path loss once coherent reflections are present. In aerial scenarios a specular ground reflection can produce constructive or destructive interference periodically with distance. The *coherent two-ray* model captures this by adding the direct and reflected fields before converting to dB [35, 5]:

$$E_{\mathrm{total}} = E_{\mathrm{direct}} + \Gamma \cdot E_{\mathrm{reflected}} \cdot e^{j\Delta\phi}, \tag{2.2}$$

where Γ is the reflection coefficient and $\Delta\phi$ is the phase difference determined by the path-length difference. A calibrated two-ray model fitted to the CKM dataset achieves approximately 1.75 dB RMSE on line-of-sight (LoS) pixels; the deterministic radial structure explains most LoS variance without any neural network.

2.2.2 Empirical and semi-empirical models

Empirical models such as the Okumura-Hata and COST-231/Hata formulas extend FSPL by incorporating frequency, antenna heights, and city-type correction terms [35, 6]. While computationally cheap, they are calibrated on specific frequency bands and morphology types, limiting accuracy when the geometry departs from the fitting dataset.

The 3GPP Technical Report 38.901 [1] formalises stochastic channel models for frequencies from 500 MHz to 100 GHz for urban macro (UMa), urban micro (UMi), rural macro (RMa), and indoor scenarios. It provides separate path-loss polynomials for LoS and non-LoS (NLoS) conditions, breakpoint distances, and shadow-fading standard deviations, which serve as reference baselines throughout this thesis.

2.2.3 UAV and air-to-ground channel models

Classical terrestrial channel models assume the transmitter is at rooftop height or below. Unmanned aerial vehicle (UAV) base-station scenarios introduce a third dimension: the UAV can fly from near-ground level to several hundred metres. Al-Hourani et al. [11] showed that the LoS probability increases monotonically with elevation angle and depends on city morphology type (suburban, urban, dense urban, high-rise). The A2G modelling literature [11, 19] commonly parameterises this dependence through logistic functions of the elevation angle $\theta = \arctan\big((h_{\mathrm{tx}} - h_{\mathrm{rx}})/d_{2D}\big)$, where h_{tx} and h_{rx} are transmit and receive heights respectively and d_{2D} is the horizontal distance. 3GPP TR 38.901 [1] is used here as standardized LoS/NLoS large-scale channel-model context, not as the direct source of that elevation-logistic A2G formula.

Recent work by Saboor and Vinogradov [36] makes this height dependence especially relevant for the present thesis: in simulated 26 GHz urban A2G links, LoS path loss remains close to a free-space exponent, whereas NLoS path-loss exponents and shadow fading evolve with ABS height and with urban layout. Vinogradov et al. [43] also propose

3D shadow projections as a fast way to generate spatially consistent A2G LoS maps without full point-by-point ray tracing. These results support two design choices used throughout this work: treating the LoS/NLoS mask as a physical input rather than as a cosmetic label, and conditioning every dense prediction on UAV height.

A key finding from the CKM dataset is that LoS path-loss residuals after two-ray subtraction form almost perfect radial rings around the transmitter. This confirms that buildings act primarily as a *visibility mask* in LoS conditions rather than as the main source of residual path-loss variance, validating the prior-subtraction philosophy of the final path-loss prior.

2.2.4 Delay spread and angular spread in standard channel models

Beyond path loss, full channel characterisation requires second-order statistics. The *root-mean-square delay spread* τ_{rms} measures the time-domain dispersion of multipath arrivals and drives inter-symbol interference in wideband systems. The *azimuth angular spread* σ_{AS} at the receiver quantifies the spatial diversity of arriving paths.

3GPP TR 38.901 [1] models both quantities as log-normal random variables conditioned on the LoS/NLoS state and distance. WINNER II follows the same broad log-domain large-scale-parameter convention [20]. In practice, the distributions of delay spread and angular spread are right-skewed with a heavy exponential tail and often include a spike near zero representing the dominant LoS path. The calibrated spread-prior analysis in this thesis empirically confirms this *spike-plus-exponential* shape for all topology classes in the CKM dataset, with skewness exceeding 5 and kurtosis exceeding 40 in several morphology classes. A Gaussian model cannot represent this shape faithfully, motivating the mixture and distributional approach used by the final residual GMM-head model.

2.3 Deep learning for radio-map estimation

2.3.1 Image-to-image translation baselines

The radio-map prediction problem can be cast as an image-to-image regression: given an input topology map (and optionally a transmitter location channel), the model predicts a path-loss map of the same spatial extent. The pix2pix framework [14] established a general recipe for such tasks using a conditional GAN with a U-Net generator and a PatchGAN discriminator. Early iterations of this thesis used this formulation. In practice the adversarial objective sharpens spatial texture but does not address the height-dependent radial pattern or the heavy NLoS tails that dominate the error budget in the CKM dataset.

RadioUNet [24] is the canonical convolutional path-loss-map baseline before the PM-Net challenge family. In its accurate-map settings it reports grey-level RMSE values around 0.0203–0.0384; using the paper's 80 dB grey-level conversion gives an approximate 1.6 dB–3.1 dB path-loss scale. RadioGUNet [48] extends this outdoor RadioMapSeer line with group-equivariant convolutions, reporting 1.304 dB on the easiest DPM setting, 1.936 dB on the IRT setting, and 1.392 dB with cars. These papers are useful path-loss scale references, but they do not report dense delay-spread or angular-spread maps and do not use the CKM city-holdout, continuous-UAV-height contract.

2.3.2 PMNet and the ICASSP outdoor challenge family

PMNet [21] was the winner of the first ICASSP 2023 Outdoor Pathloss Radio Map Prediction Challenge [50]. It uses a deep encoder-decoder with skip connections, trained and evaluated on the RadioMap3DSeer dataset [49]. RadioMap3DSeer consists of city maps derived from urban 3D geometry with rooftop transmitters placed 3 m above the selected rooftop in the dataset description, and building interior pixels masked to zero error.

The challenge metric is RMSE in normalised image units. As an approximate dB-scale reference, the RadioMap3DSeer image mapping clips path gain to the range $[\mathrm{PL}_{\mathrm{thr}}, \mathrm{PL}_{\mathrm{max}}] = [-111, -75]$ dB, yielding a 36 dB dynamic window under that interpretation. Multiplying normalised RMSE by 36 gives the following rough dB-scale references:

Table 2.1: *Approximate dB-scale RMSE references for ICASSP 2023 outdoor challenge methods. The normalised scores are taken from the challenge family and converted with the* 36 dB *RadioMap3DSeer image window described in the dataset paper [50, 49].*[1]

Method	RMSE (norm.)	Approx. RMSE (dB)
PMNet [21]	0.0383	1.38
PPNet baseline [50]	0.0507	1.83

These numbers are not directly comparable to the CKM evaluation for two structural reasons: (i) fixed Tx height eliminates the continuous UAV height-conditioning challenge; (ii) the 36 dB window is much narrower than CKM's effective 65 to 130 dB range, intrinsically compressing RMSE values. The table is therefore useful as a scale reference for modern image-to-image radio-map predictors, not as evidence that the final CKM task is solved by those architectures. In particular, the final model must predict over unseen real-city geometries, at native 513×513 resolution, with a continuously varying UAV altitude and with metrics computed only over valid ground receiver pixels.

A separate transfer test performed in this thesis with official PMNet checkpoints placed official PMNet checkpoints on CKM samples. The best zero-shot result was approximately 84 dB RMSE, with most models collapsing to a nearly constant prediction. After oracle linear calibration from test data the minimum achievable RMSE was still approximately 42 dB. This confirms that PMNet's strong challenge score does not transfer to the structurally harder CKM setting.

2.3.3 Transformers, state-space models, and attention-based encoders

The RMTransformer [26] evaluates a hybrid MaxViT/CNN encoder–decoder on the USC/PMNet dataset rather than on the ICASSP challenge test set. It reports a normalised RMSE of 0.007148 versus 0.01046 for PMNet under a random 90/10 split on 256×256 maps. The paper normalises received power/path loss from -254 to 0 dBm, so the reported pixel RMSE corresponds to roughly 1.82 dB on its own scale, and the reported channel

[1] This conversion should be read as an approximate scale reference, not as a like-for-like physical RMSE. It uses the path-gain clipping window $[\mathrm{PL}_{\mathrm{thr}}, \mathrm{PL}_{\mathrm{max}}] = [-111, -75]$ dB associated with the normalised target image, so the physical span is $-75 - (-111) = 36$ dB. Multiplying the normalised RMSE values reported in the challenge paper [50, 49] by this factor converts them to physical dB, which gives $0.0383 \times 36 \approx 1.38$ for PMNet.

prediction error of 0.008099 corresponds to roughly 2.06 dB. This supports the use of attention for long-range urban context, but it is not inserted in Table 2.1 because it is not measured on RadioMap3DSeer or under the ICASSP challenge contract.

Recent attention and foundation-model variants extend the effective receptive field of radio-map predictors [40, 47]. They are attractive when global context is necessary, such as tracking a shadowing corridor across an entire 513×513 map, but they usually require substantially more data and compute than convolutional methods to reach the same local detail quality.

2.3.4 Foundation models for radio maps

The most recent shift in the literature is towards *wireless channel foundation models* pre-trained on large heterogeneous datasets before task-specific fine-tuning. WiCo-PG [40] uses dual VQGANs and a transformer with frequency-guided mixture-of-experts. It exploits RGB drone camera images as an auxiliary modality and achieves an NMSE of 0.012 on its constructed multi-modal U2G benchmark. RadioLAM [27] targets fine-grained 3D radio-map estimation from ultra-low sampling rates using a large generative model pipeline. FM-RME [47] frames radio map estimation as a foundation modelling problem with self-supervised pre-training on simulated propagation data.

These foundation approaches represent the long-term direction of the field but require massive training corpora and compute budgets well beyond the scope of a single CKM dataset. They are cited here as the horizon against which this thesis positions its own contributions.

2.4 Physics-informed and hybrid neural approaches

2.4.1 Geometry-assisted and diffraction-aware deep learning

Purely data-driven models must implicitly learn diffraction geometry from the topology map. An alternative is to pre-compute a physics-based channel prior and provide it as an additional input channel, reducing the learning problem to residual estimation.

Shadow-projection methods [43] occupy an intermediate point between stochastic LoS-probability models and full ray tracing. They use 3D building geometry to project blocked ground regions from the aerial transmitter, producing a deterministic LoS map that is spatially consistent along user trajectories. The CKM dataset already provides a ray-traced LoS mask, so this thesis does not need to generate the mask itself; nevertheless, the method is directly relevant because it explains why a binary visibility channel can carry so much of the structure that the neural network otherwise has to infer from the height map alone.

Geometry-assisted diffraction features [4] use virtual obstacles, multi-screen knife-edge propagation, and scattering-aware local geometry cues to inject physically meaningful NLoS information into a deep model. This is closely related to the motivation behind the thesis diffraction-prior experiments, although the implementation here is a lightweight per-map channel rather than the full geometry model-assisted learning framework of that paper. The final pipeline ultimately relies on the stronger calibrated NLoS prior rather than on a standalone knife-edge ablation.

2.4.2 Physics-informed neural networks

The ReVeal architecture [39] operates as a physics-informed neural network (PINN) by incorporating a second-order PDE residual as an auxiliary loss term. This penalises predictions with inconsistent local spatial curvature under the derived propagation constraint, reducing corner-transition artefacts. ReVeal achieves 1.95 dB RMSE in rural scenarios using only 30 training points by leveraging the physical constraint as a strong regulariser. The PMHHNet diagnostic phase in this thesis implemented a lightweight analogue: a masked Laplacian penalty on the residual prediction, which penalises locally non-smooth outputs without solving the full ReVeal operator.

2.4.3 Prior-residual decomposition

A principled hybrid approach is to *freeze* a calibrated analytic prior and train the neural component to predict only the structured residual. This design reduces the effective learning problem from the full path-loss range to a narrower residual distribution, improving data efficiency and interpretability.

The final path-loss prior implements this philosophy: an enhanced two-ray model fitted per height bin provides a prior map $\hat{y}_{\mathrm{PL}}(\mathbf{x}) = \mathrm{TwoRay}(d, h_{\mathrm{tx}})$, and the neural component predicts the residual $\epsilon = y - \hat{y}_{\mathrm{PL}}$. The final spread prior extends the idea to delay spread and angular spread by using a ridge-regression model in log space with TX-centred geometric features and local morphology statistics. The final residual GMM-head model unifies all three targets under a single shared network anchored to both priors.

2.5 Channel statistics prediction: delay spread and angular spread

2.5.1 Empirical models for spread parameters

Standard channel models characterise delay spread and angular spread through log-domain large-scale parameters tied to propagation state and scenario. TR 38.901 [1] provides scenario-specific tables and formulae for these parameters, with dependencies on variables such as distance, frequency, terminal heights, and LoS/NLoS state across UMa, UMi, and RMa scenarios. These parameterisations are limited to the morphology class and frequency band of the underlying measurement campaigns.

Log-domain modelling is physically motivated: spread parameters are multiplicative quantities in physical space, so their logarithm is more Gaussian-like and additive-noise assumptions are better justified. The calibrated spread prior adopts this convention by working in $\log(1 + y)$ space throughout, consistent with the 3GPP empirical modelling framework.

2.5.2 Deep learning for wideband channel statistics

Work on deep learning for delay and angular spread is sparser than for path loss. Most published methods either predict spread globally per sample (a single scalar) rather than per pixel, or treat it as an ancillary output of a multitask path-loss model. Per-pixel dense prediction of delay spread and angular spread over 513×513 maps is, to the best of the author's knowledge, not addressed by published methods at the time of writing.

The closest external quantitative references for spreads are therefore scalar channel-characteristic predictors rather than dense CKM-map benchmarks. Huang et al. [12] are not used in the results comparison tables because their Wireless InSite campus dataset predicts scalar A2G link characteristics from coordinates and link features, not dense CKM maps from city topologies.

The main challenge is the spike-plus-exponential distribution shape of spread targets. A model trained only with mean-squared error (MSE) collapses to the conditional mean, which lies in the interior of the spike and the exponential tail and is therefore a poor representative of either regime. The distribution-first spread predecessor addressed this with a Gaussian mixture model (GMM) distribution head, and the final residual GMM-head model incorporates the log-domain ridge prior as an anchor before training the residual neural head.

2.5.3 Distributional mismatch and the spike-plus-exponential problem

The spike-plus-exponential shape, consisting of a Dirac-like concentration near zero (the LoS-dominated case) plus a heavy exponential tail (NLoS multipath), was confirmed empirically on the CKM dataset. Skewness exceeds 5 and excess kurtosis exceeds 40 for both delay and angular spread in several topology classes. A single Gaussian or even a symmetric loss function does not fit this family.

Mixture models are the natural solution, but the required number of components depends on what is being modelled. The histogram study selected the best family for the raw target distributions, where path loss and the spread targets often needed four or five components to match the full marginal shape. The final model uses three components for a different object: the residual distribution after the frozen priors have already explained the dominant LoS, NLoS, delay-spread, and angular-spread structure. This is a small design change rather than a contradiction: the prior removes much of the target-side multi-modality, so the residual head only needs a compact mixture to represent under-correction, near-zero correction, and over-correction modes. This motivates the Stage-A distribution head in the path-loss and spread predecessors, and the lighter GMM-3 residual head carried into the final model.

2.6 Distribution-aware and multi-target modelling

2.6.1 Heteroscedastic uncertainty

Kendall and Gal [18] established the framework for heteroscedastic neural networks in computer vision: predict both a mean output $\mu(x)$ and a log-variance $\log \sigma^2(x)$, optimising the negative log-likelihood (NLL)

$$\mathscr{L} = \frac{(\mu - y)^2}{\sigma^2} + \log \sigma^2. \tag{2.3}$$

Pixels with high aleatoric uncertainty (deep NLoS shadows, building edges) receive lower effective weight, while the network is penalised for overestimating uncertainty everywhere. A PMHHNet uncertainty diagnostic applied this loss to path-loss prediction, with σ^2 predicted as a second output channel of PMHHNet.

2.6.2 Gaussian mixture model heads

Beyond heteroscedastic Gaussians, the GMM head allows multi-modal output distributions. Mixture-density networks have already appeared in wireless channel modelling:

Saleh et al. [38] use an MDN to predict probabilistic mmWave path-loss distributions, Garcia Marti et al. [29] learn a stochastic Gaussian-mixture channel model for physical-layer design, and Lee et al. [23] derive conditional received-power PDFs with a mixture-density network. These works support the use of mixture outputs for wireless propagation, but they operate at the level of scalar or channel-sample distributions rather than dense radio-map generation.

The distribution-first path-loss predecessor implements a two-stage architecture: Stage A produces per-sample GMM parameters $\{\pi_k, \mu_k, \sigma_k\}_{k=1}^{K}$ via a shallow convolutional encoder with sinusoidal FiLM conditioning on UAV height; Stage B is a small U-Net that produces per-pixel component assignments $p_k(i, j)$ and within-component deviations $z(i, j)$. The final prediction is

$$\hat{y}(i, j) = \sum_{k=1}^{K} p_k(i, j) \cdot \left(\mu_k + z(i, j) \cdot \sigma_k\right). \tag{2.4}$$

This formulation avoids the mode-collapse failure of MSE regression while maintaining a compact architecture. The same two-stage GMM design is used in the spread predecessor for delay and angular spread, with modified parameters for the spike-plus-exponential prior structure. The innovative aspect is therefore not the isolated use of a Gaussian mixture, but its role as a global distribution head that tells the dense decoder what value distribution the map should contain before the model solves the harder pixel-placement problem.

2.6.3 Stochastic diffusion models

Conditional diffusion models applied to radio maps [44] generate physically plausible stochastic variations in deep shadow zones rather than predicting a single blurry mean. The iterative denoising procedure can be conditioned on topology maps, transmitter locations, and obstacle prompts. RadioDiff reports lower normalised RMSE and higher SSIM/PSNR than CNN and sequence-model baselines on static and dynamic RadioMapSeer settings, but its metrics are image-normalised rather than physical dB values. These models are computationally expensive at inference time and are listed as a future-work horizon for this thesis.

2.6.4 Multitask and joint prediction

Training a single shared backbone across multiple output targets can improve sample efficiency and consistency between targets. Path loss, delay spread, and angular spread are physically correlated: a deep shadow corresponds to high delay spread and potentially wide angular spread from diffracted paths. A joint model can therefore use the intermediate feature maps learned for path loss to partially explain spread patterns, reducing the effective learning problem.

The final residual GMM-head model instantiates this idea explicitly: one shared encoder processes the topology map and UAV height, and three parallel decoder heads predict path-loss residual, delay-spread residual, and angular-spread residual respectively, all anchored to their corresponding physical priors. The implementation defines a composite per-target objective: probabilistic NLL, KL and moment terms are active in the preceding large-capacity residual stage, while the reported final checkpoint is selected after an RMSE/MAE-dominant fine-tuning stage.

2.7 Training stabilisation and regularisation techniques

2.7.1 Stochastic weight averaging

Stochastic weight averaging (SWA) [15] averages model weights along the training trajectory at a low, approximately constant learning rate. This finds wider optima in the loss landscape that generalise better to unseen data. The CKM city-holdout evaluation makes generalisation particularly important since validation and test cities are never seen during training. The PMHHNet diagnostic phase incorporated SWA starting from 60 % of the training budget.

2.7.2 Exponential moving average weights

Separate from SWA, exponential moving average (EMA) tracking of the model weights at a decay rate of 0.9975 was useful in the earlier PMHHNet diagnostic branch because it produced smoother validation checkpoints. In that older branch, EMA weights were used for validation scoring and the smoothed score fed the ReduceLROnPlateau scheduler. The selected Try 80 model is different: it does not use EMA or ReduceLROnPlateau, and instead selects the checkpoint by the validation RMSE/MAE score under an AdamW warmup-plus-cosine `LambdaLR` schedule.

2.7.3 Feature-wise linear modulation

Feature-wise linear modulation (FiLM) [34] conditions the activations of a neural network on a scalar or vector input by applying learned affine transforms $\gamma(c) \cdot \mathbf{h} + \beta(c)$ to intermediate feature maps. Sinusoidal positional encoding of the UAV height scalar (using log-spaced frequencies over the full 12 to 478 m range) is fed into FiLM layers at multiple encoder depths. This ensures that the UAV altitude continuously modulates the activation statistics rather than being handled as a discrete bin, which is critical when the effective propagation regime changes substantially with height.

2.8 Key benchmarks and fair-comparison protocol

2.8.1 Why headline numbers cannot be compared directly

The most widespread mistake in radio-map literature reviews is comparing headline RMSE values without verifying whether the underlying tasks are equivalent. The following evaluation axes are the minimum required for a valid comparison:

1. **Split:** city-holdout vs. in-distribution random split vs. same-scene split.
2. **Transmitter height:** fixed rooftop, discrete bins, or continuous UAV altitude.
3. **Building pixels:** excluded from loss, excluded from metrics, set to zero error, or included.
4. **Target quantity:** path loss (dB), path gain (dBm), received power, delay spread, angular spread.
5. **Dynamic range and clipping:** determines the RMSE floor and ceiling independently of model quality.
6. **Inference extras and calibration source:** test-time augmentation (TTA), SWA, EMA weights, and whether calibration is train-only, validation-based, or fitted

from test/field measurements.

This thesis uses a strict city-holdout split, continuous UAV height, ground-only metrics (building pixels excluded from loss and metrics), native 513 × 513 pixel maps, and no test-data or field calibration after the train-calibrated priors and selected final model are frozen. Calibration is therefore not shown as a binary column in Table 2.2: the relevant question is not whether a method has fitted parameters, but whether it adapts to the evaluation scene using validation, test, or field measurements. Table 2.2 summarises the structural axes that can be compared compactly.

Table 2.2: *Evaluation condition comparison across representative methods.*

Method	**Unseen city**	**Cont. UAV height**	**≥512 px**	**Comparable?**
PMNet / ICASSP 2023 [21]	No	No	No	No
RMTransformer [26]	No	No	No	No
RadioGUNet [48]	Partial	No	No	Partial
AIRMap [37]	Partial	No	No	Partial
Gao et al. [9]	No	No	Partial	Partial
Tarhouni et al. [42]	No	No	No	Physical ref.
PathFinder [53]	Partial	No	No	Partial
Final model (this thesis)	**Yes**	**Yes**	**Yes**	**Reference**

Two consequences follow from this checklist. First, method rankings can change substantially when the split changes from random maps to unseen cities. A model that has learned the texture, street orientation, and height statistics of a known city may still fail when the same transmitter-height range is evaluated on a new morphology. For this reason, this thesis treats city holdout as a core part of the task definition rather than as an optional robustness test.

Second, path loss, delay spread, and angular spread should not be treated as three copies of the same regression problem. LoS path loss often contains a strong deterministic radial component, whereas NLoS path loss depends on blocked-link geometry and local morphology. Delay and angular spreads are even more distributional: many pixels are near zero, but physically important tails remain. A fair comparison therefore has to report the target, regime, masking policy, and split together; otherwise an apparently lower RMSE can simply mean that an easier quantity was measured.

2.8.2 Tarhouni et al.: measured suburban path-loss prediction

Tarhouni et al. [42] study machine-learning-based path-loss prediction in a sub-6 GHz suburban campus. It is useful because the errors are in physical dB and tied to a concrete environment, but it is terrestrial, lower-frequency, and not a dense UAV CKM with delay- and angular-spread targets. Its role here is qualitative: real-environment generalisation tends to move back toward multi-dB errors, which supports keeping city holdout as part of the task definition.

2.8.3 AIRMap: adaptive digital-twin radio maps

AIRMap [37] is an adaptive U-Net autoencoder processing fixed 200 × 200 elevation-map tensors with variable spatial resolution (2.5–15 m/pixel, corresponding to 500 m–3 km per side) for ultra-low-latency digital-twin synchronisation. It reports sub-4 dB RMSE against ray-traced path-gain maps, and a separate lightweight calibration step using 20 % of field measurements reduces the median measurement-domain error to about five percent. Its strong numbers are not directly comparable to CKM because the target is path gain rather than absolute path loss, the input is a single elevation channel, and the transmitter configuration is not the continuous UAV-height setting used in this thesis. Its reported ~4 ms inference time on an NVIDIA L40S is nevertheless a strong deployment reference: the value of a CKM surrogate is not only accuracy, but producing dense maps fast enough for planning loops.

2.8.4 Gao et al.: multi-layer segmentation with corridor weighting

Gao et al. [9] propose a ResNet-based outdoor path-loss predictor with Tx depth maps, Rx depth maps, a distance map, and a weighting map that emphasises the direct Tx-Rx corridor. Using the ICASSP 2023 outdoor data, Table III of the paper reports their method at 5.59 dB RMSE, compared to 8.09 dB for PPNet, 9.56 dB for RPNet, and 8.72 dB for a twelve-layer Vision Transformer baseline; Table II reports 12.19 dB RMSE on the ITU Challenge 2024 large-area subset, also ahead of the same baselines, with roughly 60% fewer FLOPs. This formalises an internal lesson of the CKM work: the important geometry is not the whole map uniformly, but the subset near the propagation corridor.

The PMHHNet diagnostic phase implemented only a weaker Tx-centred radial proxy of that idea. The final path-loss prior then solves the LoS part more cleanly with a two-ray radial prior, while the final residual model uses the calibrated prior as an input channel. A full Gao-style per-Rx corridor map remains future work because generating one for every receiver pixel at 513×513 resolution is substantially more expensive than using the current Tx-centred distance and obstruction features.

2.8.5 PathFinder: multi-transmitter domain adaptation

PathFinder [53] tackles distribution shift via disentangled feature encoding and mask-guided low-rank attention, reporting the lowest MSE of 0.1068 and RMSE of 0.3263 on unseen rural datasets. On the RadioMap3DSeer DS-RPP split, its normalised RMSE of 0.0331 can be approximately converted with the same 36 dB RadioMap3DSeer image-window convention, giving about 1.19 dB. The unseen-rural score would be about 11.75 dB under the same window, but this is only an approximate scale reading because the paper and code report normalised PNG image units and do not state a separate rural physical target window. The explicit domain-generalisation focus aligns with the CKM city-holdout spirit, though PathFinder's setup involves multi-transmitter inputs and a different normalisation convention. Techniques from PathFinder such as transmitter-oriented mixup augmentation are listed as medium-priority future work. The final model is not multi-transmitter, but the same concern motivates its city-holdout split: a CKM surrogate is useful only if it works on new urban layouts, not only on shuffled samples from the same city.

2.8.6 ICASSP 2025 indoor challenge

The ICASSP 2025 Indoor Pathloss Challenge [2] presents a structurally different problem, dense wall-penetration NLoS effects in bounded interior spaces, but its public results are a useful warning against over-interpreting sub-2 dB outdoor challenge numbers. The official result page reports that the winning SIP2Net submission obtains a weighted-average RMSE of 9.411 dB; IPP-Net reaches 9.501 dB; TerRaIn reaches 10.325 dB; and TransPathNet reaches 10.397 dB [13, 28, 8, 25]. These are not outdoor UAV results, but they show that when NLoS physics becomes harder and less directly observable from the input map, modern neural architectures again hit a roughly 9 dB to 10 dB error band. The comparison to the 9.41 dB PMHHNet reference is diagnostic, not competitive: it shows why the final stage moved from direct regression toward strong priors, regime separation, and residual distribution modelling.

2.9 Emerging techniques and position of the final model

Across the benchmarks reviewed here, the recurring 9 to 10 dB error band in harder NLoS-heavy settings reflects the difficulty of inferring volumetric propagation from a 2D building-height map. The most plausible routes beyond that wall are richer geometry and diffraction priors [4], PDE-style physics losses such as ReVeal [39], sparse target-domain adaptation such as RadioPiT [41], diffusion refinement [44], and larger radio-map foundation models [47]. In this thesis those options were filtered into a narrower single-GPU design: calibrated train-only priors, explicit LoS/NLoS and topology diagnostics, distribution-aware residual heads, and strict city-holdout reporting.

The final model is therefore novel less because of one isolated component than because of the combination: dense prediction of path loss, delay spread, and angular spread on 513×513 maps; frozen physics/statistical anchors; continuous UAV-height conditioning; and a residual GMM-head that keeps the sample-level distribution visible before decoding per-pixel maps [38, 29, 23]. It does not claim to replace ray tracing in arbitrary real deployments, to match RadioMap3DSeer numbers under a different benchmark contract, or to act as a frequency-general foundation model. Its claim is narrower: under the CKM dataset contract, it reproduces unseen-city maps with physical-unit metrics and clearer regime-level diagnostics than the earlier direct-regression family.

Table 2.3 summarises that position against the most relevant published work across the comparison dimensions introduced in this chapter.

The final-model number in Table 2.3 is the selected final-model test result on the held-out city split. The prior-anchored design lets the neural residual head converge to a narrower error floor than the direct-regression PMHHNet family, particularly in LoS-dominated morphology classes where the physics anchor already achieves sub-2 dB error, while still preserving a strict city-holdout evaluation contract.

Table 2.3: *Positioning of the final model against the most relevant published methods.*

Benchmark / method	Reported RMSE / score	City holdout	Interpretation
RadioUNet [24]	approx. 1.6–3.1 dB equivalent path-loss RMSE	No	Classic CNN radio-map scale reference; no spread targets.
ICASSP 2023 winner PMNet [21, 50]	approx. 1.38 dB	No	Approximate conversion from 0.0383 normalised RMSE using the RadioMap3DSeer 36 dB image-window convention.
RadioGUNet [48]	1.304 dB DPM; 1.936 dB IRT; 1.392 dB DPM with cars	Partial	Outdoor RadioMapSeer path-loss maps with held-out test layouts, but no UAV-height or spread targets.
ReVeal PINN [39]	1.95 dB outdoor rural/suburban radio-environment RMSE	No	Real outdoor sparse-measurement mapping; not dense CKM generation.
RMTransformer [26]	0.007148 normalised pixel RMSE, approximately 1.82 dB under its 254 dB scaling window	No	Recent dense radio-map path-loss reference, but random 90/10 split on 256 px USC/PMNet maps.
AIRMap [37]	path-gain < 4 dB vs. ray tracing; field calibration reported as median percentage error	Partial	Useful deployment-scale reference, but the reported target is path gain rather than this thesis's calibrated path-loss RMSE.
Gao et al. [9]	5.59 dB on ICASSP 2023; 12.19 dB on ITU Challenge 2024	No	Recent outdoor PL-map reference with Tx/Rx depth and corridor weighting; the final PL RMSE is numerically lower, but the dataset, split and feature contract differ.
PathFinder [53]	0.0331 normalised RMSE on RM3D DS-RPP; 0.3263 on unseen rural	Partial	Useful outdoor/unseen-domain reference, but reported primarily in normalised image units.
Tarhouni et al. [42]	multi-dB, sub-6 GHz	No	Physical outdoor reference, not dense UAV CKM.
Final model (this thesis)	**1.65 dB on unseen test cities**	**Yes**	**Final prior-anchored residual GMM-head model; also improves delay-spread and angular-spread RMSE over the frozen priors.**

CHAPTER 3

Methodology / project development

This chapter is structured as the route that would be followed if the project were started again with the final evidence already known. The detailed chronology is preserved in Appendix A, but the main methodology now follows the fastest defensible path: fix the evaluation contract, use the early neural models as diagnostics, identify the target distributions, calibrate frozen priors, and train a bounded residual model around those priors. The emphasis is on decisions that affect reproducibility: masking, splitting, conditioning, and the distinction between raw deep prediction and prior-anchored residual prediction.

3.1 Dataset protocol

Each sample is a complete 513×513 pixel CKM. The spatial resolution is 1 m×1 m per pixel, so each map covers a 513 m×513 m square. The UAV transmitter is horizontally fixed at the centre of the map, while its antenna height h_{tx} varies from sample to sample and is treated as a primary conditioning variable. Every non-building pixel is an active receiver location, with the receiver antenna placed exactly at 1.5 m above ground. Building pixels are not receiver positions and are excluded from the training loss and reported metrics rather than being counted as easy background pixels. The model never receives a random crop as an independent training city. Instead, the split is performed at city level. For the distribution-first and final neural models the split uses `split_seed = 42`, with 15% validation and 15% test samples after city assignment. This avoids the most common optimistic error in spatial radio prediction: training on one part of a city and testing on another visually similar part of the same city.

The transmitter-height samples are not uniformly distributed. Comparing the stored HDF5 histogram with the tested affine beta-mixture variants shows that the closest nominal sampler is $h_{\text{design}} \approx 10 + 490X$, where

$$X \sim 0.75\,\text{Beta}(3, 23) + 0.25\,\text{Beta}(4, 4).$$

However, the finite HDF5 database is the relevant distribution for training and reporting. Reading the stored `uav_height` field for all 16180 samples gives an empirical range of 11.61 m to 477.54 m, usually rounded in the text as about 12 m to 478 m; 477.54 m is the largest sampled HDF5 value, not a hard endpoint of the approximate beta law. This is statistically normal: under $10 + 490X$, $P(h_{\text{tx}} > 478\,\text{m}) = 3.19 \times 10^{-5}$ per draw, so only 0.52 such samples are expected in 16180 draws and the probability of drawing none is about 0.60. A reproducible formulaic version of the realized database distribution is the 3 m empirical density. Let $\Delta h = 3\,\text{m}$, $B_k = [10 + 3k,\ 13 + 3k)$, and let h_i be the stored height of sample i. Then

$$\hat{f}_{\Delta h}(h) = d_k, \quad h \in B_k,$$
$$d_k = \frac{n_k}{N\Delta h}, \quad n_k = \sum_{i=1}^{N} \mathbf{1}\{h_i \in B_k\}, \quad N = 16180. \tag{3.1}$$

When the density is visualized, the plotted curve linearly interpolates between the bin-centre values (c_k, d_k), with $c_k = 11.5 + 3k$. The all-database height counts are 3972 samples below 50 m, 8456 from 50 m to 150 m, 2490 from 150 m to 300 m, and 1262 above 300 m. In the actual training cities, the same bins contain 2676, 5654, 1644, and only 866 samples, respectively. This matters for interpreting the height-binned results: the higher-altitude bins contain fewer training examples, so their lower RMSE later in the thesis is evidence that those cases are physically easier, not that they are better represented in the training set.

The prior artifacts use the same city-holdout contract as the final model: `val_ratio = 0.15`, `test_ratio = 0.15`, and `split_seed = 42`. The Try 78 path-loss prior and Try 79 spread priors are fitted on the 10840 training samples. Standalone prior audits can then use the full held-out validation+test pool of 5340 samples, while final model-versus-prior comparisons use the 2590-sample final test split. This keeps training cities, validation cities, and final test cities separated throughout the prior and residual-model evaluations.

Table 3.1: *City-holdout assignment used by the distribution-first and final neural models.*

Split	Samples	Cities	City names
Train	10840	37	Abidjan, Abu Dhabi, Alexandria, Antigua Guatemala, Athens, Bangalore, Bangkok, Bath, Berlin, Bratislava, Brugge, Buenos Aires, Canberra, Chefchaouen, Dalian, Hanoi, Ho Chi Minh City, Hyderabad, Kinshasa, Leuven, Ljubljana, Luanda, Lyon, Manaus, Minsk, Nagpur, Paris, Rio de Janeiro, Santiago, Shanghai, Siem Reap, St. Louis, Thane, Toledo, Valparaiso, Venice, Vienna.
Validation	2750	15	Bilbao, Chiang Mai, Cleveland, Dhaka, Dubrovnik, Fortaleza, Marrakesh, Mexico City, Munich, Nazareth, Nice, Salvador, Segovia, Tunis, Victoria.
Test	2590	14	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver.

The receiver mask is:

$$m_{\text{ground}}(i,j) = \begin{cases} 1, & \text{if topology}(i,j) = 0 \text{ (non-building receiver pixel)}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.2}$$

Equivalently, the CKM supervision grid contains one receiver at every non-building pixel, all at the fixed receiver height $h_{\text{rx}} = 1.5$ m. The mask only removes building pixels from the loss and metrics; it does not subsample the remaining ground pixels. The LoS/NLoS visibility masks used later by the priors and the final residual model are geometric support inputs rather than target radio maps. In the CKM experiments they are read from the dataset when available; for topology-only deployment they can be generated from topology and UAV antenna height using geometric ray-casting, although generated masks are not expected to be bit-identical to the CKM reference masks. This makes the visibility signal explicit without requiring a previous prediction of the target radio maps. All RMSE values in this thesis are pixel-weighted over valid ground pixels:

$$\text{RMSE} = \sqrt{\frac{\sum_{s,i,j} m_s(i,j)\,(\hat{y}_s(i,j) - y_s(i,j))^2}{\sum_{s,i,j} m_s(i,j)}}. \tag{3.3}$$

This is stricter than averaging per-sample RMSEs because large valid maps contribute proportionally to the number of receiver pixels.

3.2 Diagnostic baselines retained from development

The first implementation used a U-Net generator with a PatchGAN discriminator. The generator predicted channel maps from topology-related inputs, while the discriminator encouraged realistic local patches. This was useful as a software prototype but not as a final scientific direction. The adversarial term did not improve the physical errors that dominated validation RMSE, and it made diagnosis harder.

The next phase focused on controlled engineering changes. Transposed convolutions were replaced by bilinear upsampling and projection to reduce checkerboard artifacts. Multi-scale supervision was added so that the model could learn both global radial structure and local map details. The combined decoder/multiscale run became the strongest clean pre-prior path-loss baseline in this family.

After the masking correction, building pixels were excluded from both training loss and validation metrics. This changed the interpretation of earlier numbers: any method that implicitly benefited from easy building pixels was no longer acceptable. The ground-only rule became an invariant for later models.

3.3 Methodology structure after the development lessons

The work was not a single clean jump from U-Net to the final residual GMM-head model. The detailed numbering is kept in Appendix A; the methodology only retains the lessons that affect the final design. If the project were repeated from the current endpoint, the efficient route would be:

1. define the city-holdout split and ground-only metric before comparing any architecture;
2. use simple U-Net and cGAN baselines only to verify the data pipeline and expose why visual realism is not enough;
3. separate LoS/NLoS, topology, and UAV-height regimes before interpreting the errors;
4. diagnose target histograms before choosing the output head;
5. calibrate the path-loss and spread priors as train-only estimators;
6. for path loss, reproduce the final prior-anchored residual GMM-head setup rather than restarting another raw backbone search;
7. for delay and angular spread, add an explicit high-tail or outlier diagnostic before final fine-tuning, because the remaining errors are concentrated in rare spread events;
8. train the final network as a bounded residual corrector over the frozen priors rather than as a raw map generator.

The earlier models are therefore not deleted from the thesis narrative. They remain as controlled evidence for why the final method is organised around the evaluation contract, distribution shape, and prior-anchored residual learning instead of around a larger generic backbone. This is a change in exposition, not a change in the experimental contract: the dataset, the city-holdout split, the ground-only mask, the final prior-anchored architecture, and the detailed path-loss-prior, spread-prior, and residual-GMM formulas remain the same. The older version of the methodology read more like a chronological development diary; this reduced version keeps the same evidence but presents it as the design path that now looks defensible after the negative results are known. The chronology still matters because it explains why each simplifying decision was made: U-Net and cGAN runs validated the pipeline, PMNet and PMHHNet diagnosed the limits of stronger backbones, the distribution-first attempts exposed the target marginals, and the final prior-anchored model combined the surviving pieces.

3.4 Physical priors and PMHHNet

The middle development phase moved from direct map prediction to residual learning, as formulated in Eq. 3.4. This phase was heavily influenced by the `PMNet` architecture [22], which established the viability of using U-Net-like backbones for path-loss map prediction in terrestrial cellular networks.

$$\hat{\mathbf{Y}} = \mathbf{P}_{\text{phys}} + f_\theta\left(\mathcal{T}, h, \mathbf{P}_{\text{phys}}\right), \tag{3.4}$$

where $\mathbf{P}_{\text{phys}}$ is the analytical or calibrated prior map and f_θ predicts only the residual correction that the prior cannot explain.

3.4.1 Mathematical Evolution: From PMNet to PMHHNet

The `PMHHNet` (Path-loss Map with High-frequency and Height-awareness Network) was developed as a specific refinement of the `PMNet` philosophy tailored for the UAV-enabled Air-to-Ground (A2G) scenario.

High-Frequency (HF) Stem. Standard convolutional encoders tend to smooth out sharp building transitions during downsampling. To counter this, `PMHHNet` introduces a dedicated high-frequency stem. Let $\mathcal{T}$ be the topology map. We compute the discrete Laplacian $|\nabla^2\mathcal{T}|$ to identify building edges and shadow boundaries. The initial feature map $\mathbf{X}_0$ is computed as:

$$\mathbf{X}_0 = \text{Stem}(\mathcal{T}) + \text{Proj}(|\nabla^2\mathcal{T}|) \tag{3.5}$$

where Proj is a 1×1 convolution that maps the edge-detector output to the stem's feature depth. This ensures that building geometry is strongly injected into the earliest layers of the encoder.

Continuous Sinusoidal FiLM Conditioning. While `PMNet` was designed for fixed base-station heights, UAVs require continuous height generalization. `PMHHNet` utilizes Feature-wise Linear Modulation (FiLM) [34] to inject the scalar height h. The height is first mapped to a sinusoidal embedding $\mathbf{e}(h) \in \mathbb{R}^K$ (Eq. 3.6). For any feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ at depth l, the conditioned map $\mathbf{X}'$ is:

$$\mathbf{e}(h) = \left[\sin(\omega_1\tilde{h}), \cos(\omega_1\tilde{h}), \dots, \sin(\omega_{K/2}\tilde{h}), \cos(\omega_{K/2}\tilde{h})\right], \qquad \tilde{h} = \frac{h - h_{\min}}{h_{\max} - h_{\min}}, \tag{3.6}$$

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}_l(\mathbf{e}(h))) + \mathbf{b}_l(\mathbf{e}(h)) \tag{3.7}$$

where $\mathbf{g}_l$ and $\mathbf{b}_l$ are learned linear projections mapping the embedding to the channel dimension C. This pattern, inspired by time-step conditioning in diffusion models [7], allows the network to modulate its internal representation dynamically based on the UAV's altitude.

3.4.2 Limitations and Legacy

Despite these improvements, the early residual phase plateaued at about 9.4 dB. The primary bottleneck was the "soft" nature of the residual learning; the network often under-compensated for the physical propagation law in NLoS regions.

> *Legacy of PMHHNet:* While the full architecture was eventually superseded by the final prior-anchored GMM approach, its core components, the HF stem and sinusoidal FiLM, remain highly effective for point-prediction tasks or real-time implementations where the distributional complexity of a GMM is not required.

3.5 Distribution-first modelling

The decisive late-stage diagnosis was that NLoS path loss was not merely a harder regression target; it possessed a fundamentally different error distribution. This diagnosis came from comparing predicted and ground-truth histograms: the maps showed where the model was wrong, but the histograms showed *what* the model had misunderstood. The central idea of the late pipeline was therefore to identify the target distribution before asking a network to perform pixel-by-pixel prediction. The distribution-first path-loss model addressed this by emitting a Gaussian Mixture Model (GMM)-parameterized residual head rather than only a direct scalar map.

The distribution-first spread model applied the same two-stage idea to delay spread and angular spread, but changed the mixture shape to match the targets: a near-zero LoS spike plus a heavy NLoS tail. The architecture is therefore not redrawn as a second full diagram; it follows the same routing and Stage-A / Stage-B structure as Figure 3.1, with the path-loss LoS/NLoS experts replaced by delay-spread and angular-spread experts.

For path loss, the distribution-first model utilized 12 specialized experts: six topology classes partitioned by LoS and NLoS regions. As shown in Figure 3.1, Stage A acts as a distribution head, predicting the global GMM parameters (π_k, μ_k, σ_k) from map features and altitude conditioning. Stage B then decodes these priors spatially, predicting per-pixel soft component assignments and within-component offsets. The final map reconstruction can be written as:

$$\hat{y}(i,j) = \sum_{k=1}^{K} p_k(i,j)\,(\mu_k + z_k(i,j)\sigma_k)\,. \tag{3.8}$$

The spread counterpart extended this distribution-first philosophy to delay spread and angular spread. These targets are not cleanly Gaussian; their empirical distributions resemble a near-zero spike (representing the LoS direct path) combined with a heavy tail (representing NLoS scattering). Standard Mean Squared Error (MSE) objectives tend to underrepresent these rare but physically significant heavy-tail spread values, necessitating a dedicated mixture head.

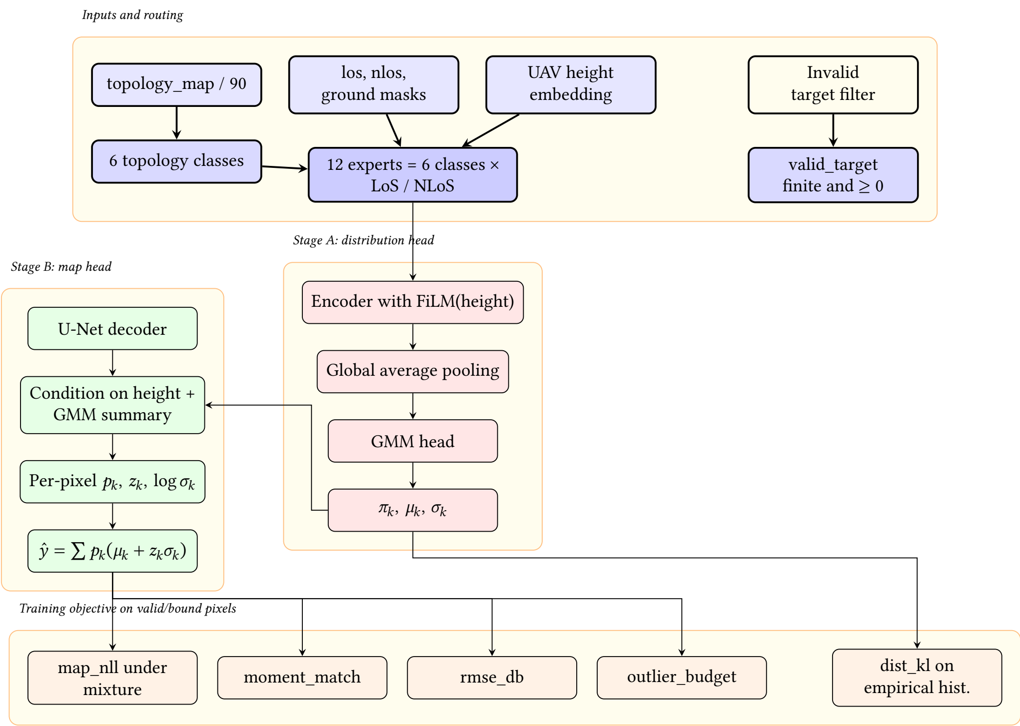

Figure 3.1: *Distribution-first path-loss system. Stage A learns global GMM parameters* (π_k, μ_k, σ_k) *from pooled features conditioned on UAV height; Stage B produces per-pixel soft assignments and within-component offsets anchored to those parameters.*

3.5.1 How to construct a good deep-learning model for radio maps

The main lesson from the deep-learning experiments is that architecture is secondary to the question: *what statistical object is the network being asked to predict?* If the project were restarted with the final evidence available, the first step would be to define the city-holdout, ground-only metric contract; the second would be to inspect the target distributions; and only then would the backbone, loss, prior, and routing design be chosen. The project did not discover this in one step. It moved from direct image translation, to residual learning over physical estimates, to topology experts, to distribution-first heads, and finally to frozen priors plus bounded neural residuals.

This evolution reframes the role of priors. Midway through the project, the best models were residual-over-prior systems, which made the prior appear to be the essential ingredient. The distribution-first path-loss model changed that interpretation. It used no COST-231 prior, no enhanced two-ray prior, no knife-edge channel, and no residual-over-prior decomposition, yet it reached NLoS results close to the final prior family by modelling the NLoS marginal directly. Therefore, the prior is useful but not magic: it is not external knowledge that solves the task by itself. It is a stable and auditable estimator of large-scale structure, especially when that structure is simpler to fit than to relearn with a high-capacity network.

The final pipeline reintroduces priors without contradicting the distribution-first lesson. The histogram diagnosis showed that the PMHHNet family was failing because the output distribution was wrong, not simply because one more physical channel was missing. The first response was therefore to fix the deep-learning side and prove that NLoS path loss can be modelled without a prior when the head represents the target distribution. Once that was clear, the question became practical: which parts of the prediction are cheapest, most stable, and easiest to audit? For LoS path loss, a calibrated two-ray/radial model is more efficient than asking a CNN to relearn the same ring structure. For blocked NLoS path loss and the spread targets, the neural model is still valuable because the problem is no longer just estimating a smooth mean; it is placing local residuals and rare tails in the right parts of the map.

Path loss also cannot be treated as one homogeneous target. LoS is not only the easy subset: it contains a deterministic spatial law driven by distance, height, and coherent two-ray structure. A convolutional network can learn a smooth radial trend from distance and topology channels, but it does not naturally learn the calibrated ring pattern if the objective rewards only the average pixel error. NLoS is different. In valid CKM ground pixels its marginal is concentrated around a blocked-link attenuation level, but it is noisy, topology-dependent, and hard to place spatially. The model must answer two questions at once: which NLoS value range belongs to this map, and which pixels belong to the high-error tail. The old PMHHNet family exposed this mismatch through severe NLoS mode collapse, with predicted NLoS means around 36.5 dB against target means around 107.1 dB.

Delay spread and angular spread are even more distributional. In LoS regions, the direct path creates a near-zero spike: small delay spread and narrow angular spread are physically expected. In NLoS regions, the spread targets often carry the useful channel information because reflected, diffracted, and scattered paths dominate the received energy. A high delay spread or wide angular spread is therefore not merely an error-prone outlier; it is evidence that multipath has become important. This is why a plain RMSE or Huber head tends to miss the engineering-relevant tail: it can improve the common near-zero region while underpredicting rare high-spread areas. The late histogram diagnostics that motivated this shift are kept in Appendix D; the methodology keeps their design consequence: spread maps need explicit spike-plus-tail or high-tail diagnostics, not just smoother images.

The resulting rule set is compact but chronological. First, identify whether the target is smooth, narrow, multi-modal, spike-plus-tail, or heavy-tailed before choosing MSE, Huber, GMM, NLL, or residual losses. Second, treat LoS and NLoS as semantic regimes, not only as reporting columns. Third, keep masks, units, city holdout, topology class, antenna height, and tail-risk policy visible during validation. Finally, use visual realism only as a qualitative check: cGAN sharpness and larger PMNet/PMHHNet-style backbones did not solve the physical RMSE problem until the target distribution itself was represented.

3.6 GMM component count and distribution choice

The Stage-A distribution heads in the distribution-first models represent each per-sample target marginal as a parametric family. Choosing the family and its number of components K is not a free hyperparameter; it is constrained by the empirical shape of the CKM histograms. A dedicated distribution-ranking diagnostic fits a fixed shortlist of parametric models to the aggregated validation histograms of every target under each topology class and LoS/NLoS split, and ranks them by the KL divergence $\mathrm{KL}(p_{\mathrm{empirical}} \parallel p_{\mathrm{model}})$. The shortlist contains Gaussian, Laplace, skew-Normal, log-Normal, Gamma, Weibull, a spike-plus-exponential model for near-degenerate targets, and Gaussian mixtures with

$K \in \{2, 3, 4, 5\}$.

Table 3.2 summarises the winning family for the four targets used in this thesis, aggregated across all cities. The table uses the target-side (ground-truth) histograms, because the distribution-head objective is defined against the ground-truth marginal, not the current prediction.

Table 3.2: *Best-fit parametric family for each target distribution on the CKM validation histograms, across all cities. Based on the validation KL-ranking study.*

Target	Best family	Shape notes
Path loss, LoS	GMM-4	Single dominant mode near 97 dB with a mild low-tail; GMM-4 wins on the raw target histogram, but a Gaussian is already competitive.
Path loss, NLoS	GMM-5	Narrow peak near 107 dB plus a secondary low-variance mode; a single Gaussian is too wide.
Delay spread	GMM-5	Near-zero spike plus a long exponential tail; the spike-plus-exponential family is a close second.
Angular spread	spike-plus-exp.	Dirac-like spike near 0° plus a heavy tail; GMM can approximate it but spike-plus-exp is the native family.

Three consequences follow. First, a single Gaussian (or any unimodal head) is never the best family for NLoS path loss or for the two spreads. This is exactly the regime in which the earlier PMHHNet family produced mode collapse with a single-output MSE head. Second, the number of components that wins is consistently $K \in \{4, 5\}$ for the raw path-loss and spread targets, not $K = 3$. The Stage-A heads in this thesis therefore use $K = 5$ by default on the target-distribution experts, even though the code supports $K \in \{2, \dots, 5\}$. The final `num_components` field configuration ($K = 3$) controls a different quantity: the shared backbone's residual distribution head after the frozen priors have already explained the main LoS, NLoS, delay-spread, and angular-spread structure. In that residual space, three modes are sufficient to model negative correction, near-zero correction, and positive correction, while avoiding a larger mixture that would be harder to stabilize during the final multi-task training. Third, the angular-spread target is better served by a native spike-plus-exponential head than by a GMM, because its marginal is effectively degenerate at 0°; this is implemented directly in the distribution-first angular-spread experts.

3.7 Loss functions of the final pipeline

The detailed definitions of the final objective are given in Sec. 3.13.4. This section keeps only the training contract needed to compare the model families. For each target $t \in \{\mathrm{PL}, \mathrm{DS}, \mathrm{AS}\}$, the implementation defines a composite masked loss

$$\begin{aligned}\mathcal{L} = \sum_t w_t \big(&\lambda_{\mathrm{NLL}} \mathcal{L}_{\mathrm{NLL},t} + \lambda_{\mathrm{KL}} \mathcal{L}_{\mathrm{KL},t} + \lambda_{\mathrm{mom}} \mathcal{L}_{\mathrm{mom},t} \\ &+ \lambda_{\mathrm{anchor}} \mathcal{L}_{\mathrm{anchor},t} + \lambda_{\mathrm{guard}} \mathcal{L}_{\mathrm{guard},t} + \lambda_{\mathrm{out}} \mathcal{L}_{\mathrm{out},t} \\ &+ \lambda_{\mathrm{RMSE}} \mathcal{L}_{\mathrm{RMSE},t} + \lambda_{\mathrm{MAE}} \mathcal{L}_{\mathrm{MAE},t}\big) .\end{aligned} \tag{3.9}$$

All sums are restricted to valid receiver pixels.

Table 3.3: *Active loss settings in the final residual GMM-head model stages.*

Term group	Large-capacity residual stage	Final residual fine-tuning stage	Interpretation
NLL, KL, moment and outlier budget	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0	Active in the preceding large-capacity stage, then disabled in the selected deterministic fine-tuning stage.
RMSE / MAE	1.50 / 0.10	1.00 / 0.05	Supervised pixel losses in native output units; the final fine-tuning stage makes them the dominant active terms.
Anchor / prior guard	0.05 / 0.10	0.01 / 0.02	Keep residual corrections small and penalize model states that become worse than the frozen priors, with lighter regularization in the fine-tuning stage.
Task weights	PL 1.00, DS 1.00, AS 1.00	PL 1.00, DS 0.25, AS 0.25	Equal target weighting before the final fine-tuning stage; then path loss becomes the main optimization target while spread supervision remains in the joint model.

This is a deliberate simplification relative to the earlier branches. The PMHHNet diagnostic runs used masked MSE with multi-scale and Laplacian-style auxiliaries; the distribution-first models removed those structural losses and focused on target shape. The final model keeps the residual GMM-head machinery available: it still emits mixture parameters and uses the local-weighted model-space mixture mean as the deterministic point prediction; for delay and angular spread this mean is computed in $\log(1 + y)$ space and inverse-transformed before native RMSE/MAE are reported. The probabilistic NLL, KL, moment and outlier terms were active in the preceding large-capacity training stage, but the selected final model is chosen after a deterministic RMSE/MAE-dominant fine-tuning stage because the frozen priors already explain the main large-scale structure. Consequently, the final claim is an RMSE improvement over the frozen priors, not a calibrated probabilistic prediction claim. Checkpoint selection in this final stage uses the validation composite $\mathrm{PL}_{\mathrm{RMSE}} + 0.30\,\mathrm{DS}_{\mathrm{RMSE}} + 0.25\,\mathrm{AS}_{\mathrm{RMSE}}$, matching the `score_weights` field of the reported configuration.

3.8 Training hyperparameters

Table 3.4 summarises the training configuration of the final prior-anchored pipeline. The values for the distribution-first predecessors and the final residual GMM-head model are reported as methodological settings rather than as file-path artifacts. Older runs are reported more compactly because the final thesis line inherits only the architectural lessons from them, not the training stack.

Two patterns deserve emphasis. First, the move from PMHHNet diagnostics to the distribution-first and final prior-anchored stages is a simplification, not a complication: SWA, EMA, CutMix, label noise and the older auxiliary stabilisation tricks are removed. The distribution-first experts use AdamW with ReduceLROnPlateau, whereas the final joint GMM uses AdamW with the warmup-plus-cosine `LambdaLR` schedule implemented in `train_try80.py`. This is consistent with the distribution-first philosophy: once the output head and target distribution are correct, the classical training-stabilisation tricks become unnecessary. Second, the final model uses a smaller learning rate than the distribution-first experts (5×10^{-5} vs 3×10^{-4}) because it is fine-tuning a residual head

Table 3.4: *Training configuration of the final pipeline. Unless otherwise stated, all runs use AdamW, image size* 513×513*, batch size 1 per GPU, and gradient accumulation over 8 steps.*

Config	Distribution-first experts	Final joint GMM	PMHHNet diagnostics
Input channels	topology, LoS, NLoS, ground mask (4 + scalar height)	topology + frozen priors + masks (9)	topology, LoS, distance, formula prior, knife-edge (5 to 6)
Optimizer	AdamW (β_1=0.9, β_2=0.999)	AdamW	AdamW
Learning rate	3×10^{-4}	5×10^{-5}	1×10^{-3} to 3×10^{-4} (cosine or plateau)
Weight decay	0.01	0.01	0.015 to 0.030
Batch size	1	1	1
Grad accumulation	8	8	16 (68) / 8 (69)
Warmup epochs	2	1	2 to 5
Epochs	120	40 fine-tune epochs	80
Scheduler	ReduceLROnPlateau, patience 15	cosine `LambdaLR` after warmup	ReduceLROnPlateau (post-April 2025 patch)
SWA	Off	Off	On, start fraction 0.60
Label noise	Off	Off	On, $\sigma_{\mathrm{dB}} = 0.5$
CutMix	Off	Off (D4 augmentation only)	On, $p = 0.25$ to 0.45, FiLM-safe
EMA weights	Off	Off	On, decay 0.9975
Dropout	0.10 (decoder only)	0.10 (decoder only)	0.12 (69) to 0.20 (67)
Height encoding	Sinusoidal FiLM, 16 log-freqs over $[12, 478]$ m	Sinusoidal FiLM at encoder and decoder scales	Sinusoidal FiLM, 8 injection points
GMM components K	5 (Stage A, path/delay), spike+exp (angular)	3 (residual head on anchored output)	not used (MSE/Huber heads)

whose inputs already include the strong frozen priors; pushing the learning rate higher pulls the residual head away from the anchors faster than `prior_guard` can catch it.

3.9 Final residual GMM-head model training progression

The final residual GMM-head model was not obtained in a single run. Training began with the joint prior-anchored residual design, but the first complete runs produced only small improvements over the frozen priors. As a diagnostic, this showed that the priors were already strong, but also that the residual model was initially too constrained to correct the remaining errors, especially for path loss.

The training path then followed four stages:

1. **Wide residual model.** The first full version used a shared multi-task U-Net with 96 base channels and a 128-dimensional FiLM condition vector. Its residual caps were conservative, especially for path loss (±2 dB in LoS and ±4 dB in NLoS), and the objective retained the complete probabilistic bundle: GMM likelihood, distribution matching, prior anchoring, prior guard, RMSE, and MAE. This stage mainly tested whether a residual network could improve the frozen priors without degrading them. The multi-task sharing was deliberately placed in the encoder-decoder representation; the PL, DS and AS residual heads remained task-specific.
2. **Wide residual model with relaxed residual bounds.** The second stage kept the same network capacity but loosened the allowed correction range. In the selected fine-tuned configuration, the path-loss caps are ±7 dB in LoS and ±25 dB in NLoS, the delay-spread caps are ±20 ns in both regimes, and the angular-spread caps are $\pm 17^\circ$ in LoS and $\pm 15^\circ$ in NLoS. This made the residual head less constrained and produced a clearer, but still limited, improvement.

3. **Large-capacity residual model.** The third stage increased model capacity to 128 base channels and a 160-dimensional conditioning vector, while keeping the relaxed residual philosophy. It also placed more weight on the deterministic RMSE term. This stage tested whether the remaining gap was capacity-limited rather than only cap-limited. The larger model helped the spread tasks, but path-loss improvement remained too small while the full probabilistic objective was still active.
4. **Final residual fine-tuning stage.** The final stage kept the large-capacity architecture and relaxed caps, lowered the learning rate, and changed the objective into a short RMSE-dominant fine-tuning stage. The probabilistic terms were disabled, the prior guard and anchor were kept only as light stabilizers, and path loss received the strongest task weight. This selected final model is used for the reported final results: the GMM structure remains the residual output head, but model selection is driven by deterministic validation performance.

For this reason, the methodology does not treat training curves themselves as evidence for the final claim. The relevant evidence is the same city-holdout validation and test evaluation used throughout the thesis, reported in Chapter 4. Training curves were used only as diagnostics for deciding which stage should be continued or replaced.

3.10 Final priors: overview and literature basis

This section provides the compact entry point for the ML-calibrated physical path-loss and spread priors and the final joint residual GMM-head model architecture. The detailed mathematical derivations and implementation-level fitting notes follow in the same methodology chapter, so the design and the audit trail remain together.

3.10.1 High-level architecture

The central design decision in the final model is *not* to learn path loss, delay spread, and angular spread from scratch. Instead:

- the path-loss prior provides a frozen prior for `path_loss`,
- the spread-prior module provides frozen priors for delay spread (`delay_spread`) and angular spread (`angular_spread`),
- the final residual GMM-head model learns bounded corrections around those priors.

The deterministic or interpretable structure is kept in the prior, and the deep model is used only for the residual part that the prior cannot explain. Figure 3.2 illustrates the overall pipeline.

This prior phase is not presented as a new universal closed-form propagation law. Its value is more practical: it turns physics, morphology, visibility, and height-dependent calibration into a strong auditable baseline. The final model then follows a divide-and-conquer split of responsibility: calibrated priors remove the stable component of the targets, and the neural residual model focuses on the local errors that remain.

The term *prior* is not used here in the Bayesian sense of a probability distribution over model parameters or targets, and no posterior inference is performed in the prior stage. It denotes deterministic, frozen estimator maps computed before the neural residual correction. The word *prior* is therefore used for three concrete estimators, not for a vague physical hint:

- **LoS path-loss prior.** This is the clearest physical branch: $\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{2\mathrm{ray}} + b + r$. FSPL uses the 3D direct Tx–Rx distance; $C_{2\mathrm{ray}}$ comes from a coherent direct-plus-ground-reflected field sum; b is a fitted height-dependent dB offset; and r is a smoothed radial residual table. The UAV is always centred horizontally, so the radial coordinate is unambiguous, but h_{tx} varies by sample: $\rho(h_{\mathrm{tx}})$, $\phi(h_{\mathrm{tx}})$, $b(h_{\mathrm{tx}})$ and $r(d_{2D}, h_{\mathrm{tx}})$ are fitted in 5 m height bins, smoothed, and linearly interpolated at inference. It is not a generic free-space formula and it is not a learned linear regressor.

- **NLoS path-loss prior.** This starts from the raw blocked-link propagation estimate $\mathrm{PL}^{(0)}$, which includes $\mathrm{PL}_{\mathrm{NLoS,raw}}$ on obstructed pixels, then calibrates it with local morphology: building density, building height, NLoS support, the elevation-dependent shadowing proxy σ_{shadow}, elevation angle, and interactions. It uses antenna-height regimes (`low_ant`, `mid_ant`, `high_ant`) because the obstruction geometry changes qualitatively when the centred UAV rises above the local skyline. It is deliberately separate from the LoS branch because blocked pixels do not follow the same two-ray mechanism.

- **Delay- and angular-spread priors.** These are fitted in $\log(1 + y)$ space. The raw estimate captures propagation state, distance, elevation, and local morphology; the final ridge calibration adds the 23-feature vector of Sec. 3.12. That vector includes continuous normalized UAV height, density-weighted transmitter clearance over nearby building pixels, and the local window-area fraction occupied by buildings taller than the transmitter. This is why the spread prior can fit the near-zero LoS mass and the NLoS tail without letting the largest tail values dominate the whole calibration, unlike a native-space global average.

The notation intentionally reuses ρ in two different blocks: $\rho(h_{\mathrm{tx}})$ is the fitted LoS reflection amplitude, whereas ρ_k is a local building-density fraction in a $k \times k$ window. They are unrelated quantities; the argument or subscript identifies which one is meant. The same convention is used for b: $b(h_{\mathrm{tx}})$ is a LoS dB bias, $b(x)$ is the binary building mask, b_{topology} is a spread-prior log-domain offset, and b_{41} is the blocker-depth feature. These symbols are local to their definitions and should not be read as one shared physical variable.

Table 3.5: *Term-by-term interpretation of the main frozen-prior quantities.*

Term	Prior block	Interpretation
d_{LoS}	LoS path loss	3D direct Tx–Rx distance. It sets the FSPL baseline and uses the fixed $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$ receiver height.
d_{ref}	LoS path loss	Image-source reflected distance. It is the length of the single ground-reflected path used in the coherent two-ray correction.
C_{2ray}	LoS path loss	Direct-plus-reflected field correction in dB. Negative values mean constructive gain over FSPL; positive values mean destructive fading.
$\rho(h_{\mathrm{tx}})$	LoS path loss	Height-interpolated effective reflection amplitude. It is fitted from CKM rather than assumed from one fixed ground material.
$\phi(h_{\mathrm{tx}})$	LoS path loss	Height-interpolated effective reflection phase. It absorbs the phase offset of the simplified one-reflection geometry.
$b(h_{\mathrm{tx}})$	LoS path loss	Height-dependent dB bias remaining after the coherent two-ray fit.
$r(d_{2D}, h_{\mathrm{tx}})$	LoS path loss	Smoothed radial residual table. It corrects distance-dependent errors that remain after FSPL, two-ray interference, and height bias.
$\mathrm{PL}^{(0)}$, $\mathrm{PL}_{\mathrm{NLoS,raw}}$	NLoS path loss	Raw hybrid formula map and its obstructed-pixel branch before map-morphology calibration.
ρ_k	NLoS PL and spreads	Fraction of building pixels in a $k \times k$ neighbourhood; a local density proxy.
h_k	NLoS PL and spreads	Local building-height summary in a $k \times k$ neighbourhood; a proxy for nearby reflectors and blockers.
n_k	NLoS PL and spreads	Local NLoS-support fraction; indicates how much of the neighbourhood is geometrically blocked.
σ_{shadow}	NLoS path loss	Elevation-dependent shadowing proxy used by the NLoS linear calibration.
θ, θ_{norm}, θ_{inv}	PL and spreads	Elevation-angle terms. Low elevation links interact with more buildings, so θ_{inv} is especially useful for spread tails.
b_{41}, c_{41}, t_{41}	Spreads	Blocker depth, transmitter clearance, and local taller-building fraction; they separate open overhead links from obstructed street-canyon-like links.

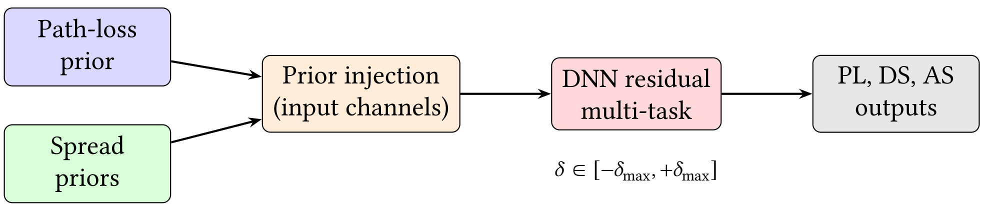

Figure 3.2: *Overall prior-anchored pipeline. The frozen priors are injected into the DNN, which learns a bounded correction δ around them.*

3.10.2 Prior design rules

The main lesson from the prior phase is that a useful prior is not just a physical-looking equation. It is a complete estimator: the formula, fitted calibration, feature units, LoS/NLoS mask, topology features, and evaluation mask must be identical between fitting and inference. Earlier prior experiments used many of the same physical ingredients as the final path-loss prior, but they did not always keep those pieces aligned. The final prior became strong because it made the prior self-contained and auditable under the same city-holdout contract used for the final model.

The distinction matters most for NLoS. A calibration fitted for one physical formula cannot reliably correct another; local height and density features must have the same units when fitted and when applied; and regime assignment should come from the sample geometry rather than only from a broad city label. Once those constraints were enforced, the prior could separate deterministic large-scale structure from the genuinely local residual that the final model later learned.

This does not mean that all earlier NLoS failures were only implementation mismatches. The no-prior and weak-prior neural branches also failed because single-output point regression was a poor match for blocked, multi-modal, heavy-tailed NLoS and spread targets. That is why the thesis keeps both lessons: the final priors fix the prior-estimator problem, while the distribution-first predecessors motivate distribution-aware heads and residual uncertainty.

Table 3.6: *Compact design rules extracted from the prior experiments.*

Rule	Effect on the final pipeline
Fit and apply the same formula	The calibration corrects the actual runtime prior, not a neighbouring variant.
Keep units explicit	Distances, heights, topology values, and normalized features have one shared definition across fitting, validation, and test.
Use the metric mask during fitting	The prior is optimized for valid receiver pixels rather than for building cells that are excluded from the final metric.
Hard-separate LoS and NLoS	The two regimes keep different physics, feature sets, and calibration parameters when the dataset provides a reliable mask.
Let literature define axes, not all coefficients	Distance, height, frequency, and elevation come from propagation theory; effective reflection and local morphology corrections are fitted from CKM.
Learn bounded residuals after the prior	The final model is allowed to correct local errors without overwriting the calibrated large-scale structure.
Treat calibration artifacts as part of the model	Large JSON and coefficient artifacts are not bookkeeping only; they store the height-, density-, visibility-, and morphology-specific decisions that make the prior a competitive baseline.

3.10.3 Literature ingredients and thesis adaptations

The final pipeline is a hybrid between classical propagation models, distribution-aware regression, and dataset-specific calibration. Table 3.7 keeps the mapping explicit; the detailed equations and fitting procedures are given in the following path-loss-prior, spread-prior, and residual-GMM methodology sections.

Table 3.7: *Literature-to-design mapping used by the final prior-anchored model.*

Block	Literature basis	CKM adaptation	New role in this thesis
LoS path loss prior	FSPL and coherent two-ray propagation.	Effective complex reflection, height-bin interpolation, and radial residual fitting on CKM LoS pixels.	Reconstructs the radial interference pattern and supplies the frozen LoS anchor.
NLoS path loss calibration	COST-231/Hata-style attenuation, A2G elevation envelopes, and LoS/NLoS separation.	Map-derived density, height, NLoS-support, elevation, and shadowing features with regime-wise calibration.	Turns a broad empirical curve into a local, topology-aware NLoS prior.
Delay and angular spread priors	Log-normal spread statistics from 3GPP/WINNER-style channel models and UAV elevation trends.	Log-domain ridge regression with local morphology features and a sparse-regime fallback hierarchy.	Provides stable spread anchors without pretending that spreads are smooth deterministic path-loss maps.
Final residual model	U-Net-style dense prediction, FiLM conditioning, probabilistic regression, and mixture-density ideas.	Prior channels, explicit masks, residual caps, anchor gates, and a shared multi-task trunk.	Learns only the bounded local correction and uncertainty left after the calibrated priors.

3.11 Path Loss Prior

Reader map

This section is intentionally detailed because it is the mathematical core of the final path-loss result. A reader only interested in the modelling logic can follow it in four passes. First, the LoS branch starts from FSPL and adds an enhanced two-ray field sum, which explains the radial ring structure observed in the CKM maps. Second, the NLoS branch uses a calibrated regime-wise model over topology and obstruction features, because a LoS-shaped formula cannot explain blocked streets by itself. Third, both branches are fitted only on the training split and then evaluated with the same pixel-weighted ground-mask RMSE used elsewhere in the thesis. Finally, the prior output becomes the frozen path-loss anchor consumed by the final residual GMM-head model.

The most important results to retain while reading the details are the LoS waterfall in Table 4.2 and the final path-loss numbers in Table 4.4: the enhanced two-ray branch reduces LoS error to about 1.75 dB, and the full hybrid prior reaches about 1.92 dB overall on the held-out validation+test cities.

3.11.1 LoS branch: enhanced two-ray model

The LoS prior is not a pure FSPL prior. FSPL is kept because it is the direct-ray normalization term of the coherent two-ray model: it gives the path loss that would be observed if only the direct ray existed. The deployed LoS branch then adds the fitted ground-reflection interference term, a height-dependent bias, and a smoothed radial residual profile. This differs from the NLoS branch, which uses a longer pixel-feature vector and a regime-wise linear calibration.

Geometry and distances

Consider a UAV transmitter at height h_{tx} (metres above the ground-plane reference) and a receiver at height h_{rx}. For each map pixel x at 2D horizontal distance $d_{2D}(x)$ from the transmitter, two propagation distances are defined.

The *direct-path* (LoS) distance is:

$$d_{\text{LoS}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} - h_{\text{rx}})^2} \tag{3.10}$$

This is simply the Euclidean distance between the two antennas in 3D space. Because $d_{2D}(x) \geq 0$ and the squared height difference $(h_{tx} - h_{rx})^2$ is non-negative, $d_{LoS}(x) \geq d_{2D}(x)$, with equality only when both antennas are at the same height.

The *reflected-path* (image) distance is:

$$d_{ref}(x) = \sqrt{d_{2D}(x)^2 + (h_{tx} + h_{rx})^2} \tag{3.11}$$

This is the distance from the image source (the UAV mirrored below the ground plane at $-h_{tx}$) to the receiver. Since $(h_{tx} + h_{rx}) \geq |h_{tx} - h_{rx}|$ for any non-negative heights, we always have $d_{ref}(x) \geq d_{LoS}(x)$. In the present UAV-to-ground setting with $h_{tx} > 0$ and $h_{rx} > 0$, the reflected path is strictly longer; in the limiting geometric case where one antenna is exactly on the ground plane, it would be equal rather than longer.

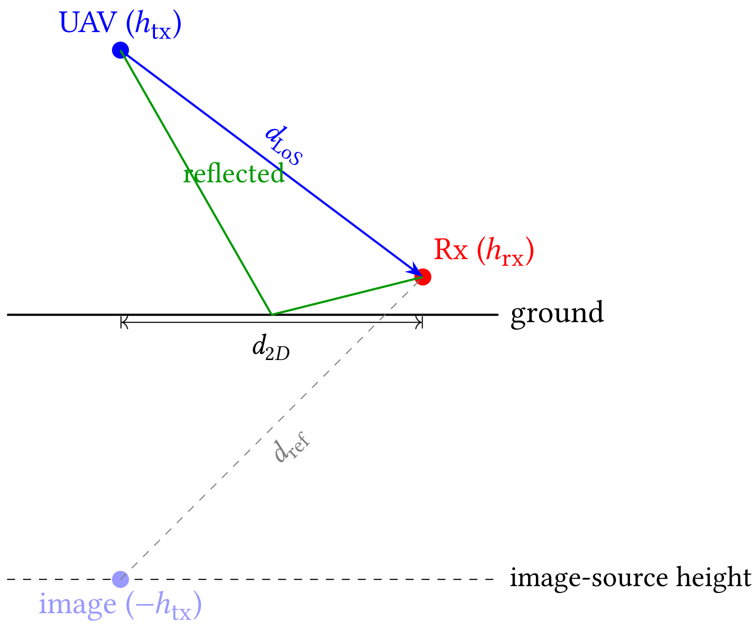

Figure 3.3: *Two-ray geometry. Direct path (blue) has length d_{LoS}; reflected path (green) has the same total length as the image-source path (gray dashed), which is d_{ref}.*

Free-space path-loss baseline

The free-space path loss in dB is defined using the standard formula at frequency f_{MHz} (in MHz):

$$\text{FSPL}(x) = 32.45 + 20\log_{10}\left(\frac{d_{LoS}(x)}{1000}\right) + 20\log_{10}(f_{MHz}) \tag{3.12}$$

Derivation. FSPL in linear power ratio is $(4\pi d/\lambda)^2$ where $\lambda = c/f$. In dB:

$$\begin{aligned}
\text{FSPL}_{dB} &= 20\log_{10}\left(\frac{4\pi d f}{c}\right) \\
&= 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c) \\
&\approx 20\log_{10}(d_{km} \cdot 1000) + 20\log_{10}(f_{MHz} \cdot 10^6) + 20\log_{10}(4\pi) - 20\log_{10}(3 \times 10^8) \\
&= 20\log_{10}(d_{km}) + 20\log_{10}(f_{MHz}) + 32.45 \quad [\text{dB}]
\end{aligned} \tag{3.13}$$

The constant $32.45 = 20\log_{10}(4\pi\cdot 10^3\cdot 10^6/(3\times 10^8)) \approx 32.45$ dB. In Eq. (3.12), $d_{\mathrm{LoS}}(x)/1000$ converts metres to kilometres.

Physical meaning. FSPL grows quadratically with distance ($\propto d^2$) and quadratically with frequency. It captures only the geometric spreading of the wavefront, with no ground interaction, scattering, or diffraction.

Enhanced two-ray correction

In the two-ray model, the received field is the superposition of the direct path and a ground-reflected path. Let $k = 2\pi f/c$ be the free-space wavenumber. The complex sum of the two waves at the receiver is:

$$E_{\mathrm{total}} \propto \frac{e^{-jkd_{\mathrm{LoS}}}}{d_{\mathrm{LoS}}} + \Gamma\,\frac{e^{-jkd_{\mathrm{ref}}}}{d_{\mathrm{ref}}} \tag{3.14}$$

where Γ is the complex ground reflection coefficient. In the standard textbook derivation, Γ is computed from the Fresnel equations for the ground material. Here, it is replaced by an *effective* complex parameter:

$$\Gamma(h_{\mathrm{tx}}) = \rho(h_{\mathrm{tx}})\,e^{-j\phi(h_{\mathrm{tx}})} \tag{3.15}$$

where $\rho \in [0, 1.5]$ is the effective amplitude (allowing over-unity values during the calibration search) and $\phi \in [-\pi, \pi]$ is the effective phase offset. This range is a robustness allowance for an effective CKM-fitted parameter, not a claim that a passive Fresnel reflection coefficient exceeds unity; the deployed smoothed values reported later remain below one. Both parameters depend on the UAV height bin.

The two-ray path-loss correction term in dB is:

$$C_{\mathrm{2ray}}(x) = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}(x)}{d_{\mathrm{ref}}(x)}e^{-j\left(k\left(d_{\mathrm{ref}}(x)-d_{\mathrm{LoS}}(x)\right)+\phi(h_{\mathrm{tx}})\right)}\right| \tag{3.16}$$

Term-by-term interpretation.

- $\rho(h_{\mathrm{tx}})$: amplitude of the ground reflection. A value of $\rho = 0$ recovers pure FSPL. The fitted values are $\rho \approx 0.46$ to 0.56 for heights 10 to 100 m, consistent with a moderate ground-reflection coefficient.
- $d_{\mathrm{LoS}}(x)/d_{\mathrm{ref}}(x)$: path amplitude ratio. This is always ≤ 1 since the reflected path is longer; it corrects for the $1/d$ amplitude decay of the reflected ray relative to the direct ray.
- $k(d_{\mathrm{ref}}(x) - d_{\mathrm{LoS}}(x))$: the phase difference accumulated due to the extra path length of the reflected wave.
- $\phi(h_{\mathrm{tx}})$: an additional phase offset capturing phase shifts upon reflection (e.g., phase shift at the boundary). The fitted values are $\phi \approx 0$ for all height bins, meaning the fitted extra phase offset is near zero under this sign convention.
- $-20\log_{10}|\cdot|$: the total field amplitude at the receiver, normalized to the direct-path field. When the two waves add in phase, the field amplitude is > 1, giving a *negative*

correction (the signal is *stronger* than FSPL alone); when they cancel, the correction is *positive* (fading loss).

This spatial interference pattern creates the characteristic ring-like behavior in the LoS path-loss maps: concentric rings of constructive and destructive interference around the transmitter.

Comparison with literature two-ray models

The field-sum in Eq. (3.16) is the standard *exact* coherent two-ray equation from the literature: a direct ray plus one ground-reflected ray summed before conversion to dB. The practical adaptation in this thesis is not an extra term inside the two-ray equation, but the way the reflection coefficient is parameterized and calibrated. We refer to this fitted variant as the *enhanced two-ray* model throughout this work: the underlying equation is the textbook coherent two-ray model, but the effective reflection coefficient is enhanced beyond the Fresnel range and allowed to take amplitudes up to $\rho \leq 1.5$ in order to absorb constructive multipath effects from the surrounding urban scene that a single-Fresnel coefficient could not represent. The coefficients are fitted independently in 5 m UAV height bins and then linearly interpolated across neighboring bins at inference time.

Physical vs. effective parameters. In classic texts such as Rappaport [35] and Goldsmith [10], the complex reflection coefficient Γ is derived from the Fresnel equations for a smooth dielectric surface:

$$\Gamma_\perp = \frac{\sin\theta_g - \sqrt{\epsilon_c - \cos^2\theta_g}}{\sin\theta_g + \sqrt{\epsilon_c - \cos^2\theta_g}}, \qquad \Gamma_\parallel = \frac{\epsilon_c\sin\theta_g - \sqrt{\epsilon_c - \cos^2\theta_g}}{\epsilon_c\sin\theta_g + \sqrt{\epsilon_c - \cos^2\theta_g}} \tag{3.17}$$

where θ_g is the grazing angle and $\epsilon_c = \epsilon_r - j\sigma/(\omega\epsilon_0)$ is the complex relative permittivity of the ground. Here ϵ_r is the relative dielectric constant, σ is the ground conductivity, $\omega = 2\pi f$ is the angular frequency, and ϵ_0 is the vacuum permittivity. This requires assuming physical constants for the ground (typically $\epsilon_r \approx 15$ for urban soil). The symbol is deliberately written as θ_g here because the later A2G and morphology features use $\theta(x)$ for the elevation angle of the UAV-to-receiver link.

In contrast, our enhanced two-ray implementation keeps the same coherent field-sum equation but replaces the angle-dependent $\Gamma(\theta)$ with a height-dependent effective coefficient $\Gamma(h_{\text{tx}}) = \rho e^{-j\phi}$. We do not assume a flat dielectric plane; instead, ρ and ϕ are fitted to the CKM dataset in 5 m UAV-height bins, with linear interpolation between bins during prediction. This allows the model to capture aggregate scattering effects from the real, non-flat urban terrain that a pure Fresnel calculation would miss.

The d^4 approximation. Many literature sources [35, 10] simplify the two-ray model for large distances ($d \gg h_{\text{tx}}, h_{\text{rx}}$) into the classic fourth-power law:

$$P_r(d) \approx P_t G_t G_r \frac{h_{\text{tx}}^2 h_{\text{rx}}^2}{d^4} \tag{3.18}$$

Here P_t is the transmitted power, G_t and G_r are the transmitter and receiver antenna gains, and $P_r(d)$ is the received power at distance d. This approximation assumes a perfect reflection ($\Gamma = -1$) and small grazing angles. Our implementation uses the *exact* field sum (Eq. 3.14) without the d^4 approximation. This is still a standard literature form; it simply avoids a far-field approximation that is not appropriate when UAV height is not negligible relative to horizontal distance.

Search range vs. fitted values. The calibration grid search (Sec. 3.11.2) allows $\rho \in [0, 1.5]$ for robustness. However, the smoothed deployed values reported in Table 3.9 and the stored calibration remain below unity, so the deployed model does not rely on an over-unity reflected ray. In practice, the main difference from a purely physical textbook instantiation is therefore the fitted height-binned effective coefficient, not an additional propagation term.

Extra calibrated terms outside the two-ray model. The quantities $b(h_{\text{tx}})$ and $r(d_{2D}, h_{\text{tx}})$ in Eq. (3.19) are not part of the literature two-ray formula itself. They are post-fit calibration terms added on top of the standard two-ray prediction to absorb residual systematic errors.

Calibrated LoS prior

The full calibrated LoS prior is:

$$\mathrm{PL}_{\mathrm{LoS}}(x) = \mathrm{FSPL}(x) + C_{2\mathrm{ray}}(x) + b(h_{\text{tx}}) + r(d_{2D}(x), h_{\text{tx}}) \tag{3.19}$$

where:

- $b(h_{\text{tx}})$ is a height-dependent bias (dB): a constant correction that shifts the overall two-ray prediction for a given UAV height.
- $r(d_{2D}(x), h_{\text{tx}})$ is a radial residual correction: a smooth function of horizontal distance (and height bin) that captures systematic over-/under-prediction of the two-ray model as a function of range. It is obtained by smoothing the empirical mean residual in each distance bin after the two-ray fit.

The four terms in Eq. (3.19) are applied sequentially: FSPL gives the geometric baseline; the two-ray correction adds the interference pattern; the bias corrects for a systematic global offset per height; and the radial term corrects for range-dependent systematic errors. The implementation then clips the frozen LoS prior to the $[20, 180]$ dB range used by the final model. For the CKM dataset used here, valid calibrated LoS values are not expected to fall outside this interval, so this clipping is a defensive range-consistency guard rather than a correction that should materially change the prior.

For compact notation, the LoS branch can be summarized by the following deterministic diagnostic vector:

$$\mathbf{x}_{78}^{\mathrm{LoS}}(x) = \begin{bmatrix} \mathrm{FSPL}(x) \\ d_{\mathrm{LoS}}(x)/d_{\mathrm{ref}}(x) \\ k(d_{\mathrm{ref}}(x) - d_{\mathrm{LoS}}(x)) \\ \rho(h_{\mathrm{tx}}) \\ \phi(h_{\mathrm{tx}}) \\ C_{\mathrm{2ray}}(x) \\ b(h_{\mathrm{tx}}) \\ r(d_{2D}(x), h_{\mathrm{tx}}) \end{bmatrix}. \tag{3.20}$$

This vector is not multiplied by a learned coefficient vector at inference time. Instead, it exposes the calibrated quantities used by the nonlinear two-ray expression $\mathrm{PL}_{\mathrm{LoS}}(x) = g_{\mathrm{LoS}}(\mathbf{x}_{78}^{\mathrm{LoS}}(x)) = \mathrm{FSPL}(x) + C_{\mathrm{2ray}}(x) + b(h_{\mathrm{tx}}) + r(d_{2D}(x), h_{\mathrm{tx}})$. The entries are computed as follows:

- $\mathrm{FSPL}(x)$ is computed from the 3D direct distance $d_{\mathrm{LoS}}(x)$ and the carrier frequency. It is the direct-ray path loss, so it sets the absolute distance and frequency scale before any reflected-path correction is added.
- $d_{\mathrm{LoS}}(x)/d_{\mathrm{ref}}(x)$ is computed from the direct-path length and the image-source reflected-path length. Since the reflected path is longer, this ratio is normally below one and gives the relative amplitude of the reflected field after $1/d$ propagation loss.
- $k(d_{\mathrm{ref}}(x) - d_{\mathrm{LoS}}(x))$ is the geometric phase advance of the reflected ray, with $k = 2\pi f/c$. This is the term that makes the correction oscillatory in space: small changes in extra path length can move the direct and reflected fields from constructive to destructive interference.
- $\rho(h_{\mathrm{tx}})$ is the effective reflection amplitude used in the coherent field sum. During calibration, each 5 m UAV-height bin is fitted by grid search over $\rho \in [0, 1.5]$ and the phase grid; for each candidate pair, the best constant bias is removed and the pair with lowest LoS RMSE is selected. The stored ρ curve is then smoothed across neighboring height bins and, at inference time, linearly interpolated to the actual h_{tx}. Thus ρ is not a Fresnel coefficient assumed from a ground material, but a CKM-calibrated effective reflected-field amplitude.
- $\phi(h_{\mathrm{tx}})$ is fitted in the same height-binned grid search, using a uniform grid over $[-\pi, \pi)$. It absorbs the effective phase shift of the reflection and any systematic phase offset caused by the simplified one-reflection geometry. As with ρ, the stored phase curve is smoothed across height bins and linearly interpolated during prediction.
- $C_{\mathrm{2ray}}(x)$ is then computed by substituting $d_{\mathrm{LoS}}/d_{\mathrm{ref}}$, the geometric phase, $\rho(h_{\mathrm{tx}})$, and $\phi(h_{\mathrm{tx}})$ into Eq. (3.16). This is the actual coherent direct-plus-reflected correction in dB. A negative value means constructive gain relative to FSPL; a positive value means a fading loss.
- $b(h_{\mathrm{tx}})$ is the constant dB offset fitted together with ρ and ϕ for each height bin. For a fixed candidate (ρ, ϕ), the calibration computes the mean residual $y - (\mathrm{FSPL} + C_{\mathrm{2ray}})$ over the sampled LoS pixels in that height bin and uses it as the optimal bias for

that candidate. The final stored b is smoothed over height and linearly interpolated at inference time. Its role is to remove the remaining height-dependent offset after the interference pattern has been fitted.

- $r(d_{2D}(x), h_{\text{tx}})$ is fitted after the smoothed two-ray model is fixed. The residual target-minus-prediction is accumulated in integer radial bins around the transmitter, separately for each 5 m height bin; empty bins fall back to a global radial profile, the profile is smoothed, and the applied correction is clipped to ±2.5 dB. This term removes range-dependent structure that remains after the coherent two-ray model and the height bias.

Stored / computed term	**Mathematical role**	**Interpretation**
$\text{FSPL}(d_{\text{LoS}}, f)$	Direct-ray baseline.	Gives the geometric free-space loss before any reflected path is added. It is a component of the two-ray prior, not the final LoS model by itself.
$\rho(h_{\text{tx}})$	Height-interpolated reflected-ray amplitude.	Controls how strongly the ground-reflected wave contributes relative to the direct wave.
$\phi(h_{\text{tx}})$	Height-interpolated phase offset.	Captures the effective reflection phase after fitting to CKM instead of assuming a fixed Fresnel coefficient.
$C_{2\text{ray}}(x)$	Coherent interference correction.	Converts the complex direct-plus-reflected field sum into an oscillatory dB correction around FSPL.
$b(h_{\text{tx}})$	Height-interpolated bias.	Removes a remaining per-height offset after the coherent two-ray fit.
$r(d_{2D}, h_{\text{tx}})$	Smoothed radial residual profile, clipped to ±2.5 dB.	Corrects systematic distance-dependent residuals not captured by the two-ray equation.
$\text{clip}_{[20,180]}$	Output guard.	Keeps the frozen prior inside the physical path-loss range used by the final model.

Table 3.8: *Term-by-term inventory of the LoS prior used by the final model. Unlike the NLoS branch, this is not a long pixel-feature regression vector; it is a height-interpolated two-ray calibration plus a radial residual table.*

3.11.2 Calibration of the LoS two-ray model

Data preparation

All training samples are used. For each sample, valid LoS pixels are selected as those satisfying:

1. topology label = 0 (ground pixel),
2. $m_{\text{LoS}}(x) > 0$ (marked as LoS),
3. the measured path loss is finite and ≥ 20 dB.

If a sample has more than 2 500 valid LoS pixels, a random subsample of 2 500 pixels is drawn (with a fixed seed). Samples are then grouped by UAV height into height bins of 5 m width:

$$h_{\text{bin}} = 5 \cdot \text{round}\left(\frac{h_{\text{tx}}}{5}\right) \tag{3.21}$$

Each bin is further subsampled to a maximum of 8 000 pixels total.

Grid search for (ρ, ϕ)

For each height bin, (ρ, ϕ) are found by grid search:

$$\phi_{\text{grid}} = \left\{-\pi + \frac{2\pi k}{N_\phi} \;:\; k = 0, 1, \dots, N_\phi - 1\right\}, \quad N_\phi = 48 \tag{3.22}$$

$$\rho_{\text{grid}} = \{0.00,\ 0.05,\ 0.10,\ \dots,\ 1.50\} \tag{3.23}$$

For each pair $(\hat{\rho}, \hat{\phi}) \in \rho_{\text{grid}} \times \phi_{\text{grid}}$, and for each pixel x_i in the height bin, define the two-ray prediction without bias:

$$\hat{p}_i(\hat{\rho}, \hat{\phi}) = \text{FSPL}(x_i) - 20 \log_{10} \left| 1 + \hat{\rho}\, \frac{d_{\text{LoS}}(x_i)}{d_{\text{ref}}(x_i)} e^{-j\left(k(d_{\text{ref}} - d_{\text{LoS}}) + \hat{\phi}\right)} \right| \tag{3.24}$$

The optimal bias for each $(\hat{\rho}, \hat{\phi})$ pair is solved analytically as the empirical mean of residuals:

$$\hat{b}(\hat{\rho}, \hat{\phi}) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) \right) \tag{3.25}$$

where y_i is the measured path loss at pixel x_i and N is the number of pixels in the bin. The biased prediction is $\hat{p}_i + \hat{b}$, and the RMSE for the pair is:

$$\text{RMSE}(\hat{\rho}, \hat{\phi}) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) - \hat{b}(\hat{\rho}, \hat{\phi}) \right)^2} \tag{3.26}$$

The best pair is:

$$(\rho^*, \phi^*) = \arg\min_{(\hat{\rho}, \hat{\phi})} \text{RMSE}(\hat{\rho}, \hat{\phi}) \tag{3.27}$$

Post-fit Gaussian smoothing across height bins

After fitting all bins independently, a Gaussian-weighted smooth is applied to stabilize sparse high-altitude bins. Let $\{h_k\}$ be the fitted height bin centers, $\{(\rho_k^*, \phi_k^*, b_k^*)\}$ the per-bin fitted values, and $\{N_k\}$ the pixel counts used in fitting. The implementation weights each bin by $\sqrt{N_k}$ rather than by N_k itself, so very populated bins stabilize the curve without completely dominating neighboring sparse bins. The smoothed values are:

$$\tilde{\rho}(h) = \frac{\sum_k w_k(h) \sqrt{N_k}\, \rho_k^*}{\sum_k w_k(h) \sqrt{N_k}}, \qquad w_k(h) = \exp\left(-\frac{(h - h_k)^2}{2\sigma_h^2}\right), \quad \sigma_h = 10 \text{ m} \tag{3.28}$$

and similarly for ϕ and b. The implementation also stores a suspicious-bin mask for bins where both ρ_k^* and b_k^* differ noticeably from the smoothed global trend, typically high-altitude bins with few samples. That mask is diagnostic metadata: the runtime calibration arrays used by the final residual model are the smoothed ρ, ϕ, and b curves themselves, not a mixture of raw values for ordinary bins and smoothed values only for flagged ones. At inference time the UAV height is not forced to be exactly one of the fitted 5 m bin centers. The implementation linearly interpolates $\rho(h_{\text{tx}})$, $\phi(h_{\text{tx}})$, $b(h_{\text{tx}})$, and the radial residual profile between the two nearest height bins, and clamps to the nearest endpoint outside the fitted range. This keeps the prior continuous with respect to UAV height instead of introducing discrete jumps every 5 m.

Radial residual correction

After fitting the two-ray model, a residual correction is computed separately per height bin and integer radius:

$$r(r_{\text{px}}, h_{\text{tx}}) = \text{clip}\Big(\text{GaussSmooth}_{\sigma_r}\Big(\mathbb{E}_{h\text{-bin},\, r_{\text{px}}}\Big[\text{clip}\big(y - \hat{y}_{\text{2ray}}, -2.5, +2.5\big)\Big]\Big), -2.5, +2.5\Big) \tag{3.29}$$

where r_{px} is the rounded 2D pixel radius from the transmitter, $\sigma_r = 1.5$ pixels is the Gaussian smoothing radius, and the ± 2.5 dB clip is applied both to individual residual samples during accumulation and to the interpolated profile used at inference. Radius bins with no samples in a height bin fall back to the global radial profile before smoothing. At inference, the profile is linearly interpolated across the stored height bins, then indexed by the pixel-radius map and added to the coherent two-ray prediction.

Smoothed deployed parameter values

Table 3.9 shows representative LoS two-ray parameters after the post-fit Gaussian smoothing across height bins. These are the deployed calibration values used by the prior at inference time, not the unsmoothed per-bin grid-search optima.

Table 3.9: *Representative smoothed deployed LoS two-ray parameters by height bin.*

h_{tx} (m)	$\tilde{\rho}$	$\tilde{\phi}$ (rad)	$\tilde{b}$ (dB)
10	0.5605	0.000	−0.037
20	0.5242	0.000	−0.042
30	0.4950	0.000	−0.042
50	0.4632	0.000	−0.047
80	0.4958	0.000	−0.013
100	0.4926	0.000	−0.011
140	0.5308	0.000	−0.028
460	0.8966	−0.030	+1.432

The dominant deployed pattern is $\tilde{\rho} \approx 0.46$ to 0.56 with $\tilde{\phi} \approx 0$ for heights 10 to 100 m, meaning a moderate-amplitude ground reflection with a near-zero fitted extra phase offset. Biases are near-zero (< 0.05 dB) for these bins. At very high altitudes, the smoothed deployed $\tilde{\rho}$ and bias increase because training data is sparser; those bins are stabilized by Gaussian smoothing before deployment.

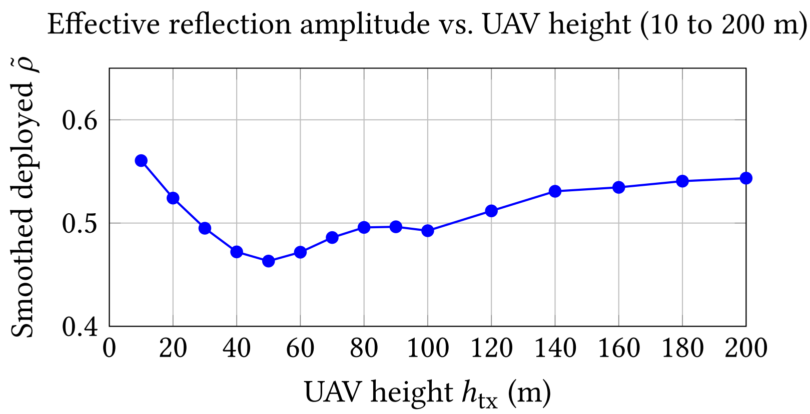

Figure 3.4: *Effective ground-reflection amplitude $\tilde{\rho}$ as a function of UAV height after post-fit smoothing across 5 m height bins. Values are stable in the range* 0.46 *to* 0.56 *for 10 to 200 m, then rise at the sparsely sampled very-high-altitude bins used by the calibration.*

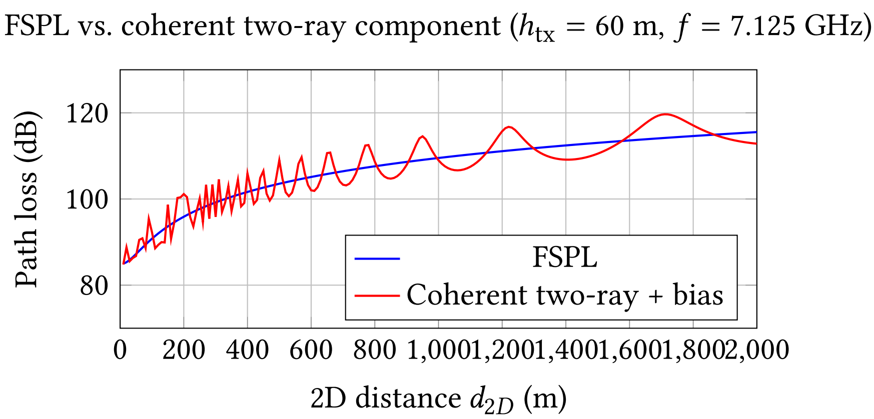

Figure 3.5: *Schematic comparison of FSPL and the calibrated coherent two-ray component for a 60 m UAV at 7.125 GHz (h_{rx} = 1.5 m). The plotted red curve uses the code-loaded 60 m calibration values before adding the radial residual profile, so it shows the constructive/destructive interference term rather than the full final LoS prior.*

3.11.3 NLoS branch

The NLoS path-loss branch used in the final model is not a pure literature formula. It is a hybrid engineering model built in five stages.

Stage 1: raw NLoS propagation prior

Two coarse physical estimates are combined.

COST-231 term. The COST-231 / Hata extension for urban macro cells gives the empirical large-scale path-loss expression used below [6]. It is included because it is one of the standard references for urban NLoS macrocell loss, even though the CKM scenario is outside its original frequency and base-station height range. None of the COST-231 coefficients is therefore treated as a physically transferred absolute predictor at 7.125 GHz; the equation is only a log-distance and height-sensitive basis that the train-only NLoS calibration can bend:

$$\begin{aligned}\mathrm{PL}_{\mathrm{COST231}}(x) = 46.3 + 33.9\log_{10}(f_{\mathrm{MHz}}) - 13.82\log_{10}(h_{\mathrm{tx}}) - a(h_{\mathrm{rx}}) \\ + \big(44.9 - 6.55\log_{10}(h_{\mathrm{tx}})\big)\log_{10}(d_{\mathrm{km}}) + 3\end{aligned} \tag{3.30}$$

where $d_{\mathrm{km}} = \max(d_{2D}(x)/1000, 0.001)$ and the implemented mobile antenna correction is the COST-231 small/medium-city correction:

$$a(h_{\mathrm{rx}}) = \big(1.1\log_{10}(f_{\mathrm{MHz}}) - 0.7\big)\,h_{\mathrm{rx}} - \big(1.56\log_{10}(f_{\mathrm{MHz}}) - 0.8\big) \tag{3.31}$$

Equation (3.30) should therefore be read as a calibrated *feature basis*, not as a trusted absolute predictor or as a claim that COST-231 itself transfers to this 7.125 GHz UAV geometry. COST-231 contributes the expected log-distance growth, transmitter-height dependence and receiver-height correction of an urban empirical model. The subsequent train-only calibration stage is what adapts that shape to the ray-tracing setting used here.

Interpretation of Eq. (3.30).

- $46.3 + 33.9\log_{10}(f_{\mathrm{MHz}})$: frequency-dependent offset. The coefficient 33.9 (compare to 20 for FSPL) reflects stronger frequency dependence in urban scattering.
- $-13.82\log_{10}(h_{\mathrm{tx}})$: as the base station / UAV gets higher, path loss decreases (better clearance above rooftops). The -13.82 coefficient is a fitted constant from the Hata empirical dataset.
- $-a(h_{\mathrm{rx}})$: accounts for the receiver height using the implemented small/medium-city Hata correction. For typical pedestrian heights ($h_{\mathrm{rx}} \approx 1.5$ m), this is a small offset relative to the distance and height terms.
- $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))\log_{10}(d_{\mathrm{km}})$: distance-dependent term. The effective path-loss exponent is $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))/10$, which decreases as the UAV climbs (effectively reducing NLoS scattering losses at longer distances from higher altitudes).
- $+3$: the COST-231 metropolitan correction constant (dB), corresponding to the dense-urban correction in the original model [6]. In this thesis it is retained only as an implementation basis constant, together with the small/medium-city receiver-height correction in Eq. (3.31). This deliberate mixture is not a canonical COST-231 deployment recipe; it is one empirical axis later re-fitted by the regime-wise calibration.

Elevation-aware UAV envelope. The A2G NLoS envelope is used as a conservative elevation-dependent shape term for urban air-to-ground path loss. UAV channel models motivate elevation angle as an important modelling axis because obstruction and shadowing statistics vary with elevation [19, 1]. The particular constants below, however, are not used as a transferred monotonic law. With the positive exponential term, the raw envelope can be larger near overhead; it is therefore only one conservative basis branch that is later combined with COST-231 through a maximum and corrected by the train-only regime-wise NLoS calibration.

$$\mathrm{PL}_{\mathrm{A2G,NLoS}}(x) = \lambda_0(h_{\mathrm{tx}}, f) + \beta_{\mathrm{NLoS}} + A_{\mathrm{NLoS}}\exp\!\left(-\frac{90 - \theta^{\circ}(x)}{\tau_{\mathrm{NLoS}}}\right) \tag{3.32}$$

where:

$$\lambda_0(h_{\text{tx}}, f) = 20 \log_{10}\left(\frac{4\pi h_{\text{tx}} f}{c}\right) \tag{3.33}$$

The fixed A2G-envelope constants used by the implementation are: $\beta_{\text{NLoS}} = -16.16$ dB, $A_{\text{NLoS}} = 12.0436$, $\tau_{\text{NLoS}} = 7.52$ deg. These A2G constants are implementation constants in the Try 78/80 prior code rather than entries in the calibration JSON files; the JSON artifacts store the fitted LoS two-ray curves and the regime-wise NLoS calibration coefficients.

Interpretation of Eq. (3.32).

- λ_0: a free-space-like offset that scales with UAV height and frequency [19, 5]. The CKM dataset uses $f = 7.125$ GHz (FR3 6G frequency band); for $h_{\text{tx}} = 50$ m at that frequency, $\lambda_0 \approx 83.5$ dB.
- $\beta_{\text{NLoS}} = -16.16$ dB: a fixed baseline offset relative to λ_0 in the raw A2G envelope.
- $A_{\text{NLoS}} \exp(-\ldots)$: an exponential modulation with $(90 - \theta^\circ)$. When the elevation angle θ° is high (UAV nearly overhead, $\theta^\circ \to 90^\circ$), the exponent vanishes and the term goes to A_{NLoS}. When θ° is low (grazing incidence), the exponent is large and the term approaches zero. With the current positive $A_{\text{NLoS}} = 12.0436$, this raw envelope therefore adds a larger A2G offset at high elevation. The later max(COST231, A2G) switch and the regime-wise NLoS calibration decide whether that branch is active in the final raw prior; it should be read as a fitted conservative envelope, not as a standalone monotonic law saying all overhead NLoS links have lower loss.
- $\tau_{\text{NLoS}} = 7.52^\circ$: the characteristic angular scale. A low value means the transition from grazing to overhead happens sharply within a small elevation range.

The raw NLoS prior is then the conservative (higher) of the two estimates:

$$\text{PL}_{\text{NLoS,raw}}(x) = \max\left(\text{PL}_{\text{COST231}}(x), \text{PL}_{\text{A2G,NLoS}}(x)\right) \tag{3.34}$$

The maximum in Eq. (3.34) is deliberately pessimistic. If the empirical COST-231 curve predicts stronger attenuation than the A2G envelope, the model keeps the urban-macro value; if the elevation-aware envelope predicts stronger attenuation, the model keeps the A2G value. This prevents the raw prior from becoming unrealistically optimistic in NLoS streets where neither source model is fully valid on its own.

Stage 2: raw hybrid prior assembly

The NLoS construction above defines only the obstructed branch. The full raw formula map used by the NLoS calibration stage is assembled only after both legacy branch estimates are available: the auxiliary LoS blend from Eq. (3.38), and the conservative NLoS branch from Eq. (3.34). This auxiliary LoS blend is not the final enhanced two-ray LoS prior; it is retained only because $\text{PL}^{(0)}$ is one input feature of the calibrated NLoS branch.

The legacy LoS path term used by this auxiliary branch is:

$$\mathrm{PL}_{\mathrm{LoS,path}}(x) = \begin{cases} \mathrm{FSPL}(x), & d_{\mathrm{LoS}}(x) \le d_c, \\ 40\log_{10}(d_{\mathrm{LoS}}(x)) - 20\log_{10}(h_{\mathrm{tx}}) - 20\log_{10}(h_{\mathrm{rx}}), & d_{\mathrm{LoS}}(x) > d_c, \end{cases} \tag{3.35}$$

with

$$d_c = \max\left\{\frac{4\pi h_{\mathrm{tx}} h_{\mathrm{rx}}}{\lambda}, 1\ \mathrm{m}\right\}, \qquad \lambda = \frac{c}{f}. \tag{3.36}$$

Equation (3.35) switches from FSPL to the classical asymptotic two-ray d^4 path-loss law after the crossover distance d_c. It is an older helper used only to construct the raw $\mathrm{PL}^{(0)}$ feature for the NLoS calibration, not the final enhanced coherent two-ray LoS prior of Eq. (3.19).

The legacy LoS blend is then defined as:

$$\mathrm{PL}_{\mathrm{A2G,LoS}}(x) = \lambda_0(h_{\mathrm{tx}}, f) - 20\log_{10}\left(\max\{\sin\theta(x), 10^{-4}\}\right) \tag{3.37}$$

This auxiliary LoS envelope uses the same λ_0 term as Eq. (3.33); the sine clamp matches the implementation and avoids a singularity at grazing elevation. It is a legacy helper for $\mathrm{PL}^{(0)}$, not the final LoS prior.

$$\mathrm{PL}_{\mathrm{LoS,blend}}(x) = 0.7\,\mathrm{PL}_{\mathrm{LoS,path}}(x) + 0.3\min\left(\mathrm{PL}_{\mathrm{LoS,path}}(x), \mathrm{PL}_{\mathrm{A2G,LoS}}(x)\right) \tag{3.38}$$

The $0.7/0.3$ blend in Eq. (3.38) gives 70% weight to the legacy LoS path model and 30% weight to the lower-loss of the legacy path model and the A2G LoS envelope. Because lower dB means a stronger received signal, the $\min(\cdot)$ operation should not be read as a conservative guard against low path-loss peaks; it is simply the older low-loss auxiliary blend inherited by the raw $\mathrm{PL}^{(0)}$ feature. Here $\mathrm{PL}_{\mathrm{LoS,path}}(x)$ denotes the *legacy* LoS path term defined in Eq. (3.35). It is not the calibrated enhanced two-ray map of Eq. (3.19).

The full raw prior combines the LoS and NLoS estimates using the binary LoS mask $m_{\mathrm{LoS}}(x) \in \{0, 1\}$:

$$\mathrm{PL}^{(0)}(x) = m_{\mathrm{LoS}}(x)\,\mathrm{PL}_{\mathrm{LoS,blend}}(x) + (1 - m_{\mathrm{LoS}}(x))\,\mathrm{PL}_{\mathrm{NLoS,raw}}(x) \tag{3.39}$$

Since $m_{\mathrm{LoS}}(x)$ is binary, Eq. (3.39) is a hard switch: it selects the auxiliary LoS expression on LoS pixels and the NLoS expression on NLoS pixels. The auxiliary blend in Eq. (3.38) was used in older variants to keep the LoS formula below an A2G envelope. In the final frozen path, LoS output pixels come directly from the separate enhanced two-ray prior $\mathrm{PL}_{\mathrm{LoS}}$ of Eq. (3.19); the raw hybrid map $\mathrm{PL}^{(0)}$ in Eq. (3.39) is still computed only as an input feature for the calibrated NLoS path-loss branch. The LoS/NLoS mask used here is the explicit CKM mask provided with the dataset and treated as an available side channel. A deployment version would need to obtain an equivalent visibility mask from deterministic geometry or another inexpensive pre-processing step; replacing the supplied mask is not evaluated in this thesis.

Stage 3: map-derived morphology features

The raw hybrid prior $\mathrm{PL}^{(0)}(x)$ defined in Eq. (3.39) is the scalar formula input to the calibrated NLoS path-loss branch. It combines the legacy FSPL/two-ray LoS path, the A2G envelopes and COST-231, but it still knows nothing about the detailed local morphology around each receiver. The feature vector $\mathbf{x}_{78}(x)$ defined below is *not* a competing path-loss model and *not* an alternative two-ray formulation. It is the per-pixel input row of a calibrated linear regressor whose role is to take the raw formula prediction and absorb its residual using context that the formula cannot see (urban morphology, NLoS visibility, shadowing and elevation geometry). Concretely, $\mathrm{PL}^{(0)}(x)$ is fed in as the dominant feature, plus a quadratic of itself so that the linear fit can bend the prior, and every other entry adds a quantity that is invisible to the raw formula.

From $\mathrm{PL}^{(0)}(x)$ and the map topology, a 15-element feature vector is computed:

$$\mathbf{x}_{78}(x) = \begin{bmatrix} \mathrm{PL}^{(0)}(x)^2 \\ \mathrm{PL}^{(0)}(x) \\ \log(1 + d_{2D}(x)) \\ \rho_{15}(x) \\ \rho_{41}(x) \\ h_{15}(x) \\ h_{41}(x) \\ \rho_{41}(x)\log(1 + d_{2D}(x)) \\ n_{15}(x) \\ n_{41}(x) \\ n_{41}(x)\log(1 + d_{2D}(x)) \\ \sigma_{\text{shadow}}(x) \\ \theta_{\text{norm}}(x) \\ n_{41}(x)\theta_{\text{norm}}(x) \\ 1 \end{bmatrix} \tag{3.40}$$

The lowercase bold notation is intentional: $\mathbf{x}_{78}(x)$ is a per-pixel feature vector. When vectors from many pixels in one regime are stacked for calibration, they form the uppercase design matrix $\mathbf{X}_r$. Table 3.10 gives the role of each entry before the detailed definitions below.

Term	Mathematical role	Physical interpretation
$\mathrm{PL}^{(0)}(x)^2$	Quadratic basis term.	Lets the linear calibration bend the raw path-loss curve when the raw model is systematically curved.
$\mathrm{PL}^{(0)}(x)$	Linear raw-prior term.	Keeps the calibrated model anchored to the physical prior instead of relearning path loss from scratch.
$\log(1 + d_{2D}(x))$	Smooth distance basis.	Mirrors log-distance propagation laws used in classical models [35, 6].
$\rho_{15}(x), \rho_{41}(x)$	Local building-density averages.	Count how much nearby urban structure can block, reflect or diffract the signal.
$h_{15}(x), h_{41}(x)$	Density-weighted local height averages.	Distinguish low dense blocks from taller blocks with stronger obstruction.
$\rho_{41}(x)\log(1 + d_{2D}(x))$	Density-distance interaction.	Allows distance slope to increase in dense areas.
$n_{15}(x), n_{41}(x)$	Local NLoS-support fractions.	Measure how much nearby ground is geometrically shielded from the UAV.
$n_{41}(x)\log(1 + d_{2D}(x))$	NLoS-distance interaction.	Allows blocked streets to accumulate loss differently with range.
$\sigma_{\text{shadow}}(x)$	Elevation-dependent shadowing proxy.	Encodes the stronger low-elevation variability reported in A2G channel models [19, 1].
$\theta_{\text{norm}}(x)$	Normalized elevation angle.	Separates grazing links from near-overhead links.
$n_{41}(x)\theta_{\text{norm}}(x)$	NLoS-elevation interaction.	Allows the NLoS correction to depend on whether shielding occurs at low or high elevation.
1	Bias term.	Lets each regime learn a constant offset after all physical terms are included.

Table 3.10: *Term-by-term inventory of the NLoS calibration vector.*

Each term is now described in detail.

Base masks. The ground mask is:

$$g(x) = \begin{cases} 1, & \text{if topology}(x) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{3.41}$$

The building mask is $b(x) = 1 - g(x)$. The ground-NLoS support mask is:

$$n_{\text{raw}}(x) = \mathbf{1}[m_{\text{LoS}}(x) \leq 0.5] \cdot g(x) \tag{3.42}$$

This selects ground pixels that are predominantly in NLoS, i.e., pixels that are at street level and are shielded from the UAV.

Quadratic raw-prior feature. The pair $(\text{PL}^{(0)}(x)^2, \text{PL}^{(0)}(x))$ allows the linear model to fit a *quadratic* relationship in the raw prior. Including both P^2 and P as features lets the calibration curve be: $\hat{P} = \beta_1 P^2 + \beta_2 P + \cdots$, which can correct for systematic curvature in the raw prior. This pair is intentionally redundant: the squared feature is completely determined by the raw-prior feature. It is kept because the calibrated fit improves when the linear model is allowed to represent a curved mapping from raw prior to measured path loss.

Distance term. $\log(1 + d_{2D}(x))$: the logarithm of one plus the horizontal distance, providing a distance-dependent offset. The $+1$ prevents a singularity at $d_{2D} = 0$. The logarithm is consistent with the log-distance structure of empirical path-loss laws: a multiplicative increase in distance produces an additive increase in predicted dB loss [35, 6].

Per-pixel box-mean operator. All multiscale features use the same 2D box-mean operator BoxMean_k. In the prior vector this operator is not a standalone feature; it is the common construction used to compute the neighborhood terms ρ_{15}, ρ_{41} from the building mask, h_{15}, h_{41} from density-weighted building height, and n_{15}, n_{41} from the ground-NLoS support mask. For a generic input map $f : \mathbb{Z}^2 \to \mathbb{R}$ and a window half-width $p = \lfloor k/2 \rfloor$, the value at pixel $x = (i, j)$ is:

$$\text{BoxMean}_k(f)(i, j) = \frac{1}{k^2} \sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\text{padded}}(i', j') \tag{3.43}$$

where f_{padded} is f extended outside the map boundary by *reflect* (mirror) padding: $f_{\text{padded}}(-r, j) = f_{\text{padded}}(r, j)$ and similarly at all four edges. This ensures border pixels receive a full k^2-pixel average, with the map reflected back inward rather than zero-padded. Equation (3.43) is evaluated efficiently using a *2D summed-area table* (integral image). Let $I(i, j) = \sum_{i' \leq i, j' \leq j} f_{\text{padded}}(i', j')$ be the prefix-sum array. Then:

$$\begin{aligned} S_k(i, j) &= \sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\text{padded}}(i', j') \\ &= I(i + p,\, j + p) - I(i - p - 1,\, j + p) \\ &\quad - I(i + p,\, j - p - 1) + I(i - p - 1,\, j - p - 1) \end{aligned} \tag{3.44}$$

This reduces each per-pixel evaluation to four table lookups, making the full-map computation $O(HW)$ regardless of k. In the implementation, $k \in \{15, 41\}$ pixels, corresponding to $\sim$15 m and $\sim$41 m neighborhoods (1 m/pixel resolution).

Multiscale building density.

$$\rho_k(x) = \mathrm{BoxMean}_k(b(\cdot))\,(x) = \frac{1}{k^2} \sum_{x' \in W_k(x)} b(x') \tag{3.45}$$

where $W_k(x)$ is the $k \times k$ window centered at x. Since $b(x') \in \{0, 1\}$, $\rho_k(x)$ is the *fraction of pixels that are buildings* in the window. It is zero at a ground pixel surrounded only by other ground pixels, and one at a pixel deep inside a solid building block. ρ_{15} captures fine-scale building density within a ~15 m neighborhood; ρ_{41} captures a coarser ~41 m neighborhood.

Multiscale building height.

$$h_k(x) = \frac{\mathrm{BoxMean}_k(\mathrm{topology}(\cdot)\, b(\cdot))\,(x)}{H_{\mathrm{norm}}}, \quad H_{\mathrm{norm}} = 90 \text{ m} \tag{3.46}$$

This is the box mean of $\mathrm{topology}(x') \cdot b(x')$: building pixels contribute their height in metres; ground pixels contribute 0. Dividing by k^2 (inside $\mathrm{BoxMean}_k$) gives:

$$h_k(x) = \frac{1}{H_{\mathrm{norm}}\, k^2} \sum_{x' \in W_k(x)} \mathrm{topology}(x') \cdot b(x') = \frac{\rho_k(x)}{H_{\mathrm{norm}}}\, \overline{h}_{\mathrm{bldg},k}(x) \tag{3.47}$$

where $\overline{h}_{\mathrm{bldg},k}(x)$ is the *mean height of building pixels only* within the window:

$$\overline{h}_{\mathrm{bldg},k}(x) = \frac{\sum_{x' \in W_k(x)} \mathrm{topology}(x')\, b(x')}{\sum_{x' \in W_k(x)} b(x')} \quad \text{(undefined if } \rho_k(x) = 0\text{)} \tag{3.48}$$

Critically, $h_k(x) = \rho_k(x) \cdot \overline{h}_{\mathrm{bldg},k}(x)/H_{\mathrm{norm}}$, so h_k combines *both* how many buildings are present and how tall they are. A window with 50% building coverage and mean height 20 m gives the same h_k as a window with 100% coverage at mean height 10 m. The $H_{\mathrm{norm}} = 90$ m normalizer maps the joint range to approximately $[0, 1]$.

Multiscale NLoS-support.

$$n_k(x) = \mathrm{BoxMean}_k(n_{\mathrm{raw}}(\cdot))\,(x) \tag{3.49}$$

Since $n_{\mathrm{raw}}(x') \in \{0, 1\}$ (Eq. 3.42), $n_k(x)$ is the *absolute fraction of the window area* that consists of ground pixels in NLoS. Just as $h_k(x)$ conflates building coverage and building height, $n_k(x)$ conflates ground coverage and NLoS probability. We can write:

$$n_k(x) = (1 - \rho_k(x))\, \overline{n}_{\mathrm{ground},k}(x) \tag{3.50}$$

where $\overline{n}_{\mathrm{ground},k}(x)$ is the relative fraction of *available ground pixels* that are shielded:

$$\bar{n}_{\text{ground},k}(x) = \frac{\sum\limits_{x' \in W_k(x)} n_{\text{raw}}(x')}{\sum\limits_{x' \in W_k(x)} g(x')} \quad \text{(undefined if } \rho_k(x) = 1) \tag{3.51}$$

Because building pixels never contribute to n_k, this feature is zero in completely open LoS areas (where $\bar{n}_{\text{ground},k} = 0$) *and* deep inside solid building blocks (where $1 - \rho_k = 0$). It reaches its maximum in narrow urban street canyons, where ground coverage is still significant but entirely shielded from the UAV.

Elevation angle.

$$\theta(x) = \arctan\left(\frac{h_{\text{tx}} - h_{\text{rx}}}{\max(d_{2D}(x), 1)}\right) \tag{3.52}$$

The $\max(\cdot, 1)$ prevents division by zero directly below the UAV. Then:

$$\theta_{\text{norm}}(x) = \text{clip}\left(\frac{\theta^{\circ}(x)}{90}, \ 0, \ 1\right) \tag{3.53}$$

where $\theta^{\circ} = \theta \cdot 180/\pi$ is in degrees. $\theta_{\text{norm}} = 0$ at grazing incidence, $= 1$ directly overhead.

Shadowing proxy.

$$\sigma_{\text{LoS}}(x) = \alpha_{\text{LoS}} \, (90 - \theta^{\circ}(x))^{\mu_{\text{LoS}}} \tag{3.54}$$

$$\sigma_{\text{NLoS}}(x) = \alpha_{\text{NLoS}} \, (90 - \theta^{\circ}(x))^{\mu_{\text{NLoS}}} \tag{3.55}$$

with fitted values $\alpha_{\text{LoS}} = 0.0272$, $\mu_{\text{LoS}} = 0.7475$, $\alpha_{\text{NLoS}} = 2.3197$, and $\mu_{\text{NLoS}} = 0.2361$. The blended proxy is:

$$\sigma_{\text{shadow}}(x) = m_{\text{LoS}}(x) \, \sigma_{\text{LoS}}(x) + (1 - m_{\text{LoS}}(x)) \, \sigma_{\text{NLoS}}(x) \tag{3.56}$$

This acts as a spatially varying uncertainty proxy rather than a measured shadow-fading standard deviation. Higher values occur at low elevation angles, where propagation conditions are more difficult; the LoS/NLoS blend lets the same feature have a different scale in the two visibility regimes.

Interaction terms. The three pairwise products $\rho_{41}(x) \log(1 + d_{2D}(x))$, $n_{41}(x) \log(1 + d_{2D}(x))$, and $n_{41}(x)\theta_{\text{norm}}(x)$ allow the linear model to capture effects where building density or NLoS support *modifies* the slope of path loss with distance or elevation angle, rather than just shifting it. These terms are also deliberately correlated with their factors. For example, $n_{41}(x) \log(1+d_{2D}(x))$ cannot vary independently of $n_{41}(x)$ or $\log(1+d_{2D}(x))$. This cross-correlation makes coefficient interpretation less clean, but removing such terms can reduce performance because the fit then loses a simple way to express "distance matters differently in blocked streets". Determining exactly which terms are dispensable would require a large ablation sweep over pairs or subsets of features. Even a pairwise screening would scale as $O(n^2)$ runs for a vector with n terms, and each run would have to regenerate and inspect large calibration artifacts. The LoS two-ray calibration JSON used by the final model, for example, is about 200 000 lines long, mostly because it stores height-dependent radial residual profiles. Taken together, the stored calibration is height-dependent, building-topology-dependent, and density-dependent: the LoS JSON contributes height-dependent radial residuals, while the NLoS side adds morphology calibration through building density, building height, NLoS support, topology regimes, and antenna-height features. For this reason the prior should be read as a calibrated baseline with substantial stored structure, not as a single hand-written formula, and the final vector keeps some correlated terms when they improved validation behavior.

Stage 4: regime-wise linear calibration

The regime r is identified by three keys:

- *Coarse topology:* one of `open_lowrise`, `mixed_midrise`, or `dense_highrise`,
- *Propagation state:* `LoS` or `NLoS`,
- *Antenna height bin:* `low_ant`, `mid_ant`, or `high_ant`.

This gives up to $3 \times 2 \times 3 = 18$ regimes.

The stored calibration for this legacy mixed-prior map still contains both LoS and NLoS regime coefficients. However, in the final frozen path-loss assembly only the NLoS pixels from this calibrated map are used, because LoS pixels are replaced by the separate enhanced two-ray prior.

How topology class and antenna bin are computed. These are *per-map* scalar labels, not per-pixel quantities. They are derived once from the full topology tensor and UAV height.

The *map-level* building density and mean building height are:

$$\bar{\rho}_{\text{map}} = \frac{1}{HW} \sum_x b(x) \tag{3.57}$$

$$\bar{h}_{\text{map}} = \frac{\sum_x \text{topology}(x)\, b(x)}{\sum_x b(x)} \tag{3.58}$$

$\bar{\rho}_{\text{map}}$ is simply the fraction of map pixels that are buildings. $\bar{h}_{\text{map}}$ is the mean building height *over building pixels only* (the denominator excludes ground pixels). This is different from the per-pixel feature $h_k(x)$ in Eq. (3.47), which divides the same numerator by the full window area k^2 instead of by the number of building pixels.

For the calibrated path-loss prior, these two quantities are converted into three coarse city types. This is a separate regime label from the later six-class topology taxonomy used by the spread priors and by final reporting:

$$\text{class}_{78} = \begin{cases} \texttt{dense\_highrise} & \bar{\rho} \geq d_2 \text{ or } \bar{h} \geq h_2 \\ \texttt{open\_lowrise} & \bar{\rho} \leq d_1 \text{ and } \bar{h} \leq h_1 \\ \texttt{mixed\_midrise} & \text{otherwise} \end{cases} \tag{3.59}$$

with thresholds $d_1 = 0.1957$, $d_2 = 0.2549$ (density fractions), $h_1 = 10.91$ m, and $h_2 = 15.95$ m. Together with the binary LoS/NLoS label and the three UAV-height bins below, this yields the 18 stored path-loss calibration keys.

The antenna-height bin is assigned by UAV height tertiles:

$$\text{ant\_bin} = \begin{cases} \texttt{low\_ant} & h_{\text{tx}} \leq 58.12 \text{ m} \\ \texttt{mid\_ant} & 58.12 < h_{\text{tx}} \leq 103.85 \text{ m} \\ \texttt{high\_ant} & h_{\text{tx}} > 103.85 \text{ m} \end{cases} \tag{3.60}$$

The calibrated prediction for regime r is:

$$\widehat{\text{PL}}_r(x) = \mathbf{x}_{78}(x)^\top \mathbf{w}_r^{78} = \sum_{j=1}^{15} x_j(x)\, w_{r,j}^{78} \tag{3.61}$$

How $\mathbf{w}_r^{78}$ is fitted. Across all training samples whose regime label equals r, a linear regression is run:

$$\hat{\mathbf{w}}_r^{78} = \arg\min_{\mathbf{w}\in\mathbb{R}^{15}} \sum_{x\in\mathcal{S}_r} \left(y(x) - \mathbf{x}_{78}(x)^\top \mathbf{w}\right)^2 \tag{3.62}$$

where $\mathcal{S}_r$ is the set of all valid pixels in training samples belonging to regime r. This ordinary least squares problem has the closed form solution, when $\mathbf{X}_r^\top \mathbf{X}_r$ is nonsingular:

$$\hat{\mathbf{w}}_r^{78} = \left(\mathbf{X}_r^\top \mathbf{X}_r\right)^{-1} \mathbf{X}_r^\top \mathbf{y}_r \tag{3.63}$$

where $\mathbf{X}_r$ stacks the feature vectors and $\mathbf{y}_r$ stacks the measured path-loss values. The runtime code does not refit this model; it loads the stored coefficient vectors from `try78_nlos_regime_calibration.json`. At inference, if the exact antenna-height key is absent, the implementation only tries alternative antenna bins in the order current, mid, low, high while keeping the same path-loss-prior city type and LoS/NLoS label. It does not use the broader cross-topology fallback hierarchy of the spread-prior calibration.

The terms in Eq. (3.63) are standard linear least-squares quantities. The matrix $\mathbf{X}_r^\top \mathbf{X}_r$ is the *Gram matrix*; its diagonal entries measure the energy of each feature inside regime r, and its off-diagonal entries measure correlations between features. The vector $\mathbf{X}_r^\top \mathbf{y}_r$ measures how strongly each feature correlates with measured path loss. The inverse maps those feature-target correlations into the fitted coefficient vector $\hat{\mathbf{w}}_r^{78}$. This is the classical ordinary least-squares solution, but applied separately per propagation regime rather than globally. Because the feature vector intentionally contains correlated terms, the important object is the least-squares coefficient vector stored in `try78_nlos_regime_calibration.json`; if a normal-equation matrix is rank-deficient or ill-conditioned, Eq. (3.63) should be understood as the full-rank notation for the corresponding least-squares or pseudo-inverse solution, not as a requirement that every calibration block be perfectly invertible.

An example fitted coefficient vector for the `mixed_midrise|NLoS|high_ant` regime is:

$$\mathbf{w}^{78}_{\texttt{mixed|NLoS|high}} = \begin{bmatrix} -0.000338,\ -0.0266,\ 8.233,\ -1.452,\ -11.659 \\ 3.508,\ -5.611,\ 1.904,\ 0.926,\ 12.084 \\ -1.628,\ -7.487,\ -12.665,\ -10.959,\ 121.480 \end{bmatrix}^{\top}$$

The large bias weight and the sizeable morphology, elevation and raw-prior weights should be interpreted together as one fitted calibration vector, not as independent physical constants. Their magnitudes mainly reflect the scale and correlation of the 15 input features in this NLoS regime. This makes the path-loss prior an interpretable calibrated surrogate rather than a pure first-principles propagation formula: the two-ray and A2G terms set the shape, while regime-wise fitting adapts the coefficients to the CKM simulator and to the dataset's stored masks.

Stage 5: region replacement and clipping

$$\mathrm{PL}_{\mathrm{NLoS}}(x) = \mathrm{clip}\big(\mathbf{x}_{78}(x)^{\top}\mathbf{w}_r^{78}, 20, 180\big) \tag{3.64}$$

Hard clipping to $[20, 180]$ dB prevents the linear model from producing physically unreasonable values. The final hybrid path-loss prior is:

$$\mathrm{PL}_{\mathrm{prior}}(x) = \begin{cases} \mathrm{PL}_{\mathrm{LoS}}(x), & m_{\mathrm{LoS}}(x) > 0.5 \\ \mathrm{PL}_{\mathrm{NLoS}}(x), & \text{otherwise} \end{cases} \tag{3.65}$$

3.11.4 Evaluation metric: pixel-weighted RMSE

All numerical results are reported as *pixel-weighted* (pw) RMSE or MAE. Building pixels ($\mathrm{topology}(x) > 0$) are always excluded; only valid ground pixels with a finite, physically plausible target value contribute. Let $\mathscr{V}_k$ be the set of all such valid (s, x) pairs across all evaluation samples for task k. Then:

$$\mathrm{RMSE}_k^{\mathrm{pw}} = \sqrt{\frac{\sum_{(s,x)\in\mathscr{V}_k} \left(\hat{y}_k^{(s)}(x) - y_k^{(s)}(x)\right)^2}{|\mathscr{V}_k|}} \tag{3.66}$$

This is a single square root over all pooled valid pixels, not an average of per-sample RMSEs. Units are dB (path loss), ns (delay spread), and degrees (angular spread), all in the native non-transformed domain.

3.11.5 Observed path-loss-prior results

The headline path-loss numbers for the calibrated prior are reported in the results chapter (LoS waterfall in Table 4.2): the enhanced LoS two-ray prior $\mathrm{PL}_{\mathrm{LoS}}$ reaches 1.7516 dB pixel-weighted RMSE on LoS pixels, the regime-wise OLS NLoS head reaches 3.3967 dB on NLoS pixels, and the combined prior reaches 1.9246 dB overall on the held-out validation+test split.

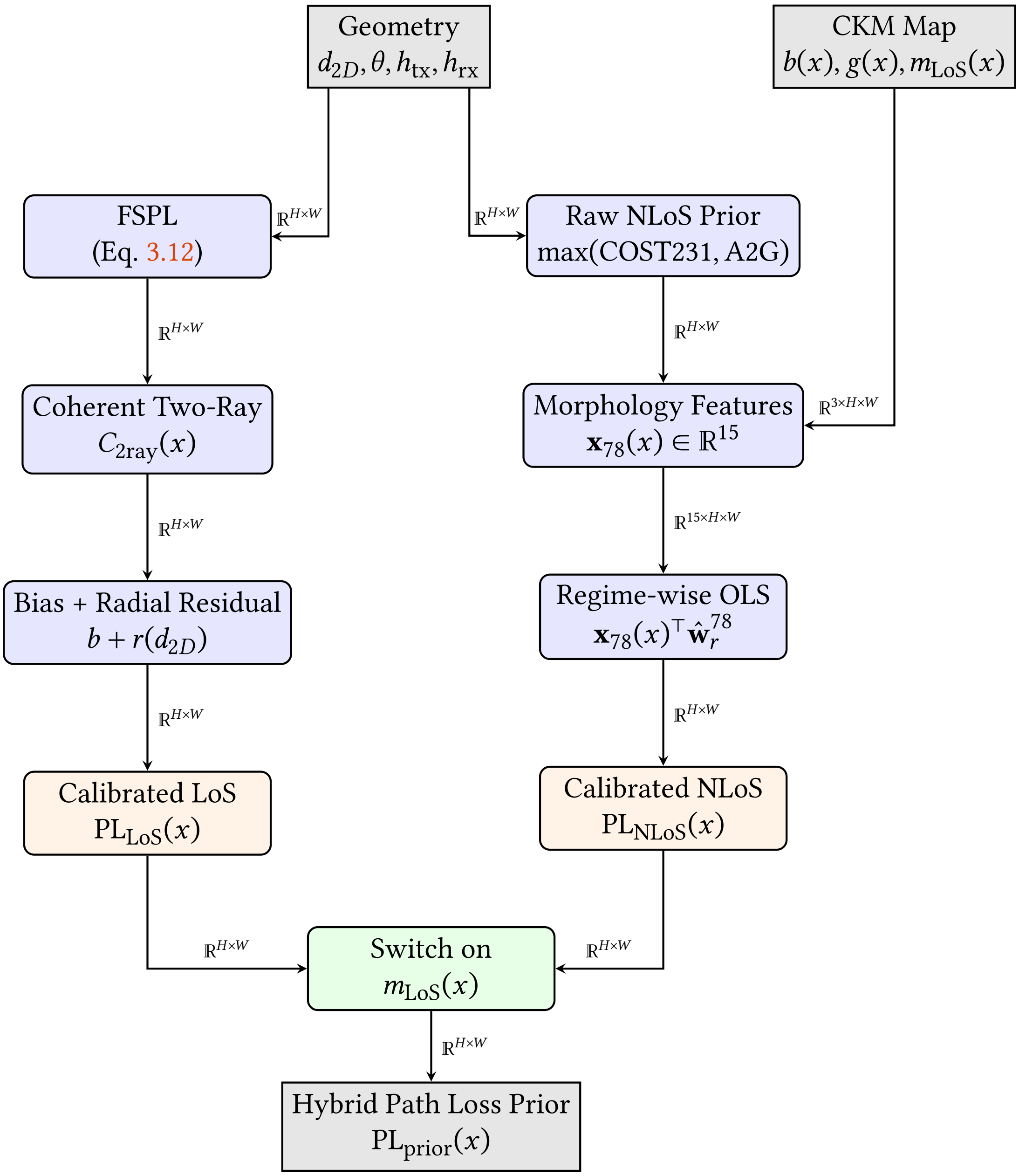

Figure 3.6: *Pipeline for the hybrid path loss prior. The LoS branch relies on an explicit two-ray interference model, while the NLoS branch calibrates a raw prior against map-derived morphology features using regime-wise OLS. The two branches are hard-switched based on the LoS mask to produce the final frozen prior.*

3.12 Delay and Angular Spread Priors

3.12.1 Physical motivation for log-domain modeling

Delay spread (DS) and angular spread (AS) are non-negative, often zero-inflated, and heavy-tailed. Their empirical distributions are commonly modeled in the logarithmic domain in standard channel models [1, 20]. For a spread variable y, define:

$$z(x) = \log(1 + y(x)), \qquad \hat{y}(x) = \exp(\hat{z}(x)) - 1 \tag{3.67}$$

The addition of 1 is not cosmetic. It keeps exact zeros finite, avoids log(0), and makes the same transform usable for both delay spread in nanoseconds and angular spread in degrees. If two pixels have native spread values y_a and y_b, their difference in transformed space is

$$z_a - z_b = \log\left(\frac{1 + y_a}{1 + y_b}\right). \tag{3.68}$$

Thus, a fixed error in z corresponds approximately to a fixed multiplicative error in $1 + y$. This is more appropriate than native-space least squares for spreads because large NLoS tails would otherwise dominate the fit and the near-zero LoS mass would be ignored. The calibrated spread prior is still a deterministic log-domain ridge estimator; it is not claiming to estimate the full spike-plus-tail probability distribution.

3.12.2 Pixel-wise notation for the spread-prior fit

Let n index a valid ground pixel from a training map. For a selected metric $m \in \{\mathrm{DS}, \mathrm{AS}\}$, define the native target $y_{m,n}$, the transformed target $z_{m,n} = \log(1 + y_{m,n})$, and the raw log-prior value $z^{(0)}_{m,n}$. The calibrated prior is linear in a 23-dimensional feature vector:

$$\hat{z}_{m,n} = \mathbf{x}_{m,n}^{\top}\mathbf{w}_{m,r(n)}, \qquad \mathbf{x}_{m,n} \in \mathbb{R}^{23}, \qquad \mathbf{w}_{m,r} \in \mathbb{R}^{23}. \tag{3.69}$$

The regime $r(n)$ is a discrete key containing the metric, topology class, LoS/NLoS state and UAV-height bin. In other words, the spread prior does not fit one global line through all pixels. It fits a family of small linear models, each one specialized to a propagation regime.

3.12.3 Raw spread prior in log domain

$$\begin{aligned} z^{(0)}(x) = \;& \log\Big(1 + b^{\text{native}}_{\text{LoS/NLoS}}(x)\Big) + b_{\text{topology}} + a_1 \log(1 + d_{2D}) + a_2\theta_{\text{inv}} \\ & + a_3\rho_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41}\theta_{\text{inv}}. \end{aligned} \tag{3.70}$$

where $\theta_{\text{inv}}(x) = 1 - \theta_{\text{norm}}(x)$ (high value at low elevation). Example coefficients for delay spread: the native anchors are $b^{\text{native}}_{\text{LoS}} = 4.0$ ns and $b^{\text{native}}_{\text{NLoS}} = 11.0$ ns, which the implementation converts to $\log(1 + 4.0)$ and $\log(1 + 11.0)$ before adding the remaining log-domain terms. The delay coefficients are $a_1 = 0.11$, $a_2 = 0.62$, $a_3 = 0.48$, $a_4 = 0.32$, $a_5 = 0.86$, $a_6 = 0.60$. After Eq. (3.70), the raw log prior is clipped to $[0, 8]$, inverse transformed with $\exp(\cdot) - 1$, and clipped to the metric upper bound. For reproducibility, the $z^{(0)}(x)$ entries used later in $\mathbf{x}_{79}$ denote the log value recovered from this clipped native raw prior, i.e., $\log(1 + \text{raw\_prior}_{\text{clipped}}(x))$, not an unconstrained copy of Eq. (3.70).

The raw prior is not a 3GPP formula copied directly into CKM. Instead, it takes the 3GPP/WINNER lesson that spread statistics should be separated by propagation state and represented in log space [1, 20], then adds CKM-specific map morphology terms. UAV measurement studies also motivate the elevation-angle dependence: low-elevation links usually interact with more rooftops and street canyons than near-overhead links [3, 16]. For reproducibility, Table 3.11 lists the fixed raw-prior constants used before the regime-wise ridge calibration.

Table 3.11: *Fixed constants used in the raw spread prior of Eq. (3.70).*

Quantity	Delay spread	Angular spread	Meaning
$b_{\text{LoS}}^{\text{native}}$	4.0 ns	3.0°	Native LoS anchor converted to $\log(1+b)$.
$b_{\text{NLoS}}^{\text{native}}$	11.0 ns	7.0°	Native NLoS anchor converted to $\log(1+b)$.
$a_1,\dots,a_6$	0.11, 0.62, 0.48, 0.32, 0.86, 0.60	0.05, 0.52, 0.60, 0.38, 0.58, 0.35	Coefficients of $\log(1+d_{2D})$, θ_{inv}, ρ_{41}, h_{41}, n_{41}, and $n_{41}\theta_{\text{inv}}$, respectively.

The topology offset b_{topology} is also fixed before ridge calibration. For delay spread, the offsets for `open_sparse_lowrise`, `open_sparse_vertical`, `mixed_compact_lowrise`, `mixed_compact_midrise`, `dense_block_midrise`, and `dense_block_highrise` are $0.00, 0.18, 0.06, 0.22, 0.24, 0.34$. For angular spread, the corresponding offsets are $0.00, 0.09, 0.05, 0.16, 0.18, 0.24$. These are log-domain offsets, not native ns/degree offsets.

Every term in Eq. (3.70) has a physical interpretation:

- $b_{\text{LoS/NLoS}}$ is the base spread level for the propagation state. NLoS starts higher because blocked paths arrive after additional reflections and diffractions.
- b_{topology} is a morphology offset. Dense and high-rise topologies receive larger offsets because they contain more reflectors.
- $\log(1+d_{2D})$ gives a smooth range dependence. Longer links have more opportunity to encounter delayed or angularly separated paths.
- θ_{inv} increases at low elevation angle, where the UAV-to-ground ray intersects more urban structure before reaching the receiver.
- ρ_{41}, h_{41}, and n_{41} describe local building density, mean building height, and NLoS support in a 41×41 window. These are direct map-derived proxies for multipath richness.
- $n_{41}\theta_{\text{inv}}$ is an interaction term. It activates when low-elevation geometry and local obstruction occur together.

3.12.4 Calibrated spread prior

Stage 2: 23-feature vector and design matrix

$$\mathbf{x}_{79}(x) = \begin{bmatrix} z^{(0)}(x)^2,\ z^{(0)}(x),\ \log(1+d_{2D}),\ \theta_{\text{norm}},\ \theta_{\text{inv}} \\ h_{\text{norm}},\ h_{\text{norm}}^2,\ \rho_{15},\ \rho_{41},\ h_{15},\ h_{41} \\ n_{15},\ n_{41},\ n_{41}\log(1+d_{2D}),\ \rho_{41}\theta_{\text{inv}},\ b_{41},\ 1 \\ c_{41},\ t_{41},\ \theta_{\text{norm}}\rho_{41},\ c_{41}\theta_{\text{inv}},\ t_{41}\rho_{41},\ t_{41}n_{41} \end{bmatrix}^{\top} \tag{3.71}$$

The vector in Eq. (3.71) can be grouped into five blocks:

$$\mathbf{x}_{79}(x) = \Big[\underbrace{\mathbf{x}_{\text{prior}}(x),}_{\text{raw prior}}\ \underbrace{\mathbf{x}_{\text{geom}}(x)}_{\text{range/elevation/height}},\ \underbrace{\mathbf{x}_{\text{morph}}(x)}_{\text{local buildings}},\ \underbrace{1}_{\text{bias}},\ \underbrace{\mathbf{x}_{\text{int}}(x)}_{\text{interactions}}\Big]^{\top}. \tag{3.72}$$

The first block lets the model bend the raw prior quadratically. The geometry block handles range and height. The morphology block measures whether there are nearby scatterers. The interaction block allows the model to express statements such as "density matters more at low elevation" or "density-weighted transmitter clearance matters differently at low elevation". As in the path-loss-prior vector, some of these terms are not independent. For instance, $z^{(0)}(x)^2$ is determined by $z^{(0)}(x)$, and $t_{41}(x)n_{41}(x)$ is tied to both the taller-building and NLoS-support features. This cross-correlation means individual fitted coefficients should not be interpreted as isolated physical constants. The terms were kept because the calibrated prior improved with them, and a rigorous feature-pruning study would require many repeated calibration runs. A simple pair-wise screening would already scale as $O(n^2)$ for an n-term vector, before considering higher-order subsets.

Term	**Mathematical role**	**Physical interpretation**
$z^{(0)}(x)^2$	Quadratic raw-prior basis.	Allows the calibration to correct curvature in the first spread estimate.
$z^{(0)}(x)$	Linear raw-prior basis.	Keeps the calibrated model tied to the log-domain physical prior.
$\log(1+d_{2D})$	Smooth distance basis.	Longer links can produce wider delay and angle supports because more scatterers can contribute.
θ_{norm}	Normalized elevation angle.	Separates overhead links from grazing links.
θ_{inv}	Inverse elevation angle.	Gives a large value to low-elevation links, where A2G models report stronger obstruction sensitivity [19, 1].
$h_{\text{norm}}, h^2_{\text{norm}}$	UAV-height basis and quadratic height basis.	Allows the model to learn a nonlinear effect of transmitter height on spread.
ρ_{15}, ρ_{41}	Fine/coarse local building density.	Measures how many nearby reflectors and blockers exist at two scales.
h_{15}, h_{41}	Fine/coarse density-weighted building height.	Distinguishes low-rise clutter from high-rise clutter.
n_{15}, n_{41}	Fine/coarse local NLoS support.	Measures how much nearby ground is geometrically shielded.
b_{41}	Blocker-depth term.	Measures how far nearby buildings protrude above receiver height.
1	Bias term.	Gives each regime its own additive offset.
c_{41}	Transmitter-clearance term.	Density-weighted window-area mean of UAV clearance above nearby building pixels.
t_{41}	Taller-building area fraction.	Fraction of the full local window occupied by buildings whose height exceeds the UAV height.

Table 3.12: *Term-by-term inventory of the direct terms in the spread calibration vector.*

Term	Mathematical role	Physical interpretation
$n_{41}\log(1+d_{2D})$	NLoS-distance interaction.	Lets range matter differently in blocked streets than in open ground.
$\rho_{41}\theta_{\mathrm{inv}}$	Density-elevation interaction.	Activates when dense morphology and low elevation occur together.
$\theta_{\mathrm{norm}}\rho_{41}$	Overhead-density interaction.	Allows dense areas to behave differently when viewed from high elevation.
$c_{41}\theta_{\mathrm{inv}}$	Clearance-elevation interaction.	Lets density-weighted rooftop clearance matter differently at grazing elevation.
$t_{41}\rho_{41}$	Tall-building-density interaction.	Highlights dense high-rise blocks where a large window area contains buildings above the UAV.
$t_{41}n_{41}$	Tall-building-NLoS interaction.	Activates in blocked regions where taller-building area is also present.

Table 3.13: *Interaction terms in the spread calibration vector.*

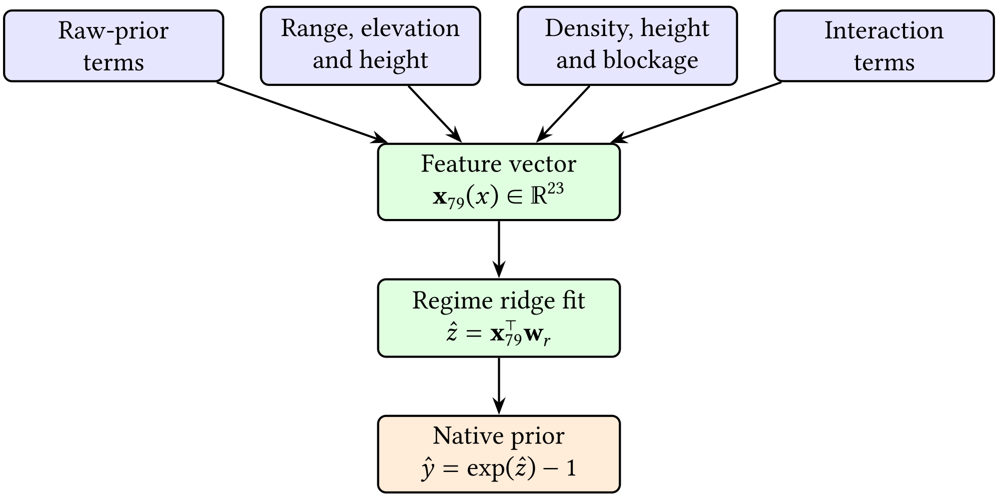

Figure 3.7: *Feature grouping used by the calibrated spread prior. The calibrated model is linear in $\boldsymbol{x}_{79}$, but the features include nonlinear transforms and interactions selected from propagation intuition.*

The new features beyond those in $\mathbf{x}_{78}$ are listed below. The shared morphology notation follows the path-loss-prior construction in Sec. 3.11.3. In particular, $b(\cdot)$ is the building mask, ρ_k is the building-density box mean, h_k is the density-weighted building-height box mean, n_k is the ground-NLoS support box mean, and θ_{norm} is the normalized elevation angle. The same reflected-padding $\mathrm{BoxMean}_k$ operator is used here, and $H_{\mathrm{norm}} = 90$ m is the building-height normalizer. This keeps the spread-prior vector self-contained while preserving the common feature definitions used by the path-loss prior.

Normalised transmitter height.

$$h_{\text{norm}} = \min\left\{1, \frac{\log(1 + \max(h_{\text{tx}} - h_{\text{rx}}, 0))}{\log(401)}\right\} \tag{3.73}$$

This compresses UAV height on a log scale and saturates it in $[0, 1]$, using $\log(401)$ as the scale associated with a 400 m height difference. Samples above that relative height are assigned $h_{\text{norm}} = 1$, matching the implementation. This cap is intentional: the transmitter-height distribution has only a small high-altitude tail, so after splitting samples into topology-specific experts there is not enough support above 400 m to estimate a separate high-altitude response reliably. The vector also includes h^2_{norm}, which is an intentionally correlated basis expansion that lets the linear model learn a curved height effect.

Blocker depth.

$$b_{41}(x) = \frac{\text{BoxMean}_{41}(\max(\text{topology}(\cdot) - h_{\text{rx}}, 0)\, b(\cdot))\,(x)}{H_{\text{norm}}} \tag{3.74}$$

Measures how much buildings in the 41×41 window protrude above the receiver height. Only building pixels contribute because of the multiplier $b(\cdot)$. A tall building directly next to the receiver therefore increases b_{41}, while flat open ground leaves it at zero.

Transmitter clearance.

$$c_{41}(x) = \frac{\text{BoxMean}_{41}(\max(h_{\text{tx}} - \text{topology}(\cdot), 0)\, b(\cdot))\,(x)}{H_{\text{norm}}} \tag{3.75}$$

This is a density-weighted clearance feature: the clearance above each building pixel is averaged by BoxMean_{41} over the full 41×41 window and then normalized by H_{norm}. Ground pixels contribute zero through the multiplier $b(\cdot)$. A large c_{41} therefore means both that nearby buildings are present and that the transmitter is well above them. A small value can mean either little local building coverage, or buildings that reach close to or above the UAV height.

Taller-building fraction.

$$t_{41}(x) = \text{BoxMean}_{41}(\mathbf{1}[\text{topology}(\cdot) > h_{\text{tx}}] \cdot b(\cdot))\,(x) \tag{3.76}$$

This is the fraction of the full 41×41 window area occupied by building pixels taller than the UAV, not the fraction of buildings after conditioning on there being a building pixel. It is zero if the UAV is above all local buildings or if the local window contains no buildings, and grows when the link is likely to be embedded in a high-rise canyon. It is especially useful for distinguishing dense but low-rise zones from dense high-rise zones.

Six-class topology definition. The spread prior uses the six topology labels produced by the final Try 80 prior code. For a map, let $\bar{\rho}_{\text{map}}$ be the fraction of non-ground pixels and let $\bar{h}_{\text{map}}$ be the mean height over non-ground pixels only. The class assignment is:

$$\text{class}_{79} = \begin{cases} \texttt{open\_sparse\_lowrise}, & \bar{\rho}_{\text{map}} \leq 0.12,\ \bar{h}_{\text{map}} \leq 12\,\text{m}, \\ \texttt{open\_sparse\_vertical}, & \bar{\rho}_{\text{map}} \leq 0.12,\ \bar{h}_{\text{map}} > 12\,\text{m}, \\ \texttt{dense\_block\_midrise}, & \bar{\rho}_{\text{map}} \geq 0.28,\ \bar{h}_{\text{map}} \leq 28\,\text{m}, \\ \texttt{dense\_block\_highrise}, & \bar{\rho}_{\text{map}} \geq 0.28,\ \bar{h}_{\text{map}} > 28\,\text{m}, \\ \texttt{mixed\_compact\_lowrise}, & 0.12 < \bar{\rho}_{\text{map}} < 0.28,\ \bar{h}_{\text{map}} \leq 12\,\text{m}, \\ \texttt{mixed\_compact\_midrise}, & 0.12 < \bar{\rho}_{\text{map}} < 0.28,\ \bar{h}_{\text{map}} > 12\,\text{m}. \end{cases} \tag{3.77}$$

These labels are map-level routing keys for the spread-prior ridge models; they are not additional per-pixel morphology features.

Stage 3: regime-wise ridge regression

The regime r for the spread prior is keyed by: metric (DS or AS), topology class (6 classes), propagation state (LoS or NLoS), antenna-height bin, giving up to $2 \times 6 \times 2 \times 3 = 72$ regimes. The antenna-height bins use the same tertile-style thresholds as Eq. (3.60): `low_ant` for $h_{\text{tx}} \leq 58.12\,\text{m}$, `mid_ant` for $58.12\,\text{m} < h_{\text{tx}} \leq 103.85\,\text{m}$, and `high_ant` above $103.85\,\text{m}$. These calibration bins are different from the broader height-reporting bins used later in the results tables.

Valid pixels are subsampled at rate $p = 0.02$ (2%). The ridge regression:

$$\hat{\mathbf{w}}_r^{79} = \left(\mathbf{X}_r^\top \mathbf{X}_r + \lambda \mathbf{D}^\top \mathbf{D}\right)^{-1} \mathbf{X}_r^\top \mathbf{z}_r, \quad \lambda = 10^{-3} \tag{3.78}$$

Here $\mathbf{X}_r \in \mathbb{R}^{N_r \times 23}$ stacks one row $\mathbf{x}_{79,n}^\top$ per sampled pixel in regime r, and $\mathbf{z}_r \in \mathbb{R}^{N_r}$ stacks the corresponding transformed targets. The coefficient vector $\hat{\mathbf{w}}_r^{79}$ minimizes the regularized least-squares problem

$$\hat{\mathbf{w}}_r^{79} = \arg\min_{\mathbf{w} \in \mathbb{R}^{23}} \|\mathbf{X}_r \mathbf{w} - \mathbf{z}_r\|_2^2 + \lambda \|\mathbf{D}\mathbf{w}\|_2^2, \tag{3.79}$$

where $\mathbf{D} = \text{diag}(d_1, \ldots, d_{23})$, with $d_{23} = 0$ and $d_i = 1$ for $i \neq 23$. Thus, the final interaction term $t_{41} n_{41}$, which is the last entry of Eq. (3.71), is left unpenalized by the ridge term in the frozen calibration used by the final model. This is an implementation detail of the stored artifact, not a separate physical assumption. The regularizer is small, but it prevents unstable coefficients when a regime contains few valid pixels or when two morphology features are strongly correlated.

Term by term, $\mathbf{X}_r^\top \mathbf{X}_r$ is the feature correlation matrix for regime r, $\lambda \mathbf{D}^\top \mathbf{D}$ adds a small diagonal stabilizer to every coefficient except the final $t_{41} n_{41}$ coefficient, and $\mathbf{X}_r^\top \mathbf{z}_r$ is the feature-target correlation vector in log-spread space. This is ridge regression: it is the ordinary least-squares fit plus a penalty that discourages very large coefficients. The penalty is important here because many morphology features are related; for example, ρ_{41}, h_{41}, b_{41} and t_{41} all increase in dense high-rise areas.

Sufficient statistics are accumulated incrementally:

$$\mathbf{A}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{X}_{r,s} \quad \forall s \tag{3.80}$$
$$\mathbf{v}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{z}_{r,s} \quad \forall s \tag{3.81}$$

so the full design matrix is never materialised in memory. This is important in practice because the calibration artifacts are already large even after compression into fitted coefficients and profiles. The path-loss LoS calibration JSON reused by the final model is approximately 200 000 lines, and the spread calibration must be rerun separately for delay spread and angular spread whenever the feature set changes.

Fallback hierarchy. During fitting, each sampled pixel updates the exact accumulator and the four broader fallback accumulators listed below. A coefficient vector is written only for accumulators with at least $4 \times 23 = 92$ sampled pixels. During inference, the implementation tries the same keys in this order and uses the first one present:

1. exact regime: (metric, topology class, LoS/NLoS, antenna bin),
2. same topology class + LoS/NLoS + *all* antenna heights,
3. same topology class + *all* LoS states + all antenna heights,
4. global topology + LoS/NLoS + all antenna heights,
5. fully global fallback: (`metric`, `global`, `all_los`, `all_ant`), where `all_los` is the implementation key for all visibility states, not LoS-only pixels.

Stage 4: inverse transform and clipping

$$\hat{y}(x) = \exp(\hat{z}(x)) - 1, \qquad \hat{z}(x) = \mathbf{x}_{79}(x)^\top \hat{\mathbf{w}}_r^{79} \tag{3.82}$$

Regime-dependent clipping is applied after the inverse transform:

$$\hat{y}_{\text{clip}}(x) = \min\big(y_{\max,r}, \max(0, \exp(\hat{z}(x)) - 1)\big), \tag{3.83}$$

with delay spread clipped to $[0, 400]$ ns and angular spread clipped to $[0, 15^\circ]$ in LoS or $[0, 90^\circ]$ in NLoS. The asymmetric angular limit is deliberate: LoS angular spread is dominated by the direct path and is physically closer to a near-zero spike plus small reflections, whereas NLoS angular spread can occupy a much wider support.

3.12.5 Spread-prior results

Table 3.14: *Delay spread: calibrated test RMSE (ns) by topology class.*

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	16.88	17.01	15.43
open_sparse_vertical	40.84	40.44	44.08
mixed_compact_lowrise	17.27	16.65	18.91
mixed_compact_midrise	37.88	38.26	37.25
dense_block_midrise	26.90	29.00	25.02
dense_block_highrise	43.52	40.16	45.08
Overall	**28.10**	**27.49**	**29.62**

Table 3.15: *Angular spread: calibrated test RMSE (°) by topology class.*

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	10.01	10.30	6.33
open_sparse_vertical	12.70	12.98	9.97
mixed_compact_lowrise	14.37	15.99	8.17
mixed_compact_midrise	15.26	17.85	9.57
dense_block_midrise	15.53	21.20	8.17
dense_block_highrise	14.64	21.33	9.78
Overall	**13.74**	**15.28**	**8.66**

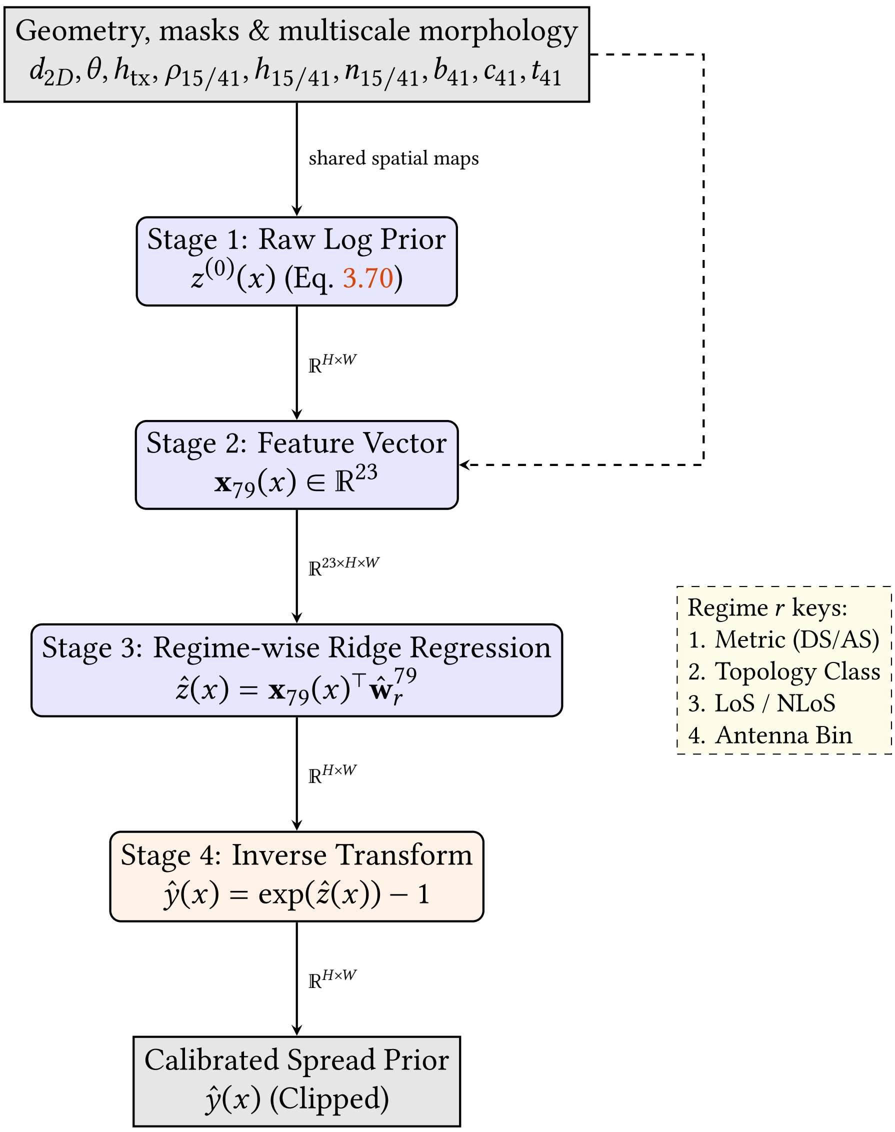

Figure 3.8: *Pipeline for the delay and angular spread priors. Both spread modalities share this identical linear architecture, predicting expected spread in the log domain via regime-wise ridge regression over 23 spatial and geometric features.*

3.13 Joint Prior-Anchored Residual GMM-Head Model

3.13.1 Experimental protocol and reporting status

The final residual GMM-head model follows the same strict split philosophy as the recent distribution-first models. The split is a city-holdout split with `split_seed = 42`, `val_ratio = 0.15`, and `test_ratio = 0.15`. Metrics and losses are computed only on valid ground pixels, using the target-specific no-data masks and the explicit LoS/NLoS masks. The model input contains nine spatial channels:

1. normalized topology,
2. LoS mask,
3. NLoS mask,
4. ground mask,
5. combined path-loss prior,
6. LoS path-loss prior,
7. calibrated Try 78 NLoS-branch map,
8. delay-spread prior,
9. angular-spread prior.

The seventh channel is named `path_loss_nlos_prior` in the code, but it should be read carefully. It is the calibrated Try 78 mixed/NLoS map computed by the regime-wise path-loss calibration, not a map that has been zeroed outside NLoS pixels. In the final combined path-loss prior and in the final output selection, only the explicit LoS/NLoS masks decide which regional branch is used.

The three mask channels are passed as binary maps. The topology channel is masked to ground pixels and divided by 90 m; the three path-loss-prior channels are divided by 180 dB; and the delay- and angular-spread priors are passed as $\log(1+\text{prior})/\log(1+400)$ and $\log(1+\text{prior})/\log(1+90)$, respectively. These scaling constants only affect the neural-network input representation; the residual targets and reported metrics remain in the native units.

The scalar UAV height is not appended as a constant image channel. It is encoded with a sinusoidal embedding and injected into the network through FiLM conditioning. The selected final configuration uses a shared encoder-decoder with `base_width = 128`, `cond_dim = 160`, three GMM components per task and region, α-gate bias -2, and fixed residual caps specified separately for path loss, delay spread and angular spread.

The selected final model reported in Chapter 4 uses these frozen path-loss and spread priors as anchors. This section therefore documents both the frozen-prior inputs and the residual architecture used by the final evaluation.

3.13.2 Prior-Anchored GMM Residual Model

The final model introduces a joint multi-task residual GMM-head architecture that emits a *Gaussian Mixture Model (GMM)*-parameterized residual head at each pixel, rather than only a direct scalar output. For each task $k \in \{\text{PL, DS, AS}\}$, region $r \in \{\text{LoS}, \text{NLoS}\}$, component $j \in \{1, 2, 3\}$, and valid pixel x, the model predicts a mixture weight, a mean and a standard deviation. Path loss is modeled in native dB space, while delay spread and angular spread are modeled in transformed $\log(1 + y)$ space inside the probabilistic head and converted back to native units for deterministic metrics. In the selected final fine-tuning stage the same GMM parameterization is kept, but the probabilistic losses are disabled; therefore the final reported claim is based on the deterministic mixture-mean map rather than on calibrated predictive densities.

The native residual caps are task- and region-dependent:

$$\delta_{\max,k,r} = \begin{cases} 7\text{ dB}, & k = \text{PL},\ r = \text{LoS}, \\ 25\text{ dB}, & k = \text{PL},\ r = \text{NLoS}, \\ 20\text{ ns}, & k = \text{DS},\ r = \text{LoS}, \\ 20\text{ ns}, & k = \text{DS},\ r = \text{NLoS}, \\ 17^\circ, & k = \text{AS},\ r = \text{LoS}, \\ 15^\circ, & k = \text{AS},\ r = \text{NLoS}. \end{cases} \tag{3.84}$$

The bounded native residual is assembled from a global map-level offset and a local pixel-level correction:

$$\varDelta_{k,r,j}(x) = \text{clip}\Big(\delta^{(g)}_{k,r,j} + \delta^{(\ell)}_{k,r}(x),\ -\delta_{\max,k,r},\ \delta_{\max,k,r}\Big). \tag{3.85}$$

The mean of each mixture component is then anchored to the frozen prior:

$$\mu^{\text{native}}_{k,r,j}(x) = y^{\text{prior}}_k(x) + \alpha_{k,r}(x)\varDelta_{k,r,j}(x), \tag{3.86}$$

where:

- $y^{\text{prior}}_k(x)$ is the per-task frozen prior selected inside the shared output loop, then broadcast to the LoS and NLoS residual branches,
- $\delta^{(g)}_{k,r,j}$ is a map-level residual offset for component j,
- $\delta^{(\ell)}_{k,r}(x)$ is a per-pixel residual refinement shared across components,
- $\alpha_{k,r}(x) \in (0, 1)$ is a learned spatial gate, initialized with bias -2, so the model starts close to the prior,
- the residual is bounded to $\pm\delta_{\max,k,r}$ to prevent the DNN from deviating catastrophically from the prior.

For the spread tasks, the native anchored mean is mapped into transformed space before the Gaussian likelihood:

$$\mu_{k,r,j}(x) = \begin{cases} \mu^{\text{native}}_{k,r,j}(x), & k = \text{PL}, \\ \log\Big(1 + \max\{\mu^{\text{native}}_{k,r,j}(x), 0\}\Big), & k \in \{\text{DS}, \text{AS}\}. \end{cases} \tag{3.87}$$

The DNN also predicts per-pixel mixture probabilities $p_{k,r,j}(x)$ and combines global and local standard-deviation terms into the component standard deviation:

$$\sigma_{k,r,j}(x) = \sqrt{\Big(\sigma^{(g)}_{k,r,j}\Big)^2 + \Big(\tilde{\sigma}^{(\ell)}_{k,r}(x)\Big)^2}. \tag{3.88}$$

The Gaussian variance used by the density is therefore $\sigma^2_{k,r,j}(x)$. Both standard-deviation terms are constrained by sigmoid transforms to lie between $\sigma_{\min} = 0.05$ and $\sigma_{\max} = 3.0$. These bounds stabilize the probabilistic head and prevent the NLL term, when it is enabled, from collapsing to unrealistically tiny uncertainty or exploding to a uselessly flat density.

The local component logits are transformed with a per-pixel softmax, so $\sum_j p_{k,r,j}(x) = 1$. The global mixture weights $\pi^{(g)}_{k,r,j}$ emitted by the global branch belong to the map-level distributional branch used when map-level KL-style terms are active; they are not multiplied into the deterministic point map in Eq. (3.89). This distinction is important for the selected final checkpoint, where the distributional losses are disabled and RMSE is computed from the local-weighted point prediction.

The architecture supports a full GMM Negative Log Likelihood (NLL), and the preceding large-capacity residual stage used non-zero probabilistic weights. The selected final model keeps the same GMM head, but the last residual fine-tuning stage is RMSE/MAE-dominant: its NLL, KL, moment and outlier-budget weights are set to zero in the resolved YAML configuration. The deterministic point prediction used for RMSE evaluation is the mixture mean in the modelled space:

$$\hat{z}_{k,r}(x) = \sum_{j=1}^{3} p_{k,r,j}(x)\mu_{k,r,j}(x), \qquad \hat{y}_{k,r}(x) = \begin{cases} \hat{z}_{k,r}(x), & k = \text{PL}, \\ \exp(\hat{z}_{k,r}(x)) - 1, & k \in \{\text{DS}, \text{AS}\}. \end{cases} \tag{3.89}$$

For path loss, this is also the native-space mixture mean because the modelled space is dB. For delay spread and angular spread, the mixture is averaged in $\log(1+y)$ space and then mapped back with $\exp(\cdot) - 1$. Therefore the reported point map should be read as the deterministic prediction used for RMSE, not as the exact native-space expectation of the probabilistic mixture. The final task map selects $\hat{y}_{k,\text{LoS}}$ or $\hat{y}_{k,\text{NLoS}}$ using the explicit CKM LoS/NLoS masks. This prevents a visually plausible NLoS correction from leaking into pixels whose geometric path is LoS, and conversely prevents LoS ring structure from being used as an explanation for blocked pixels.

The training loss also includes a prior-preservation term:

$$\mathcal{L}_{\text{guard},k} = \mathbb{E}_{x\in\Omega_k}\left[\max\left(0,\, (\hat{y}_k(x) - y_k(x))^2 - \left(y_k^{\text{prior}}(x) - y_k(x)\right)^2\right)\right] \tag{3.90}$$

This *pixel-wise* prior guard penalizes the model explicitly on any pixel where its squared error exceeds the prior's squared error. The guard is not used alone; it is part of the composite optimization objective in Sec. 3.13.4.

3.13.3 Architecture: Shared UNet with FiLM

The final model processes the map inputs through a shared multi-task convolutional encoder-decoder. In the final large-capacity configuration, scalar UAV height is encoded into a condition vector $\mathbf{c} \in \mathbb{R}^{160}$ and injected dynamically into the convolutional feature maps at multiple scales via *Feature-wise Linear Modulation (FiLM)* layers.

For a given convolutional feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, a linear projection of the height condition $\mathbf{c}$ outputs a scale vector $\mathbf{g} \in \mathbb{R}^{C}$ and a shift vector $\mathbf{b} \in \mathbb{R}^{C}$. The feature map is then modulated via an affine transformation:

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b} \tag{3.91}$$

where $\mathbf{g}$ and $\mathbf{b}$ are broadcasted spatially across the height and width. The FiLM mechanism and injection points are unchanged between the wide and large-capacity variants; only the width of the feature maps changes, so the FiLM projection emits $2C$ parameters for the larger channel counts.

The word "joint" in this model refers to shared representation learning and joint optimization, not to an autoregressive or cross-output decoder. Path loss, delay spread and angular spread all pass through the same encoder, the same height-conditioning path, the same decoder feature map and the same prior-anchored GMM assembly rule. After these shared features are computed, however, the implementation iterates over the three tasks and applies task-specific global and local heads. Therefore PL, DS and AS can exchange information through the common spatial representation, but the final residual parameters, mixture weights and frozen prior anchor are selected independently for each task.

The architecture explicitly enforces the separation of global and local effects via a dual-head design for each task:

- **Global Head (Map-Level Context)**: The deepest bottleneck feature map is globally average-pooled to produce a single vector representing the macro-context. This pooled vector, fused with the height condition, is passed through an MLP to predict the non-spatial quantities used by the distributional branch: auxiliary global mixture weights $\pi^{(g)}_{k,r,j}$, global offsets $\delta^{(g)}_{k,r,j}$, and global standard deviations $\sigma^{(g)}_{k,r,j}$.

- **Local Head (Pixel-Level Refinement)**: The high-resolution output of the decoder is passed through 1×1 convolutions to predict the spatially varying fields: local component weights $p_{k,r,j}(x)$, $\delta^{(\ell)}_{k,r}(x)$, $\alpha_{k,r}(x)$, and $\tilde{\sigma}_{k,r}(x)$.

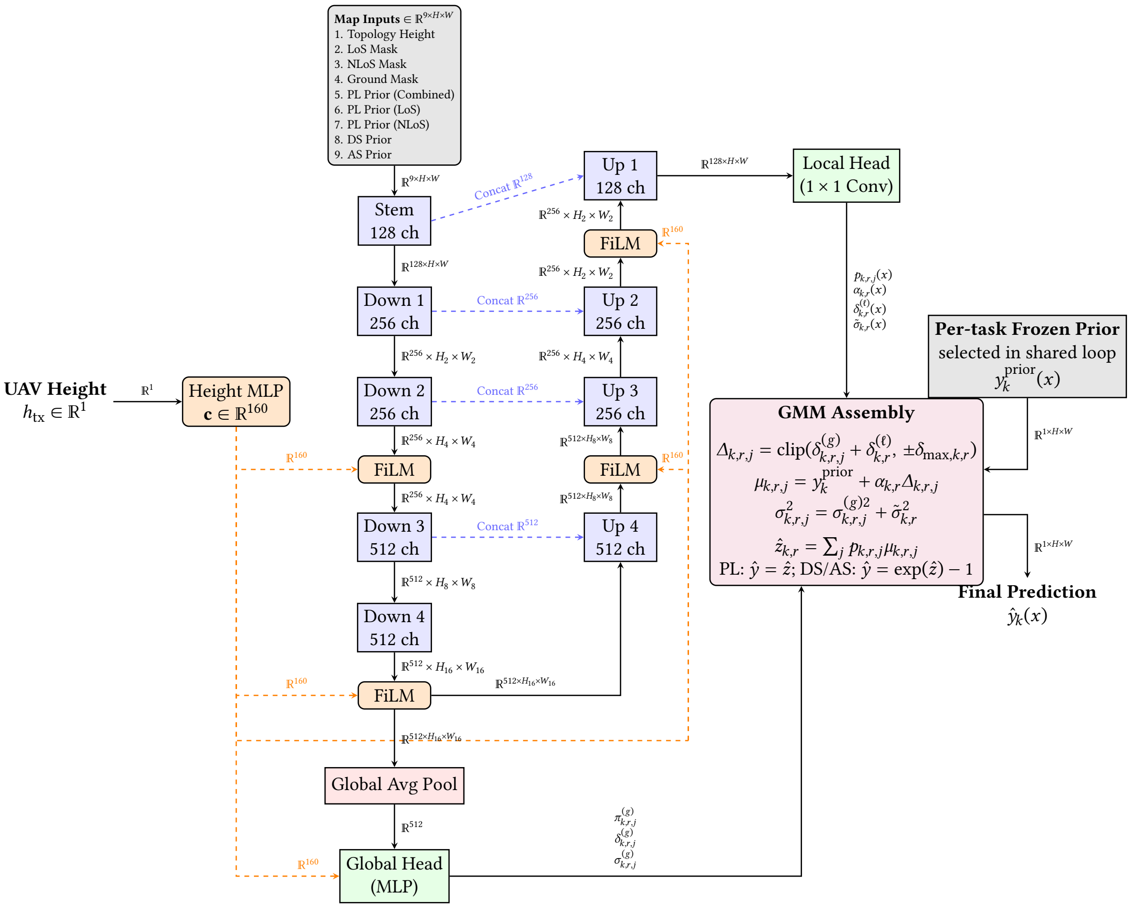

Figure 3.9: *Schematic of the final large-capacity shared UNet architecture with FiLM conditioning. The backbone and decode logic are shared, while each task uses its own heads and frozen prior.*

Detailed sinusoidal height embedding and FiLM path. The scalar UAV height h_{tx} is first converted into a 32-dimensional sinusoidal embedding:

$$f_i = \exp\!\left(\log 12 + \frac{i-1}{15}(\log 478 - \log 12)\right), \qquad i = 1, \ldots, 16 \tag{3.92}$$

$$\mathbf{e}(h_{\text{tx}}) = \left[\sin\!\left(\frac{h_{\text{tx}}}{f_1}\right), \ldots, \sin\!\left(\frac{h_{\text{tx}}}{f_{16}}\right), \cos\!\left(\frac{h_{\text{tx}}}{f_1}\right), \ldots, \cos\!\left(\frac{h_{\text{tx}}}{f_{16}}\right)\right] \in \mathbb{R}^{32} \tag{3.93}$$

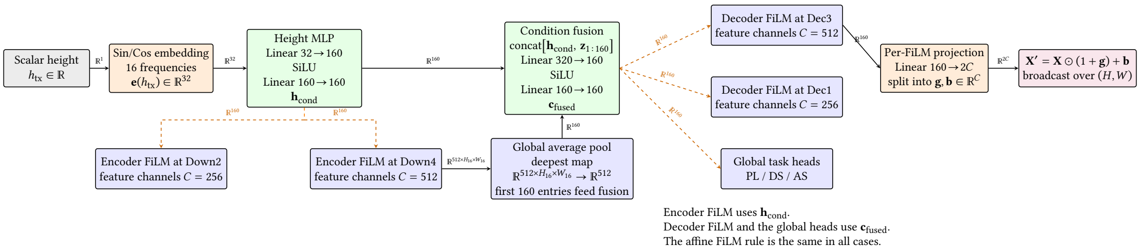

Figure 3.10: *Detailed sinusoidal height-conditioning path.*

Detailed prior-anchored GMM head. For each task k and for each region $r \in \{\text{LoS}, \text{NLoS}\}$, the model predicts a 3-component mixture. The global head outputs $\pi^{(g)}_{k,r,j}$,

$\delta^{(g)}_{k,r,j}$ and $\sigma^{(g)}_{k,r,j}$, while the local head outputs per-pixel maps $p^{(\ell)}_{k,r,j}(x)$, $\delta^{(\ell)}_{k,r}(x)$, $\alpha_{k,r}(x)$, and $\tilde{\sigma}_{k,r}(x)$. After softmax over components, $p^{(\ell)}_{k,r,j}(x)$ becomes the local probability $p_{k,r,j}(x)$ used in the regional model-space mean. The auxiliary global weights $\pi^{(g)}_{k,r,j}$ are retained for the distributional training branch and diagnostics, but the selected RMSE/MAE-dominant checkpoint uses the local probabilities for the deterministic output map.

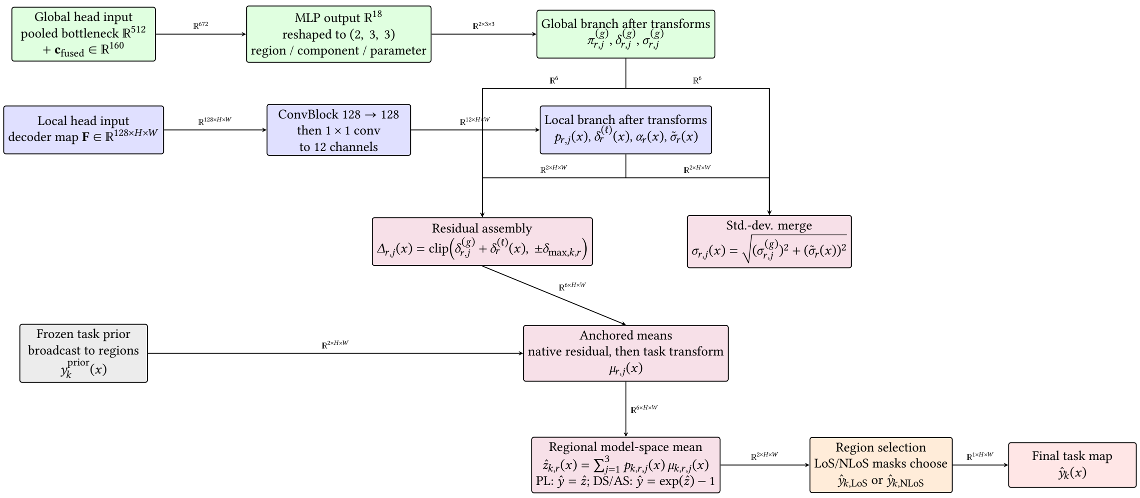

Figure 3.11: *Detailed prior-anchored GMM head used by the final model.*

3.13.4 Composite Optimization Objective

The implementation defines a composite loss that can train both the probabilistic GMM head and the deterministic residual map. For each task, let Ω_k be the valid ground-pixel set, let $q_k(z \mid x)$ be the predicted mixture density in the modelled space, with $z = y$ for path loss and $z = \log(1 + y)$ for delay spread and angular spread, and let $\hat{y}_k(x)$ be the native-space point prediction used for RMSE. The total objective is:

$$\begin{aligned}\mathcal{L} = \sum_{k\in\{\text{PL},\text{DS},\text{AS}\}} w_k \Big[&\lambda_{\text{NLL}}\mathcal{L}_{\text{NLL},k} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL},k} + \lambda_{\text{mom}}\mathcal{L}_{\text{mom},k} \\ &+ \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor},k} + \lambda_{\text{guard}}\mathcal{L}_{\text{guard},k} + \lambda_{\text{out}}\mathcal{L}_{\text{out},k} \\ &+ \lambda_{\text{RMSE}}\mathcal{L}_{\text{RMSE},k} + \lambda_{\text{MAE}}\mathcal{L}_{\text{MAE},k} \Big]. \end{aligned} \tag{3.94}$$

Table 3.3 gives the resolved weights for both the large-capacity residual stage and the final residual fine-tuning stage; the equations below define the terms used in that table.

Map negative log-likelihood. The likelihood is computed separately in LoS and NLoS regions, then averaged over the non-empty regions. For a transformed target $z_k(x)$,

$$q_{k,r}(z_k(x) \mid x) = \sum_{j=1}^{3} p_{k,r,j}(x)\mathcal{N}\left(z_k(x); \mu_{k,r,j}(x), \sigma^2_{k,r,j}(x)\right), \tag{3.95}$$

and the regional NLL is

$$\mathcal{L}_{\text{NLL},k,r} = -\frac{1}{|\Omega_{k,r}|} \sum_{x\in\Omega_{k,r}} \log q_{k,r}(z_k(x) \mid x). \tag{3.96}$$

The implemented version evaluates Eq. (3.96) using $\log \sum \exp$ for numerical stability.

Distributional KL term. The global branch predicts residual components $\delta^{(g)}_{k,r,j}$. To keep those components aligned with the empirical residual distribution of the current batch, residuals $\epsilon_k(x) = y_k(x) - y^{\text{prior}}_k(x)$ are softly binned into an empirical probability vector $\mathbf{u}_{k,r}$. The global mixture offsets define a predicted residual histogram $\mathbf{v}_{k,r}$. The loss is

$$\mathcal{L}_{\text{KL},k} = \frac{1}{2} \sum_{r\in\{\text{LoS},\text{NLoS}\}} \sum_{b=1}^{B} u_{k,r,b} \log \frac{u_{k,r,b}}{v_{k,r,b}}, \qquad B = 64. \tag{3.97}$$

This term does not force a perfect pixel-wise correction. It only encourages the map-level mixture to place probability mass where the batch residuals actually live.

Moment matching. The point prediction is also constrained to have approximately the same first two moments as the target map:

$$\mathcal{L}_{\text{mom},k} = \left(\bar{\hat{y}}_k - \bar{y}_k\right)^2 + \left(s_{\hat{y},k} - s_{y,k}\right)^2, \tag{3.98}$$

where bars denote masked means over Ω_k and s denotes masked standard deviation. When active, this term prevents a low-NLL solution from matching local density while drifting in map-average level. In the final fine-tune it is defined by the code but has zero weight.

Anchor and guard terms. The anchor penalty keeps the correction small:

$$\mathcal{L}_{\text{anchor},k} = \mathbb{E}\left[\alpha_k(x)^2 + 0.25\left(\delta^{(g)}_k\right)^2 + 0.50\left(\delta^{(\ell)}_k(x)\right)^2\right]. \tag{3.99}$$

The guard term is Eq. (3.90). Together, these two terms encode the final design philosophy: the DNN is allowed to improve the prior, but it must pay a penalty for large or harmful deviations. In Eq. (3.99), the residual variables are the same tensors used by the implementation: native-dB offsets for path loss, and the corresponding $\log(1 + y)$-space offsets for delay and angular spread after the native residual caps have been applied. The guard term, by contrast, is evaluated in the native units of each task against the frozen prior map.

Outlier budget and deterministic losses. When the outlier-budget term is active, the implementation uses the last local mixture component as a proxy for broad/outlier usage. This is an indexing convention rather than a hard architectural constraint: the model is not forced to make that component the broadest one. If the mean usage of this proxy exceeds $\tau = 0.60$, a hinge penalty is applied:

$$\mathcal{L}_{\text{out},k} = \max\left(0, \frac{1}{|\Omega_k|} \sum_{x\in\Omega_k} p_{k,\text{out}}(x) - \tau\right). \tag{3.100}$$

The RMSE and MAE terms are conventional masked deterministic losses:

$$\mathcal{L}_{\text{RMSE},k} = \sqrt{\frac{1}{|\Omega_k|} \sum_{x\in\Omega_k} \left(\hat{y}_k(x) - y_k(x)\right)^2}, \qquad \mathcal{L}_{\text{MAE},k} = \frac{1}{|\Omega_k|} \sum_{x\in\Omega_k} \left|\hat{y}_k(x) - y_k(x)\right|. \tag{3.101}$$

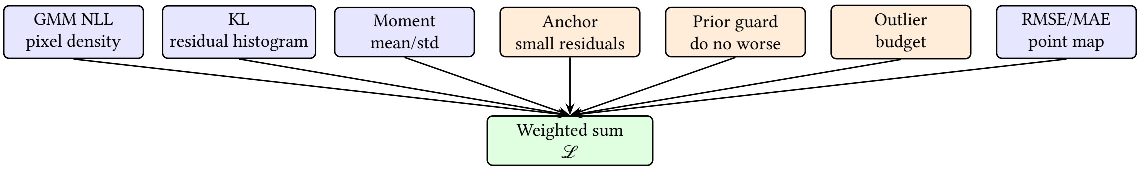

Figure 3.12: *Composite final loss family. In the final residual fine-tuning stage, the selected final model keeps the GMM head and uses RMSE/MAE, anchor and guard penalties; NLL, KL, moment and outlier-budget weights are set to zero. Earlier residual stages used non-zero probabilistic weights.*

CHAPTER 4

Results

This chapter reports the main quantitative and qualitative outcomes of the project. The standalone path-loss prior audit uses the city-holdout validation+test pool: 5340 samples from 29 unseen cities. The selected final residual GMM-head model is then evaluated on the final 2590-sample, 14-city test split. Two path-loss-prior numbers therefore appear in this chapter: 1.9246 dB is the standalone calibrated hybrid prior on the held-out validation+test pool, whereas 1.9383 dB is the frozen path-loss prior baseline used by the final residual model on the final 14-city test split. All final model-versus-prior comparisons use the latter. The final model is reported as the final prior-anchored residual GMM-head model over the three outputs: path loss, delay spread, and angular spread. The prior is therefore not a weak reference included only for comparison. It is one of the central results of the thesis: a strong, train-calibrated baseline that absorbs the stable part of the map before the residual model is asked to solve the harder local errors.

4.1 Experimental Evaluation

Table 4.1 summarizes the most important milestones. The early rows are not meant as a leaderboard because masking rules and objectives changed during the project; they document why the next stage was necessary.

Table 4.1: *Representative experimental milestones.*

Experiment family	Main lesson	Representative result
Early U-Net family	U-Net/cGAN and decoder fixes; cGAN did not solve physical error.	The combined decoder/multiscale run became the clean pre-prior baseline.
Masked-evaluation reset	Ground-only masking became mandatory.	Building pixels excluded from loss and metrics.
Prior-residual phase	Prior plus residual helped LoS but NLoS remained dominant.	PMNet residual branch: 19.78 dB overall, 3.86 dB LoS, 34.47 dB NLoS.
PMHHNet reference	PMHHNet synthesis with six experts and physical channels.	9.41 dB overall, 3.75 dB LoS, 31.48 dB NLoS.
Distribution-first GMM	Distribution-first path-loss modelling.	NLoS pixel-weighted RMSE $\approx$ 3.34 dB.
Final path-loss prior	ML-calibrated path-loss prior.	Held-out validation+test: 1.9246 dB overall, 1.75 dB LoS, 3.40 dB NLoS; final-test prior baseline: 1.9383 dB.
Final spread priors	ML-calibrated spread priors.	28.10 ns delay; 13.74° angular.
Final residual GMM-head model	Joint prior-anchored residual GMM-head model.	Final test: 1.65 dB path loss, 26.56 ns delay spread, 11.39° angular spread; RMSE improves over the frozen priors on all three outputs.

The final path-loss prior is the clearest evidence that the prior is a strong path-loss baseline in its own right. The enhanced two-ray LoS model reduced LoS RMSE from roughly 3.95 dB with FSPL to 1.75 dB. The final hybrid prior reached 1.9246 dB overall under the pixel-weighted metric on the held-out validation+test pool.

4.1.1 LoS path-loss prior progression

The LoS branch of the final path-loss prior was developed as a short waterfall of three increasingly physical models fitted on the same city-holdout split, so the gain from each modelling choice can be isolated (Table 4.2).

Table 4.2: *LoS path-loss prior progression: three priors evaluated on the same held-out validation+test pool (pixel-weighted RMSE).*

Prior	LoS RMSE (dB)	Gain over previous
FSPL only	3.9540	N/A
FSPL + radial residual	3.7086	−0.25
Enhanced two-ray	**1.7516**	**−1.96**

Two lessons follow. First, adding a purely radial correction on top of FSPL only buys about 0.25 dB. The residuals that a non-coherent model can absorb are small compared to the structured deviation from FSPL. Second, switching to the *enhanced* two-ray model with a fitted effective reflection coefficient (ρ, ϕ) per 5 m UAV-height bin removes almost 2 dB of additional error. This is exactly the magnitude expected when the dataset contains a visible ground-reflection interference pattern, because the radial-residual model cannot reproduce the oscillatory ring structure with a purely monotonic function of distance. The two-ray model is therefore not just a better fit numerically; it is a physically motivated explanation of the dominant LoS residual shape in CKM, and it is why the final model uses the two-ray output directly as a frozen anchor rather than re-learning it from pixels.

4.2 RMSE evolution and interpretation

The RMSE history should be read as a sequence of increasingly strict experiments, not as a perfectly homogeneous leaderboard. The dataset changed, the building-mask rule became stricter after the early experiments, and later runs used city-holdout discipline more consistently. Nevertheless, the representative values in Table 4.3 show the real direction of the project: from direct image regression, to prior-guided residual learning, to frozen calibrated priors and finally to the joint final residual GMM-head model.

Split-comparability note. The distribution-first predecessors, spread priors, and final residual model use the same `city_holdout` split contract with `val_ratio = 0.15`, `test_ratio = 0.15`, and `split_seed = 42`. The resulting test set is the same 14-city, 2590-sample holdout for these runs. The diagnostic grouping is different: the distribution-first predecessors report the six topology classes directly, whereas the final residual model groups them into three two-class morphology families (`open`, `mixed`, `dense`) and then separates each family again by low/mid/high UAV antenna bin. That changes the reporting grid, not the test samples. The standalone path-loss-prior audit uses the same training cities and the full validation+test held-out pool, giving 1.9246 dB over 5340 held-out samples. The final residual-model comparison remains test-only, so the frozen path-loss prior baseline in Table 4.11 is 1.9383 dB.

Table 4.3: *Representative RMSE evolution across the project. Values are reported under their own experiment contract and are used to show trend and diagnosis, not to claim every row is directly comparable.*

Milestone	Target / region	RMSE	Interpretation
Bilinear decoder	Path loss	$\approx$ 17.40 dB	Bilinear decoder alone reduced artifacts but did not solve global propagation.
Multiscale supervision	Path loss	$\approx$ 17.06 dB	Multiscale supervision helped the large-scale field.
Decoder + multiscale	Path loss	$\approx$ 16.72 dB	Best clean pre-prior baseline in the early U-Net family.
Raw physical prior	Path loss	$\approx$ 67.23 dB	Raw physical prior was far too crude without calibration.
Calibrated physical prior	Path loss	$\approx$ 24.16 dB	Regime-aware train-only calibration made the prior useful but incomplete.
PMNet residual branch	Path loss	$\approx$ 19.78 dB	PMNet-style residual learning improved over prior-only calibration.
Prior-confidence stage	Path loss	$\approx$ 18.87 dB	Prior-confidence and robust loss helped, but tail errors persisted.
PMHHNet reference	Path loss overall	9.41 dB	PMHHNet consolidated the physical-channel and expert approach.
PMHHNet reference	Path loss NLoS	31.48 dB	The headline bottleneck was no longer global; it was NLoS tail failure.
Dist.-first GMM	Path loss NLoS	$\approx$ 3.34 dB	Distribution-first modelling directly attacked NLoS multi-modality.
Final path-loss prior	Path loss overall	1.9246 dB	ML-calibrated prior became the strongest path-loss baseline.
Final DS prior	Delay spread overall	28.10 ns	Log-domain ridge calibration met the original 50 ns project target.
Final AS prior	Angular spread overall	13.74°	Regime-wise calibration met the original 20 degree target.
Final GMM test	Joint outputs	1.6519 dB PL; 26.5570 ns DS; 11.3854° AS	Selected final model on 14 unseen test cities; model RMSE improves over every frozen prior.

Several conclusions follow from this table. First, the project did not improve only because the neural network became larger. Some larger or more complex variants stalled when their objective was not aligned with the physical error. The biggest gains came from changing the formulation: masking invalid receiver pixels, separating regimes, injecting priors, and eventually making the prior itself strong enough to be a useful baseline. This is the divide-and-conquer result of the thesis: the calibration stage handles the repeatable structure, and the neural stage is evaluated only on the residual that remains after that baseline has already done substantial work.

The calibrated path-loss prior is the clearest example of this design. It is not a single generic propagation formula. It is split into a LoS branch and an NLoS branch because those pixels have different error mechanisms. The LoS branch extracts direct and reflected distances, FSPL, a coherent two-ray correction, height-bin bias terms, and a smoothed radial residual profile. The NLoS branch extracts a calibrated feature vector containing the raw NLoS prior, distance, local building density, local building height, local NLoS support, a shadowing proxy, normalized elevation angle, and interactions between those terms:

$$\begin{aligned}\mathbf{x}_{78} = \big[P_0^2,\ P_0,\ \log(1+d_{2D}),\ \rho_{15},\ \rho_{41},\ h_{15},\ h_{41},\\ \rho_{41}\log(1+d_{2D}),\ n_{15},\ n_{41},\ n_{41}\log(1+d_{2D}),\\ \sigma_{\text{shadow}},\ \theta_{\text{norm}},\ n_{41}\theta_{\text{norm}},\ 1\big]^{\top}.\end{aligned}$$

The weights are fitted using training data only and separately by propagation regime. This is why the final prior is a calibrated feature extractor, not just an analytic baseline.

Second, RMSE and MAE told different stories during the middle of the project. For example, the prior-confidence stage had moderate average absolute error on many samples, but RMSE remained high because a smaller number of hard NLoS cases produced very large misses. This is why the later work moved toward tail-aware and distribution-aware methods rather than only lowering the mean error.

Third, LoS and NLoS must be reported separately. The PMHHNet reference already had LoS error near 3.75 dB, while NLoS error remained above 30 dB. A single overall number would hide the fact that the LoS branch was nearly solved for that stage while the NLoS branch was still structurally wrong.

Table 4.4: *Final hybrid path-loss prior, pixel-weighted RMSE on the held-out validation+test split.*

Region	RMSE	Unit
LoS	1.7516	dB
NLoS	3.3967	dB
Overall	1.9246	dB

The final spread priors showed that a large part of the spread targets can also be explained by geometry, morphology, and calibration without a neural network. The fitted calibration dictionary contains 114 regime keys, but this number is not one simple Cartesian product. For each spread metric, the exact grid has 6 topology classes × 2 LoS/NLoS states × 3 antenna-height bins, or 36 exact keys. The implementation then adds fallback calibrations for sparse regimes: 12 antenna-collapsed topology and LoS/NLoS keys, 6 topology-wide keys, 2 global LoS/NLoS keys, and 1 fully global key. This gives 57 keys per spread metric and 114 keys across delay spread and angular spread. Equivalently, the dictionary contains 72 exact DS/AS regimes plus 42 fallback regimes. The point of this dictionary is not novelty by formula alone, but coverage: the baseline is strong because it stores many small regime-specific decisions about density, building height, visibility, and antenna height instead of forcing one global estimator to explain all pixels. The calibrated spread-prior module applies this idea separately to delay spread and angular spread. The raw spread estimate is only the first feature; the calibrated model then uses a 23-term vector with raw-prior curvature, range, elevation angle, UAV height, local building density at two scales, local height at two scales, LoS/NLoS support at two scales, blocker depth, transmitter clearance, taller-building fraction, and interaction terms. Both spread targets are fitted in $\log(1 + y)$ space, so the model can handle the LoS spike near zero and the long NLoS tail with one calibrated framework while still reporting metrics in native ns and degrees.

Table 4.5: *Calibrated spread-prior test results, pixel-weighted RMSE.*

Metric	Overall	LoS	NLoS
Delay spread (ns)	28.10	27.49	29.62
Angular spread (°)	13.74	15.28	8.66

4.2.1 Distribution-first path-loss reference

The distribution-first path-loss predecessor is not the final model of this thesis, but its result is quoted here because it is the piece of evidence that motivated the final path-loss design. Trained with no physical prior of any kind, no COST-231 formula, no knife-edge channel, and no residual-over-prior structure, the distribution-first family reached a pixel-weighted NLoS path-loss RMSE of approximately 3.34 dB on its own city-holdout test split. The two lessons it provided were (i) a plain MSE/Huber regressor with a single Gaussian-shaped output head cannot represent the NLoS path-loss target because that marginal is strongly multi-modal, and (ii) once the output head is a Gaussian mixture conditioned on the UAV height, the NLoS mode-collapse that dominated the error budget of the PMHHNet diagnostic phase disappears. This is a small RMSE difference compared to the ML-calibrated path-loss prior (3.40 dB NLoS), but it is a qualitatively different result because it is obtained without injecting any physics.

Per-expert breakdown (12 experts: 6 topology classes × LoS / NLoS)

The distribution-first path-loss predecessor trains twelve specialised GMM experts. A sequential DirectML evaluation of the 6 LoS and 6 NLoS trained expert models on the shared test split gives an aggregated NLoS metric of 3.34 dB pixel-weighted and 2.64 dB sample-weighted; the matching LoS aggregate is 4.43 dB pixel-weighted and 4.34 dB sample-weighted. This hides a wide spread between topology classes and a systematic asymmetry between LoS and NLoS regimes. Table 4.6 reports the DirectML test RMSE per expert together with the standard deviation of the underlying target distribution, which makes the asymmetry interpretable.

The headline finding here is counter-intuitive at first sight: **LoS path loss is consistently harder for the distribution-first path-loss model than NLoS path loss**. In Table 4.6, the LoS RMSE is higher than the NLoS RMSE in every topology class, with a class-wise gap ranging from about 0.6 dB to 1.7 dB. The same pattern survives any choice of aggregation,

Table 4.6: *Distribution-first path-loss DirectML test RMSE per expert on the shared city-holdout test split, together with the standard deviation of the corresponding target histogram. RMSE values are pixel-weighted over valid ground pixels. Source: internal DirectML evaluation JSON.*

Topology class	LoS RMSE (dB)	NLoS RMSE (dB)	LoS target σ (dB)	NLoS target σ (dB)
open_sparse_lowrise	4.54	2.87	5.25	3.82
open_sparse_vertical	4.71	3.75	5.43	4.35
mixed_compact_lowrise	4.15	2.81	5.22	3.94
mixed_compact_midrise	4.56	3.67	5.66	4.49
dense_block_midrise	4.18	3.21	5.30	4.33
dense_block_highrise	4.79	4.18	5.91	4.67
Macro mean	**4.49**	**3.42**	**5.46**	**4.27**
Pixel aggregate	**4.43**	**3.34**	N/A	N/A

and it contradicts the intuition built up during the PMHHNet diagnostic phase, where NLoS was the dominant source of error and also what made the most physical sense. This reversal motivated the later return to explicit priors for both path loss and spread.

Why LoS is harder than NLoS in the GMM regime

The explanation is a property of the marginal target distributions, not of the model. The histogram study reports the following global sanity-check statistics over all CKM ground pixels:

- **LoS path-loss target**: $\mu \approx 96.95$ dB, $\sigma \approx 5.34$ dB, skewness ≈ 0.04, excess kurtosis ≈ 0.61. This is not a perfect Gaussian: the bulk is broad, with rare high-loss outliers and topology-dependent shoulders.
- **NLoS path-loss target**: $\mu \approx 107.06$ dB, $\sigma \approx 4.32$ dB, skewness ≈ -0.08, excess kurtosis ≈ 2.25. This is much more peaked but has clearly heavier tails (kurtosis nearly four times that of the LoS marginal), and mixture fits separate a tight central obstruction-loss component from broader tail components.

Two consequences follow. First, *the absolute RMSE floor is larger for LoS*: the distribution-first model only sees $\sigma \approx 5.3$ dB of useful LoS variability and predicting near the conditional mean already implies errors of several dB. Second, *the right object is the per-expert GMM, not a single Gaussian summary*: each topology/regime expert sees a different mixture of central mass, shoulders, tails and outliers. NLoS is where this pays back most visibly: the heavy-tailed mass near 107 dB plus obstruction-driven tail components is precisely the shape that mode collapse damaged in the PMHHNet diagnostic phase, and the GMM head captures it cleanly. LoS is also not Gaussian, but its broader spread makes the remaining RMSE budget harder to collapse into a few dominant components.

The global comparison is only a first-order diagnostic: the distribution-first path-loss model trains one LoS and one NLoS GMM expert per topology class, not one collapsed pair. This also explains the topology ranking inside each regime. The LoS target standard deviation grows from 5.22 dB in `mixed_compact_lowrise` to 5.91 dB in `dense_block_highrise`, and the LoS RMSE broadly follows the same trend, with small reorderings in the middle classes (best 4.15 dB, worst 4.79 dB). Denser, taller cities scatter the LoS path loss over a wider range, so even with perfect calibration the RMSE budget grows. The same monotonic growth is visible on the NLoS side, but with a smaller absolute gap because NLoS is the regime where the GMM head pays back the most.

A useful normalisation is the ratio RMSE$/\sigma_{\text{target}}$, which estimates how much of the target variability the model fails to explain. Figure 4.1 (right panel) plots this skill metric per expert: **LoS** sits in the 0.79 to 0.87 range, while **NLoS** spans 0.71 to 0.90. In most classes, the distribution-first predecessor explains a larger fraction of the NLoS variance even though its absolute NLoS RMSE is also lower. The model is doing harder statistical work on NLoS and reaping a larger absolute benefit at the same time.

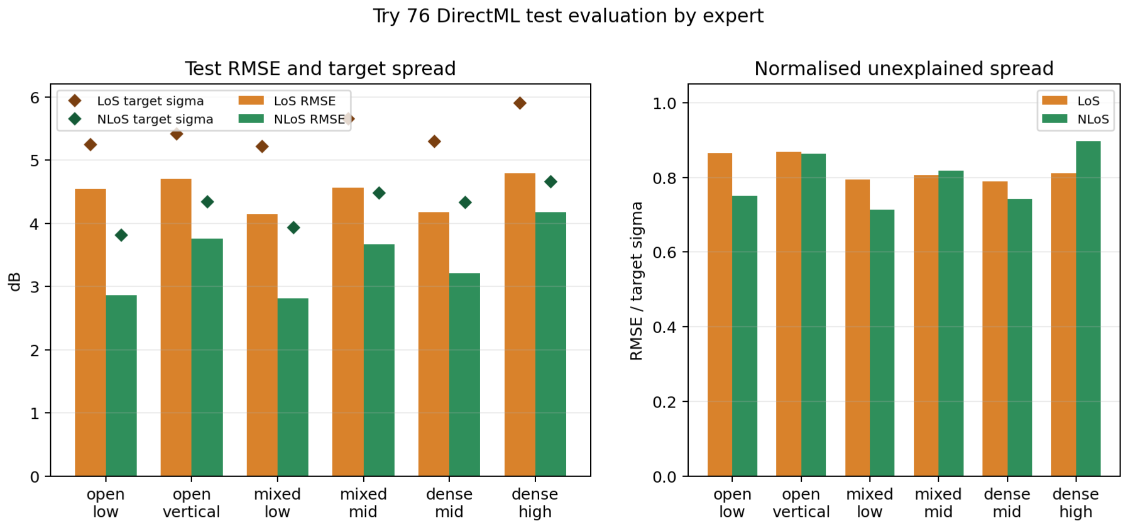

Figure 4.1: *Distribution-first path-loss DirectML test RMSE per expert.* Left*: bars are RMSE for LoS (orange) and NLoS (green); diamond markers report the standard deviation of the underlying target histogram.* Right*: the same data normalised as RMSE / target σ. LoS is harder in absolute dB* and *less well explained relative to its own variability, both consistent with the LoS marginal being a single broad bell, while the NLoS marginal is a peaked, heavy-tailed distribution that the GMM head can attack directly.*

Why priors came back after distribution-first modelling

Reading the project in chronological order, the move from distribution-first path-loss modelling to the final path-loss prior can look contradictory: the first demonstrates that a distribution-aware neural head does not need a physical prior to solve NLoS path loss, and the second goes back to an explicit analytic prior. The sequence is consistent once the order in which the evidence arrived is spelled out. The distribution-first model came *immediately after* the histogram diagnosis. That diagnosis showed that the main deficit of the PMHHNet family was not a missing physical channel, but an output-distribution mismatch. The first response was therefore to fix the deep-learning side: change the objective, change the output head, and show that NLoS path loss can in fact be modelled without extra physics. Once that point was made, the question changed. Given that the full target distribution is tractable, which part of the prediction pipeline is the cheapest and most auditable? For LoS path loss the answer turned out to be a 100 –line analytic model with a handful of fitted constants (the final path-loss prior), not a neural residual head. For delay and angular spread the answer was a log-domain ridge regression over geometry and morphology features. The final residual GMM-head model keeps the distribution-aware philosophy but anchors the neural residual head to the frozen priors rather than forcing the network to rediscover them from pixels. The priors in the final pipeline are an efficiency and stability device, not a contradiction of the distribution-first result.

4.2.2 Distribution-first spread experts

The distribution-first spread predecessor applies the path-loss mixture recipe to delay spread and angular spread. It partitions the dataset into the same six topology classes and trains two GMM experts per class (one per spread target), for a total of twelve experts. Unlike the path-loss predecessor, the LoS / NLoS split is not enforced inside this spread model because the spread targets are defined over all ground pixels (the Stage-A GMM components recover the spike-plus-tail shape themselves, as shown by the distribution-class histogram study).

Expert nomenclature. In this chapter, "six experts" denotes the legacy distribution-first topology taxonomy: *open sparse lowrise*, *open sparse vertical*, *mixed compact lowrise*, *mixed compact midrise*, *dense block midrise*, and *dense block highrise*. The analytic priors then refine this routing internally: the path-loss prior uses coarse topology, LoS/NLoS state and antenna-height bin, while the spread prior also keys the regime by spread metric. The final residual model reports a simpler final subexpert grid: three morphology groups (`open`, `mixed`, `dense`) crossed with three antenna-height bins (`low_ant`, `mid_ant`, `high_ant`), giving the nine subexperts used in the final breakdowns. These audit bins follow the calibration thresholds of Eq. (3.60): low is at or below 58.12 m, mid is above 58.12 m and at or below 103.85 m, and high is above 103.85 m. They are not the four broader reporting bins used in the height-stratified prior tables.

A sequential DirectML pass over the 12 mirrored expert models now gives test metrics on the same 2590-sample holdout used by the distribution-first path-loss predecessor and the final model. Table 4.7 reports both sample-mean and pixel-weighted RMSE. The distinction matters for spread targets because large maps with broad valid ground support can dominate the pixel-weighted aggregate and there are more outliers.

Table 4.7: *Distribution-first spread DirectML test RMSE per expert on the shared city-holdout test split. Sample columns average per-map RMSE; pixel columns aggregate squared error over all valid ground pixels. Source: internal DirectML evaluation JSON.*

Topology class	Delay sample (ns)	Delay pixel (ns)	Angular sample (°)	Angular pixel (°)
open_sparse_lowrise	11.62	18.69	8.71	9.58
open_sparse_vertical	35.92	45.65	11.49	12.56
mixed_compact_lowrise	14.69	20.71	10.42	10.96
mixed_compact_midrise	34.04	41.00	12.96	13.39
dense_block_midrise	23.56	28.96	12.29	12.66
dense_block_highrise	35.27	45.31	13.45	14.35
Expert macro mean	**25.85**	**33.39**	**11.55**	**12.25**
Sample aggregate	**22.68**	N/A	**11.21**	N/A
Pixel aggregate	N/A	**30.96**	N/A	**11.78**

The angular-spread side of the distribution-first spread predecessor is the cleanest neural result in this family: every topology class remains below the original 20° target under both sample-mean and pixel-weighted aggregation, and the global pixel-weighted RMSE is 11.78°. Delay spread is also below the original 50 ns target, with a 30.96 ns pixel-weighted aggregate, but it is less competitive with the later ML-calibrated spread prior (28.10 ns). The gap between sample-mean and pixel-weighted delay RMSE also shows that the difficult cases are not only rare maps; they occupy large valid-pixel supports in the open vertical, mixed midrise and dense highrise classes. For this reason the spread predecessor remains a useful distribution-first diagnostic and architectural precursor, while the calibrated spread prior is the deployed spread baseline used to anchor the final model.

The spread RMSE should also be read with one caveat: the CKM delay- and angular-spread targets are heavily quantised. A smooth prediction can therefore be penalised at staircase-like stored target boundaries even when the physical morphology is plausible. The reported spread RMSE remains the correct metric for the released arrays, but it can slightly overstate the physical mistake; Section 4.4 gives the quantisation floor explicitly.

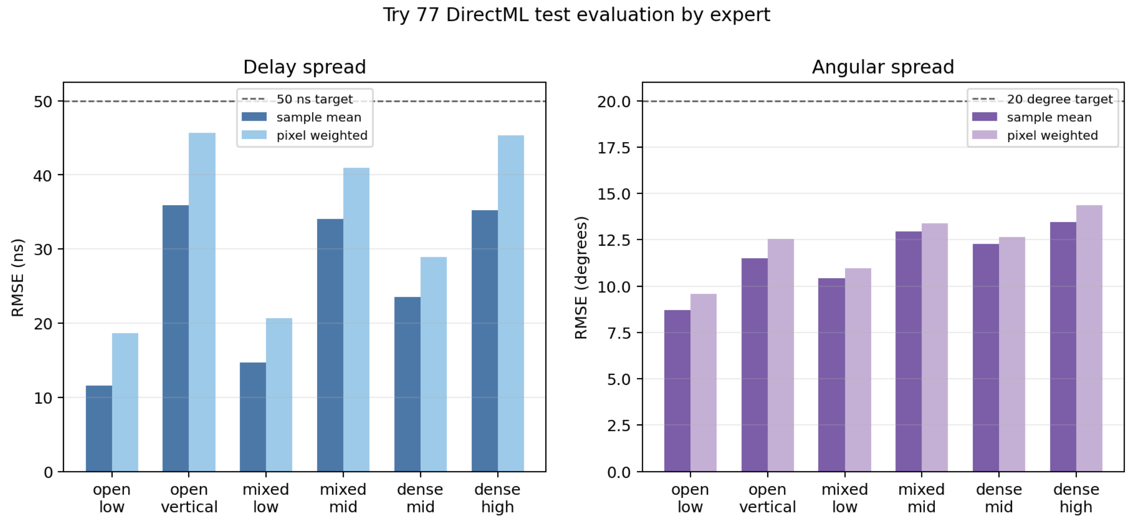

Figure 4.2: *Distribution-first spread DirectML test RMSE per expert, separated by spread target. For each topology class, paired bars show sample-mean and pixel-weighted aggregation. Dashed grey lines mark the original project targets (50 ns for delay spread, 20° for angular spread). The angular-spread experts beat the target across all topology classes; the delay-spread experts also remain below 50 ns, but their pixel-weighted errors are larger and less competitive with the final ridge calibration.*

4.2.3 RMSE by UAV transmitter height

The CKM dataset covers stored UAV heights from about 12 m to almost 478 m, while the closest nominal beta law has support up to 500 m. The absence of samples above 478 m is statistically normal for this finite draw, because the expected count above that height is only 0.52. Therefore the conditional error at a single aggregate RMSE can hide strong altitude-dependent structure. Tables 4.8 and 4.9 therefore break the path-loss prior and the spread-prior delay-spread and angular-spread priors into four UAV-height bins, using the same pixel-weighted aggregation as the headline tables.

Table 4.8: *Path-loss prior pixel-weighted RMSE by UAV transmitter height, on the held-out validation+test split.*

UAV height bin	Overall	LoS	NLoS	Samples
12 to 50	2.180	1.797	3.756	1211
50 to 150	1.936	1.785	3.228	2634
150 to 300	1.680	1.666	2.259	786
300 to 500	1.627	1.619	2.255	362

Table 4.9: *Delay-spread and angular-spread calibrated pixel-weighted RMSE by UAV transmitter height, on the city-holdout test split.*

UAV height bin	Metric	Overall	LoS	NLoS
12 to 50	Delay spread (ns)	39.90	43.04	35.13
50 to 150	Delay spread (ns)	27.17	26.72	28.29
150 to 300	Delay spread (ns)	8.70	8.79	8.13
300 to 500	Delay spread (ns)	3.70	3.80	2.76
12 to 50	Angular spread (°)	15.69	18.81	9.93
50 to 150	Angular spread (°)	14.19	15.89	8.41
150 to 300	Angular spread (°)	10.39	11.07	4.49
300 to 500	Angular spread (°)	8.85	9.34	2.85

The altitude dependence is monotonic and physically intuitive. Higher UAV transmitters see more of the ground as unobstructed LoS and encounter fewer deep shadow pockets, so both the NLoS pixel fraction and the worst-case tails shrink. Path-loss NLoS RMSE drops from 3.76 dB at low altitudes to 2.26 dB above 300 m, and delay-spread NLoS RMSE collapses from 35.13 ns to 2.76 ns in the same comparison. The opposite is true for the absolute number of samples: the dataset is dominated by the $50-150$ m band, so the global aggregate RMSE reflects that regime more than the higher-altitude extremes. Together with the empirical HDF5 histogram of Eq. 3.1 and the counts reported in Chapter 3, this means the high-altitude rows are better despite having much less training support, not because they are over-represented in the dataset. A practical consequence for the final residual model is that the neural residual head has less to correct at high altitudes, where the path-loss prior is already close to the quantisation floor. The remaining headroom lives in the low-altitude, NLoS-heavy bin, which is exactly where a distribution-aware residual can matter.

4.2.4 Path-loss prior by topology class

The headline path-loss-prior number (1.92 dB overall RMSE on the held-out validation+test split, see Table 4.4) hides the same morphological structure that the distribution-first path-loss predecessor surfaced. The per-sample evaluation records were re-aggregated by classifying each evaluated sample into the six topology classes with the same routing rule used by the distribution-first experiments. The pixel-weighted RMSE per topology class is reported in Table 4.10 and visualised in Figure 4.3.

The LoS dependency on morphology is monotonic and matches the LoS variance pattern observed in the distribution-first path-loss predecessor. The two-ray prior fits its own ground reflection ring against an LoS marginal whose width grows with city density and building height (see the LoS standard-deviation column of Table 4.6), so the LoS RMSE of the analytic model rises from 1.39 dB in the open sparse lowrise class to 2.44 dB in the dense block highrise class. The NLoS branch is less ordered by morphology: the best Try 78 NLoS topology is *mixed compact lowrise* at 2.91 dB, while the worst is *dense block highrise* at 4.09 dB. The dataset is also unbalanced in this respect: dense block highrise contributes only 104 of the 4994 topology-assigned evaluation samples, so its high RMSE has a relatively small impact on the overall pixel-weighted aggregate but is the regime

Table 4.10: *Hybrid path-loss prior, pixel-weighted RMSE on the held-out validation+test split, broken down by topology class. The overall row reproduces the RMSE values in Table 4.4; the sample count is the sum of samples assigned to the six topology classes. The aggregate Try 78 evaluator contains 4994 topology-assigned samples.*

Topology class	LoS	NLoS	Overall	Samples
open sparse lowrise	1.39	2.92	1.43	895
open sparse vertical	1.43	3.91	1.55	370
mixed compact lowrise	1.81	2.91	1.90	1240
mixed compact midrise	1.91	3.84	2.21	1060
dense block midrise	2.19	3.20	2.40	1325
dense block highrise	2.44	4.09	2.97	104
Overall	**1.75**	**3.40**	**1.92**	**4994**

most likely to limit deployment in dense urban environments.

This per-topology view also makes the central methodological claim of the thesis quantitative. The right panel of Figure 4.3 compares the distribution-first DL model against the analytic two-ray prior on the same six topology classes for the NLoS regime. Both methods agree within about 1 dB on all topology classes. The distribution-first model is slightly better in the open and mixed classes, while the analytic prior is slightly better in the dense classes. The aggregate gap quoted earlier (3.34 dB for the distribution-first NLoS model versus 3.40 dB for the path-loss prior NLoS) is therefore not an artefact of one easy class dominating the average; it is genuinely a per-class equivalence. This is what justifies the chronology of the project: once the GMM head shows that the NLoS marginal is reachable without physics, the choice between a deep GMM and an ML-calibrated NLoS prior becomes a question of cost, not of accuracy, and the cheap, auditable analytic prior wins on deployability.

4.2.5 Final joint prior-anchored residual GMM-head model test results

The final residual GMM-head model freezes the path-loss and spread priors and learns a bounded residual correction for the three radio maps. It is tested on the city-holdout test split with 2590 samples from 14 cities that were not seen in training. The final prior-anchored residual GMM-head model should be read as a two-stage final pipeline, not only as the last fine-tuning loss: the large-capacity residual/GMM stage keeps NLL, KL, moment matching and outlier-budget terms active, while the selected reported checkpoint is the validation-selected state after the deterministic RMSE/MAE-dominant fine-tuning stage. The exact weights of both stages are listed in Table 3.3. The final claim is therefore an RMSE improvement over the frozen priors rather than a calibrated probabilistic prediction claim. For a target t, group G, and valid pixel set $\Omega_{G,t}$, the reported model RMSE is

$$\mathrm{RMSE}_{\mathrm{model}}(G,t) = \sqrt{\frac{1}{|\Omega_{G,t}|}\sum_{(i,p)\in\Omega_{G,t}} \left(\hat{y}_{i,p,t} - y_{i,p,t}\right)^2}. \tag{4.1}$$

The prior RMSE uses the same formula with the frozen-prior prediction $y^{\mathrm{prior}}_{i,p,t}$ in place

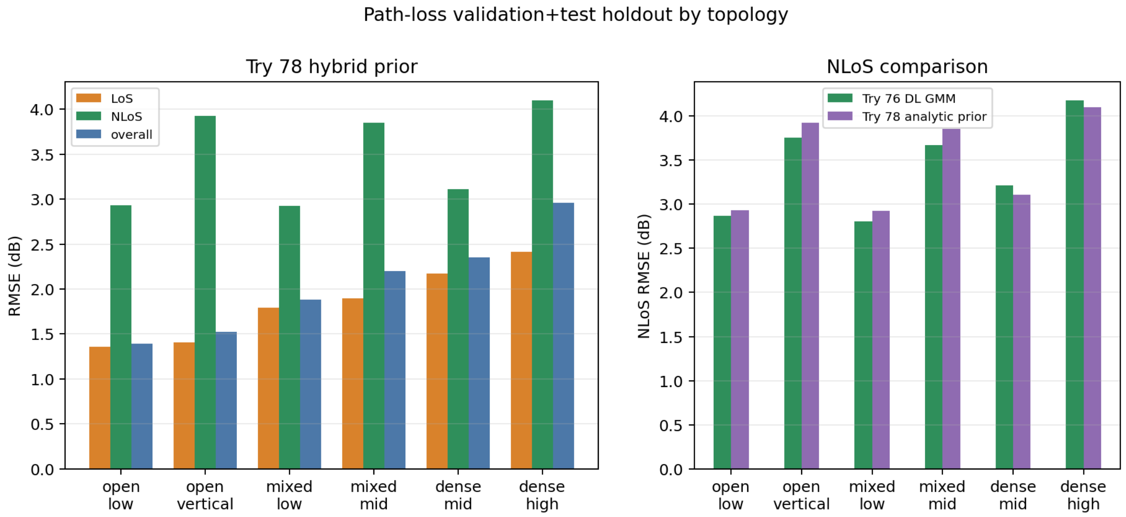

Figure 4.3: *Path-loss prior by topology class.* Left*: LoS, NLoS and overall pixel-weighted RMSE for each of the six topology classes. The LoS RMSE grows monotonically with morphological density, mirroring the growth of the LoS target standard deviation reported for the distribution-first model in Table 4.6.* Right*: NLoS RMSE side by side with the distribution-first DL model on the same six topology classes. The two methods agree within roughly* 1 dB *per class, which is what reduces the choice between the two pipelines to deployability rather than accuracy.*

of $\hat{y}_{i,p,t}$. Therefore `model_rmse` in the evaluation JSON is target-specific: it is the RMSE for path loss, delay spread, or angular spread individually, not a sum of the three RMSE values. The three targets are kept separate because their units are incompatible.

Table 4.11: *Final residual GMM-head model test results on unseen cities, compared with the frozen priors. Negative Δ means that the model improves the prior.*

Target	Model RMSE	Prior RMSE	Δ RMSE	Model MAE	Prior MAE	Δ MAE	Unit
Path loss	1.6519	1.9383	-0.2865	0.9298	1.2319	-0.3020	dB
Delay spread	26.5570	28.1023	-1.5453	7.9897	7.3033	+0.6864	ns
Angular spread	11.3854	13.7416	-2.3562	5.0088	4.9890	+0.0198	$^\circ$

Table 4.12: *Final residual GMM-head model test RMSE separated by LoS and NLoS pixels. The test cities are unseen during training.*

Target	Scope	Model RMSE	Prior RMSE	Δ RMSE
Path loss	LoS	1.4675	1.7519	-0.2844
Path loss	NLoS	3.1482	3.5143	-0.3661
Delay spread	LoS	25.4419	27.4851	-2.0432
Delay spread	NLoS	29.2045	29.6157	-0.4112
Angular spread	LoS	12.3510	15.2818	-2.9308
Angular spread	NLoS	8.4424	8.6614	-0.2191

The key point is generalisation: the test cities are completely unseen, yet the residual model reduces RMSE for path loss, delay spread, and angular spread simultaneously. Path loss is the cleanest case because both RMSE and MAE improve in train, validation, test, and all-city reporting. Angular spread improves RMSE strongly, while MAE is practically neutral overall; on the unseen test split the MAE change is only $+0.02^\circ$, and the LoS MAE even improves.

The validation metrics are not the final test claim. They are an external validation re-evaluation of the same selected checkpoint on the validation cities. The same selected model is then evaluated on the separate test cities, so the table checks validation/test drift without changing checkpoint or metric. Table 4.13 reports the two scopes side by side.

Table 4.13: *Validation-versus-test sanity check for the same selected final model. The validation row re-evaluates the selected checkpoint on validation cities; the test row is the final unseen-city generalisation result.*

Split	Samples	PL RMSE	DS RMSE	DS prior RMSE	AS RMSE	Meaning
Validation	2750	1.6157	22.1061	23.4494	11.3782	External validation re-evaluation of the selected model state.
Test	2590	1.6519	26.5570	28.1023	11.3854	Separate 14-city holdout; this is the thesis generalisation number.

The larger delay-spread gap between validation and test is mainly a city-mix effect, not a change of model or metric. The frozen delay-spread prior also worsens from 23.45 ns RMSE on validation to 28.10 ns on test, so the test cities contain harder delay-spread distributions before the neural residual is applied. In particular, Kuala Lumpur contributes only 13.2% of the valid delay-spread test pixels but 36.2% of the test delay-spread squared error, with 44.00 ns model RMSE and 46.93 ns prior RMSE. This is why the final test delay-spread RMSE is higher than the validation re-evaluation row, while still improving over the prior.

Delay spread is the only output where RMSE and MAE move in opposite directions. This is not a contradiction: RMSE weights rare high-delay errors quadratically, so suppressing the largest artefacts can improve RMSE while many tiny local shifts slightly increase MAE. The test MAE change is only +0.69 ns, while the RMSE improves by 1.55 ns. The same model also improves RMSE in both LoS and NLoS pixels for all three outputs, and the all-city subexpert audit shows improvements across every morphology/height group rather than a single easy subset.

Target vs final result

The selected final model clears the original numerical targets on the unseen-city test split: 1.65 dB path loss against the 5 dB target, 26.56 ns delay spread against the 50 ns target, and 11.39° angular spread against the 20° target. The same target check also holds on validation and on the all-city aggregation; even the worst unseen-test subexpert for each output remains below its target (2.66 dB, 36.89 ns, and 14.06°). On the all-city subexpert audit, the corresponding worst cases are 2.45 dB, 42.24 ns, and 13.97°, still below the same targets.

Comparison with closest state-of-the-art metrics

The state-of-the-art numbers cannot be used as a direct leaderboard because their datasets, height assumptions, city splits, and target definitions differ. Still, Table 4.14 places the final test result next to the closest available metrics from Section 2.9. Path loss has several external dB-scale references, but no external paper found here reports the same dense CKM city-holdout contract for delay spread and angular spread. The table is therefore limited to path-loss-like external rows plus the final-model reference row; scalar channel-parameter papers are kept out of this results table. In the table, PL means path-loss RMSE and lower RMSE is better. The better/worse ratios should be read cautiously because the rows use different data, different inputs, or different evaluation contracts. Earlier internal baselines are discussed in the ablation results instead of repeated here.

Table 4.14: *Plain-language comparison with the closest available path-loss-side metrics. No scalar indoor or link-level spread predictor is included as a table row, because those numbers are not dense CKM-style map benchmarks.*

Reference	What it reports	Verdict for our final model	Why the verdict is limited
RadioUNet [24]	1.6 dB–3.1 dB equivalent path-loss RMSE in accurate-map settings	Our final PL RMSE lies inside the reported 1.6 dB–3.1 dB dB-scale band.	2D path-loss maps only; no UAV-height conditioning or spread targets.
RadioGUNet [48]	1.304 dB DPM, 1.936 dB IRT, and 1.392 dB DPM with cars on RadioMapSeer	Our final PL RMSE is close to the reported RadioMapSeer dB-scale range, while remaining above its best DPM number.	Outdoor simulated terrestrial maps; held-out test layouts, but no UAV-height or spread targets.
PMNet / ICASSP 2023 winner [21, 50]	approx. 1.38 dB converted from the RadioMap3DSeer normalized score	Final PL is close but higher under a different benchmark contract; use only as scale context, not as a direct ranking.	Not directly comparable: ICASSP/RadioMap3DSeer testing differs from our unseen CKM-city split, continuous UAV-height conditioning, and joint PL/DS/AS target contract.
Gao et al. [9]	5.59 dB on ICASSP 2023; 12.19 dB on ITU Challenge 2024, with about 60% fewer FLOPs than PPNet	Our final PL RMSE is about 3.4× lower than the ICASSP number and 7.4× lower than the ITU number.	Recent outdoor PL-map model with Tx/Rx depth and corridor weighting, but no CKM variable-height UAV, no joint PL/DS/AS output, and different split/features.
ReVeal PINN [39]	1.95 dB outdoor rural/suburban radio-environment RMSE with sparse measurements	Final PL is in the same dB-scale range as ReVeal's sparse-measurement result.	Real outdoor measurements, but sparse RSSI radio-environment mapping rather than dense UAV-height-conditioned CKM generation.
AIRMap [37]	path-gain error reported below 4 dB against ray tracing	Our final PL RMSE is below 2 dB, but an exact ratio is not recoverable from the reported path-gain threshold.	Reports path gain, not the same path-loss calibration.
RMTransformer [26]	0.007148 normalised pixel RMSE, approximately 1.82 dB with its 254 dB USC/PMNet scaling window; 0.008099 channel prediction error, approximately 2.06 dB	Our final PL RMSE is slightly below the converted dB-scale values under that paper's normalisation window.	Recent dense path-loss radio-map reference, but on 256 px USC/PMNet maps with a random 90/10 split, no CKM city-holdout, no continuous UAV height, and no spread targets.
PathFinder [53]	0.0331 normalized RMSE on RM3D DS-RPP, approximately 1.19 dB with the 36 dB RM3D image-window convention; 0.3263 on unseen rural, approximately 11.75 dB only under the same assumption	Our final PL RMSE is above the converted DS-RPP value and far below the rural value under the same approximate 36 dB assumption.	The DS-RPP conversion follows RM3D scaling; the rural conversion is only an approximate scale check because the paper reports normalised image-space RMSE.
Final model	1.6519 dB **PL,** 26.5570 ns **DS,** 11.3854° **AS on unseen CKM cities**	**Reference row for the ratios above.**	Only row here reporting all three dense CKM targets under the final city-holdout protocol.

This comparison supports a cautious interpretation. The final path-loss number is not claimed to beat every published benchmark in absolute terms; PMNet's reported ICASSP score is lower on a different contract, and RadioGUNet is lower on the easiest RadioMapSeer setting. The path-loss result is nevertheless in the same broad outdoor simulated-map band as RadioUNet and RadioGUNet, below the PathFinder unseen-rural value under the approximate RM3D scaling, numerically well below Gao et al.'s recent corridor-weighted segmentation numbers, close to ReVeal's real outdoor sparse-measurement scale, and slightly below the converted RMTransformer value after applying that paper's own 254 dB normalisation window. For spreads, the honest external reading is that the table has no direct dense counterpart: the scalar channel-parameter papers reviewed in the state of the art are too different in input, output, and evaluation protocol to rank here. The thesis claim is therefore not that the spread maps beat external scalar predictors; it is that the model reports dense delay-spread and angular-spread RMSE on unseen CKM cities, improves the closest same-protocol spread baselines, and keeps the stricter ground-mask, LoS/NLoS, and continuous-height protocol visible.

Inference-time sanity check

The secondary runtime objective was checked on the Barcelona geometry supplied with the MATLAB ray-tracing script. The benchmark uses the same local laptop as the final generator: an AMD Ryzen 5 5600H CPU and an RTX 3050 Ti Laptop GPU with 4 GiB of VRAM. The ray-tracing run was configured at 1 m receiver spacing, so each non-building pixel of the 513×513 area is represented by a receiver at 1.5 m. Two Barcelona layouts and five UAV heights per layout give ten full-resolution CKM samples. The local MATLAB installation does not include Parallel Computing Toolbox, so the ray tracer was run on CPU with the same propagation settings as the supplied script: one reflection, no diffraction, 7.125 GHz, and extraction of path loss, delay spread, and angular spread from the ray-tracing paths.

The surrogate timing uses the same ten Barcelona topologies and antenna heights. It includes the complete non-RT generation path after the checkpoint is loaded once: geometric LoS/NLoS mask generation from the height map, the final ML-calibrated path-loss prior, the final ML-calibrated delay-spread and angular-spread priors, and mixed-precision inference with the final joint prior-anchored residual GMM-head model, including the default uncompressed `.npz` numerical archive export. The excluded costs are the one-time checkpoint load, the one-time CUDA warm-up, and optional PNG exports for masks or rendered maps.

The ten ray-traced samples contain 156,088 and 115,825 valid receiver pixels in the two Barcelona layouts, respectively. The measured average is 99.89 s per full-resolution sample for ray tracing and 0.88 s per sample for the complete final prior-anchored generator, including masks, priors, predictions, and the default numerical `.npz` archive export, but not optional PNG exports. The resulting 113.7× speed-up exceeds the original 10 to 100× runtime goal and should therefore be read as a deployment-style marginal runtime result after training and calibration, not as a comparison of the full research cost of developing the model.

The component logs explain where the remaining runtime goes without needing a second table: mask generation takes about 0.30 s, torch priors about 0.04 s, and final neural prediction plus default `.npz` export about 0.54 s. The vectorized mask path was checked

Table 4.15: *Full-pipeline runtime comparison for one* 513×513 *Barcelona CKM sample. Both rows are measured on the same Ryzen 5 5600H / RTX 3050 Ti Laptop machine. The surrogate row includes geometric mask generation, prior computation, and prediction of path loss, delay spread, and angular spread, including the default numerical `.npz` export. Checkpoint loading, one-time CUDA warm-up, and optional PNG exports are excluded.*

Method	Timed path	Resource note	Time per sample	Dataset-scale extrapolation
MATLAB ray tracing	Pixel-level receivers outside buildings; PL, DS and AS parsed from paths	Ryzen 5 5600H CPU; GPU off, PCT unavailable	99.89 s	448.96 h (≈18.7 days), 1×
Final prior-anchored generator	Generated LoS/NLoS masks, final calibrated priors, PL/DS/AS prediction, and default `.npz` export	RTX 3050 Ti Laptop GPU; mixed fp16, $b = 1$	0.88 s	3.95 h, 113.7× faster

against the original GT ray-caster with zero LoS/NLoS differences on the Barcelona samples and on 100 exported CKM samples; mixed precision changed RMSE by only 0.0014 dB, 0.0028 ns, and 0.0025° relative to fp32.

4.3 Qualitative Analysis

The qualitative evidence supports the quantitative story without needing the full diagnostic gallery in the main body. The early path-loss histograms showed NLoS mode collapse: visually plausible maps could follow the radial LoS trend while missing the blocked-link tail. The selected final model removes the worst of that collapse, reduces RMSE relative to the frozen priors across the split and LoS/NLoS checks, and applies residual corrections whose sign follows the prior-error maps. The complete histogram figures, split/regime bar plots, and representative inspection panel are kept in Appendix D.

The main lesson is that LoS path loss in this dataset has a strong radial ring structure, so a calibrated two-ray prior is more efficient than asking a CNN to learn the same pattern from pixels. The remaining qualitative failures are concentrated in sparse NLoS and high-spread cases, which is why future spread work should include an explicit high-tail or outlier diagnostic.

4.4 Limitations

The dataset is produced by ray tracing, not by field measurements. The results therefore measure agreement with a simulator and not direct over-the-air deployment performance. The path-loss target is stored as uint8, which imposes approximately 1 dB quantization. The spread targets are also stored as integer maps: delay spread as uint16 in ns and angular spread as uint8 in degrees. A more suitable dataset for the final modelling stage would preserve the continuous ray-traced targets, for example as float32 or as documented scaled integers, before any display-oriented quantisation. This should not be confused with a quantized model output: the final residual GMM-head model emits continuous PL/DS/AS values. In that light, the final path-loss MAE of 0.9298 dB on labels stored at approximately 1 dB resolution is already close to the path-loss label step, even though the RMSE remains 1.6519 dB because it is dominated by the harder tail errors.

The effect can be written explicitly. Let $y^\star$ be the continuous ray-traced value and $y_q = Q_\Delta(y^\star) = y^\star + \epsilon_q$ the stored value after rounding to step Δ. Under the usual high-resolution quantisation approximation, $\epsilon_q \sim \mathcal{U}(-\Delta/2, \Delta/2)$, so

$$\sigma_q = \sqrt{\mathbb{E}[\epsilon_q^2]} = \frac{\Delta}{\sqrt{12}}. \tag{4.2}$$

With $\Delta = 1$ in the physical units used here, the hidden uncertainty is 0.289 dB for path loss, 0.289 ns for delay spread, and 0.289° for angular spread. This is not the dominant error in the reported final model, but it is an irreducible resolution limit of the saved labels and becomes important once residual errors approach the sub-unit regime.

Table 4.16: *Approximate quantisation RMSE implied by the stored integer target precision in the current CKM HDF5 file. The table reports the actual storage used in this thesis.*

Target	Stored type	Current integer step	Current floor
Path loss	uint8	1 dB	0.289 dB
Delay spread	uint16	1 ns	0.289 ns
Angular spread	uint8	1°	0.289°

This also explains why getting very close to the best possible error is mathematically harder than the headline RMSE suggests. If model error and quantisation error are approximately independent, the observed squared error decomposes as

$$\mathbb{E}\left[(\hat{y} - y_q)^2\right] \approx \mathbb{E}\left[(\hat{y} - y^\star)^2\right] + \sigma_q^2.$$

Near σ_q, a real reduction in model error produces only a small change in total RMSE because of the square-root compression. In addition, many different continuous ray-tracing values collapse to the same stored integer, so the model cannot learn sub-bin structure from the labels even if the input geometry contains enough information. The heavy-tailed NLoS distributions add a second difficulty: MSE is dominated by rare large misses, so once the common LoS pixels are near the quantisation scale, further improvements depend on resolving sparse hard NLoS cases rather than on smoothing the whole map.

The LoS/NLoS mask used in these experiments should be interpreted as a geometric binary visibility mask, not as an augmented electromagnetic propagation label. In the CKM setup the transmitter position is fixed at the map centre and the UAV height is known for each sample, so the same binary mask can be derived by line-of-sight tracing through the building topology. It is therefore a deterministic geometric feature and branch key, not extra measured radio information unavailable at deployment.

The selected final model is evaluated on one city-holdout split and one training seed. This demonstrates generalisation to unseen cities, but not full robustness: future work should repeat selection across seeds and add field measurements, especially for delay-spread tails where quantised labels make sparse high-delay artefacts difficult to interpret.

Finally, the dataset fixes frequency and simulator assumptions. Deployment would need conditioning on carrier frequency, bandwidth, antenna pattern, and environment class; the city-holdout protocol tests urban layouts, not this broader domain adaptation.

CHAPTER 5

Sustainability Analysis and Ethical Implications

This chapter follows the ETSETB sustainability guide structure, but in a condensed form compatible with the reduced thesis. The work does not manufacture a physical product; its deployable result is a software pipeline for faster CKM prediction. The analysis therefore separates the one-time academic development cost from the possible operational use of the trained generator.

5.1 Sustainability matrix

Table 5.1: *Sustainability matrix summary for the thesis.*

Perspective	Development of BT	Project execution	Risks and limitations
Environmental	GPU training, laptop use, storage, and writing.	Reduced need for repeated ray-tracing simulations once the surrogate is trained.	Extra retraining, larger models, or poor reuse of cached priors could increase footprint.
Economic	Student and supervision time, plus shared HPC use.	Lower marginal planning cost if the surrogate replaces repeated simulations.	Real deployment needs field validation and maintenance.
Social	Skills gained in wireless modelling, ML, HPC workflows and scientific reporting.	Faster UAV coverage planning for temporary, rural or emergency networks.	Bias toward represented city morphologies and risk of overclaiming simulated results.

5.2 Environmental impact

The main environmental cost during development was computation. The final residual GMM-head model line used a four-GPU RTX 2080 Ti node for approximately 12 h; the distribution-first and prior-calibration phase used additional GPU/CPU time. Across the full exploratory project, the total exceeded roughly 400 $\mathrm{GPU}-\mathrm{h}$. This is a one-time research overhead and includes many failed or diagnostic experiments, not only the selected final run.

Several decisions reduced avoidable impact: reusing saved histories and frozen calibration artifacts, evaluating city-holdout metrics from stored checkpoints, and keeping the final priors frozen instead of recomputing them blindly inside every neural experiment. The final design is also relatively efficient at inference. On the Barcelona benchmark, the local MATLAB ray-tracing path took 99.89 s per full-resolution sample, while the complete generator path with masks, priors, neural prediction and export took 0.88 s. The sustainability advantage therefore depends on reuse: if the same calibrated generator is queried many times, the marginal cost is much lower than repeated ray tracing; if the model is retrained from scratch for every new scenario, that advantage weakens.

The life-cycle interpretation is therefore important. During the research phase, the footprint is dominated by exploration: many failed or partially successful attempts were necessary to discover that the final path should combine strong priors with a residual GMM-head model. During the use phase, the footprint is dominated by inference and by occasional recalibration. The project is most sustainable when the trained model, frozen priors, cached masks and exported checkpoints are reused across many planning queries. It would be less sustainable if the same design were scaled by simply increasing model size or retraining from zero for every city.

The final generator also improves reproducibility, which has an environmental side. A documented `.npz` output, explicit mask policy, and fixed hardware/runtime benchmark make it easier to compare changes without rerunning expensive ray-tracing baselines. Reproducible experiments are not only a scientific requirement; they reduce wasted computation caused by ambiguous settings, hidden preprocessing changes or repeated debugging runs.

5.3 Economic and social impact

Economically, the main development cost is human time: literature review, dataset handling, training/debugging, analysis, supervision and writing. The compute cost is small compared with this time cost in an academic project, but it is still relevant because the full exploration required many GPU-hours. In a planning workflow, the value of the surrogate would come from reducing repeated simulation runs and enabling faster what-if studies over UAV heights or city layouts.

Socially, faster CKM prediction can support temporary coverage planning for public events, rural extensions, emergency response or infrastructure restoration. The direct users would be network planners; the indirect beneficiaries would be users receiving more reliable wireless coverage. The main social risk is overconfidence: a simulated city-holdout result should not be treated as certified field performance. Errors should continue to be reported by LoS/NLoS state, UAV-height bin and morphology group rather than hidden behind one global RMSE.

The social value is strongest when the tool is used as decision support rather than as an automatic decision maker. A fast surrogate can help planners compare candidate UAV heights, identify weak-coverage zones, and decide where detailed simulation or measurement should be spent. It should not replace domain review: the model has been validated on simulated CKM cities and inherits the assumptions of the ray tracer, the topology representation, and the ground-mask definition.

There is also a fairness dimension in the morphology split. The model is tested on unseen cities, but the training distribution is still finite and urban morphologies are not equally represented. Dense high-rise, unusual street canyons, informal settlements, vegetation-heavy areas or non-European building patterns may behave differently from the dominant CKM regimes. For that reason, future deployment should report uncertainty or at least split diagnostics by morphology and height, instead of using the global average as the only quality indicator.

5.4 Ethical implications and SDGs

The main ethical obligation is honesty about validity. The dataset is generated by ray tracing, so the thesis demonstrates simulator-level generalization to unseen cities, not real-world deployment performance. A future operational system would require field measurements, calibration and human review, especially before public-service or emergency use.

A second ethical issue is transparency. The final predictor is not a pure black-box CNN: the path-loss and spread priors expose the geometric and morphological assumptions before the residual network is applied. This makes the system easier to audit, because a user can inspect whether an error comes from the LoS/NLoS mask, the prior, or the learned residual correction. That audit path is particularly relevant for emergency or infrastructure planning, where a wrong map can lead to misplaced confidence in predicted coverage.

The work does not process personal data, but operational radio planning can interact with sensitive contexts such as emergency response, public events or critical infrastructure. The generator should therefore be accompanied by clear metadata: input source, resolution, antenna height, mask-generation mode, checkpoint version and known limitations. Keeping that metadata with the prediction is part of the ethical responsibility of making a fast tool: speed is useful only if the output remains traceable.

The project aligns most directly with SDG 9, because it develops a faster tool for wireless-infrastructure planning; SDG 11, because it can support resilient urban and temporary connectivity; and SDG 13, because a validated surrogate can reduce repeated simulation energy. These contributions are indirect: the thesis provides a reproducible method and generator, not an operational network deployment by itself.

Overall, the sustainability conclusion is conditional but positive. The research process was computation-heavy, but it produced a reusable generator that can replace many repeated full-resolution simulations once validated for a target domain. The strongest sustainable-use case is therefore not “train a larger model”, but “reuse a documented, prior-anchored surrogate to decide where slower simulation or real measurement is actually needed”.

CHAPTER 6

Conclusions and Future Work

6.1 Conclusions

This thesis demonstrates that dense Channel Knowledge Maps for UAV-enabled networks can be predicted with varying transmitter height, provided that the problem is treated as a structured propagation task rather than as generic image-to-image translation. The final system predicts path loss, delay spread, and angular spread on 513×513 city maps under a strict city-holdout protocol, with the UAV located at the map centre and its height changing from sample to sample. The work therefore answers the original practical question: a single surrogate can produce dense CKM targets for unseen cities while conditioning on continuous UAV height.

The central lesson is that the strongest model is not a pure black-box neural network. It is a hybrid system in which calibrated physical/statistical priors explain the stable part of the channel, and a neural residual model corrects what remains. This division of labour is what made the final result reliable: the priors capture radial propagation, visibility, height effects, morphology, and spread-scale statistics; the learned model then focuses on local residual structure and distribution mismatch.

The finished pipeline meets the original numerical targets. The standalone path-loss prior achieved 1.9246 dB RMSE on the held-out validation+test split, and 1.9383 dB when reused as the frozen prior on the final 14-city unseen test split. The spread priors reached 28.10 ns delay-spread RMSE and 13.74° angular-spread RMSE on the same final test split. The final prior-anchored residual GMM-head model improved those anchors to 1.6519 dB, 26.5570 ns, and 11.3854°, respectively. These numbers are meaningful because the baseline being improved is already strong and train-calibrated, not a weak baseline.

6.1.1 What the final result means

The clearest contribution is a working CKM predictor for variable-height UAV maps. The final model receives geometry, masks, frozen priors, and a continuous height condition, and produces dense path-loss, delay-spread, and angular-spread maps jointly. The height dependence is not cosmetic: the same city footprint changes its visibility structure, two-ray interference pattern, and spread regimes as the UAV altitude changes. The final architecture handles this by combining height-aware priors with a sinusoidal height embedding injected through FiLM conditioning.

The path-loss result is the strongest success case. Most LoS path-loss structure is radial, height-dependent, and coherent enough that an enhanced two-ray prior can explain much of it before any neural correction is applied. The residual model still matters, but its role is narrower and more realistic: it improves local blocked streets, transition regions, and geometry-to-radio misalignments that the analytic prior cannot fully capture. Reducing the test RMSE from 1.9383 dB to 1.6519 dB after such a strong prior is therefore a substantial result.

Delay spread and angular spread are harder to judge by a single RMSE. Their maps contain sparse high-value regions, spike-like behaviour, and long-tailed outliers. Qualitatively, the final predictions usually recover the main spatial structure: low-spread and high-spread regions appear in the right general parts of the city, and the LoS/NLoS geometry is respected. The remaining mistakes are mostly local outlier failures. Some groups of pixels with unusually high DS or AS are missed, while some false positive high-spread patches are predicted where the target remains moderate. This explains why the spread result is less uniform than the path-loss result, while still being useful: the model improves the global structure and many local regimes, but rare spread hot spots remain the hardest cases. These RMSE gains should not be read as universal typical-error gains: delay-spread MAE increases, and angular-spread MAE is nearly unchanged, so the spread claim remains an RMSE-and-structure claim rather than a calibrated-error claim.

The most important methodological conclusion is that CKM prediction improves when the model head follows the target distribution. Earlier point-regression models could produce plausible-looking maps while missing NLoS path-loss modes or suppressing spread tails. The distribution-first stage fixed that diagnosis by using GMM-style outputs and histogram-aware evaluation. The final model keeps that lesson in its head design: it emits mixture parameters around calibrated anchors and exports their local-weighted point prediction, while the final reported claim remains the deterministic RMSE improvement selected after the RMSE/MAE-dominant fine-tuning stage.

6.1.2 Answers to the research questions

RQ1. Can a single deep-learning surrogate predict per-pixel path loss, delay spread, and angular spread on 513×513 **channel-knowledge maps, under strict city-holdout evaluation, with errors compatible with the thesis targets?** Yes, within the fixed CKM contract used in this thesis. The final model predicts PL, DS, and AS jointly on unseen cities and clears the original targets: 1.6519 dB path-loss RMSE, 26.5570 ns delay-spread RMSE, and 11.3854° angular-spread RMSE. The result should be understood under one carrier frequency, one centred UAV, valid ground receivers only, and city-level train/validation/test separation.

RQ2. Which part of the prediction should be left to a calibrated analytic prior and which part should be left to the neural network, so that the hybrid model is both accurate and interpretable? The calibrated priors should handle the most predictable propagation structure: two-ray LoS path loss, height-binned LoS calibration, morphology-aware NLoS corrections, and log-domain ridge models for DS and AS. These priors are strong enough to be useful on their own. The neural model is then valuable because it learns bounded residual corrections around them, reducing the frozen-prior test errors from 1.9383 dB, 28.1023 ns, and 13.7416° to 1.6519 dB, 26.5570 ns, and 11.3854°.

RQ3. How does the continuous UAV transmitter height (12 m **to** 478 m**) affect the prediction error, and is the same architecture able to cover the full height range without per-height retraining?** Height affects the difficulty of the prediction through visibility, blocking and spread-tail behaviour, so it must be represented explicitly. In the final model, height enters twice: through height-aware calibrated priors and through a sinusoidal FiLM conditioning path inside the neural residual model. Low-altitude cases contain more blocked links, stronger transition zones and harder spread tails, while high-altitude samples are easier even though they are less represented in the training data. This indicates that the height trend is physical, not only a sampling artifact.

6.1.3 Main contributions

The first contribution is the final variable-height CKM predictor itself: a single prior-anchored residual GMM-head model that jointly predicts path loss, delay spread, and angular spread for unseen city layouts while conditioning on continuous UAV height.

The second contribution is the calibrated prior family. The path-loss prior combines free-space/radial structure, coherent two-ray LoS behaviour, height-dependent calibration, NLoS morphology features, and fallback regimes. The spread priors extend the same philosophy to DS and AS through log-domain ridge calibration, topology/visibility grouping, height regimes, and robust fallbacks. Together, these priors put a large amount of physical and empirical knowledge into a reproducible estimator that is already competitive before the neural residual is trained.

The third contribution is the distribution-first modelling lesson. The thesis shows that target histograms, LoS/NLoS marginals, and spike-plus-tail spread behaviour are not diagnostic extras; they determine which output head is appropriate. This is why GMM-style residual prediction and histogram-aware checks became part of the final design.

The fourth contribution is the evaluation contract: city-holdout splitting, ground-only masking, LoS/NLoS reporting, topology diagnostics, and UAV-height-stratified reporting. This contract is essential because random splits or unmasked building pixels can make the same model appear stronger without improving the deployment-relevant task.

6.1.4 Practical implications and scope

The practical implication is that future CKM systems should be compared under the same contract before architectural claims are made. A new backbone is useful only if it improves held-out city metrics after applying the same receiver mask, height conditioning, LoS/NLoS reporting, topology analysis, and prior baselines. Otherwise, a lower error may simply come from solving an easier problem.

The result is strongest as a reproducible surrogate for the CKM simulation contract used here. It is not a universal replacement for ray tracing across all frequencies, antenna patterns, or cities. It does show, however, that once the dataset contract is fixed, the slow simulator can be compressed into a fast and interpretable predictor. In the Barcelona runtime benchmark, pixel-level MATLAB ray tracing took 99.89 s per sample, while the complete final-generator path took 0.88 s, a 113.7× reduction.

6.1.5 Negative results

The negative results are useful mainly because they explain why the final system looks the way it does. Pure image-translation models improved visual sharpness but not the physical metrics that mattered. Larger CNN variants, topology experts, multi-scale outputs, and uncertainty heads each helped expose some failure mode, but none solved the deeper issue by themselves. The repeated pattern was that plausible maps could still miss the target distribution, especially NLoS path-loss modes and spread tails.

This is why the thesis ends with a hybrid, distribution-aware model instead of just a larger neural network. The failures ruled out simpler explanations: the problem was not only capacity, not only regularisation, and not only decoder artifacts. The final design is stronger because it is built on those rejected hypotheses.

6.2 Future Directions

Short-medium term. The next step is to strengthen the current residual pipeline without changing the evaluation contract, then extend it toward frequency and radio-configuration conditioning. Useful additions include transmitter-oriented mixup, explicit corridor losses, and controlled ablations of each residual head and loss weight. The current dataset fixes the carrier and antenna assumptions, while a more general outdoor CKM model should learn how path loss and spreads vary with frequency, bandwidth, antenna pattern, and possibly polarization. A natural dataset for this stage would simulate the same cities across several radio configurations so that the model can separate geometry effects from configuration effects.

A nearer-term training direction is adaptation to new or under-represented topology types. The per-topology analysis suggests that dense high-rise, compact mid-rise, open low-rise, mixed urban layouts, and future morphology classes do not fail in exactly the same way. A practical continuation would therefore be to fine-tune the final prior-anchored model on newly introduced urban typologies, or to train lightweight specialist heads on top of the shared backbone when only limited examples are available. This would test whether the model can recover local structure in topology regimes beyond the original training distribution, rather than only specialising to already defined classes.

For delay and angular spreads, another short- to medium-term direction is to optimize not only pixel-wise RMSE but also spatial structure. Spread maps can look physically coherent even when small shifts across quantized target boundaries are penalized harshly by RMSE. Future fine-tuning should therefore test structural objectives, for example gradient consistency, region-wise correlation, SSIM-like terms, LoS/NLoS boundary consistency, or topology-aware ranking losses, so that the model preserves the correct spread patterns rather than only minimizing the average numeric error.

Medium-long term. The final extension is field calibration. The path-loss and spread priors are interpretable, which makes them suitable for simulator-to-reality correction using a limited number of real measurements. In the medium-long term, this could lead to a CKM predictor that starts from ray-traced or prior-generated maps, adapts to a new city with sparse measurements, and still reports uncertainty by LoS/NLoS regime, UAV-height bin, and topology class. Those safeguards should remain part of the method before considering operational use in emergency coverage, UAV network planning, or rapid deployment studies.

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APPENDIX A

Development Phases and Attempt Trace

The original appendix listed the internal attempt numbers in several long tables. For the reduced thesis, the compiled appendix keeps phases instead of eighty row-level entries. The full try-number map remains in `appendices.tex` and in the repository history, so the trace is not lost; it is simply no longer part of the main PDF.

Table A.1: *Compact attempt log grouped by methodological phase.*

Attempt range	Phase	Why it matters in the final thesis
1–22	Direct image-to-image baselines	Established the CKM pipeline, cGAN/U-Net feasibility, bilinear decoder fixes and multiscale supervision. The phase ended because visual realism did not imply physical RMSE.
23–40	Spread branches and mask reset	Added delay/angular experiments and, most importantly, the ground-only receiver mask. This changed the evaluation contract.
41–50	Prior-residual PMNet phase	Showed that raw formulas were weak, train-only calibration helped, and residual learning was useful but still NLoS-limited.
51–66	PMHHNet and topology experts	Introduced six topology experts, height FiLM, high-frequency paths, obstruction channels and no-prior controls. LoS improved, NLoS remained the bottleneck.
67–75	Stabilisation and distribution diagnosis	Tested knife-edge channels, quad heads, uncertainty and histogram-focused diagnostics. The key result was the visible NLoS and spread-tail distribution mismatch.
76–77	Distribution-first models	GMM/spike-plus-tail heads showed that modelling the target distribution before pixel assignment could remove much of the collapse.
78–79	Frozen calibrated priors	Produced the final train-only path-loss and spread priors, including height-aware LoS two-ray calibration and log-domain spread ridge calibration.
80	Final prior-anchored residual GMM model	Combined frozen priors with a shared residual GMM-head model and selected the RMSE-dominant fine-tuning stage reported in Chapter 4.

The main methodological lesson is that the project did not progress by simply making a larger network. The decisive changes were the evaluation contract, regime separation, distribution diagnostics and the return to strong frozen priors once their role had become clear.

A.1 Expanded phase notes

The compact table above is intentionally phase-based, but the thesis still needs enough chronology to show that the final model was not chosen in one step. The following notes keep the development evidence that most directly explains the final design.

A.1.1 From image translation to receiver-valid evaluation

The first part of the project treated CKM prediction as aligned image-to-image regression. This was useful for building the dataset pipeline: the topology map, LoS/NLoS support, distance field and targets all live on the same 513×513 grid. The early cGAN and U-Net variants also exposed practical issues such as checkerboard artifacts, excessive adversarial pressure and the need for multiscale supervision.

The decisive correction was methodological rather than architectural. Once the receiver mask was enforced, the task became prediction on ground receivers only: buildings and padded pixels

Table A.2: *Decisive attempt milestones kept from the detailed attempt log.*

Attempt	Public role	Lasting lesson
22	Clean early image-to-image baseline	Bilinear decoding and multiscale supervision reduced visual artifacts, but direct image realism was still a weak proxy for calibrated radio-map error.
33–40	Ground-mask reset	Building pixels had to be excluded from loss, metrics and visual error maps; scores before and after this contract are historical, not directly comparable.
41–45	First explicit prior-residual models	Raw formulas were weak, train-only calibration helped strongly, and residual learning was useful only when the prior carried real physical structure.
50	NLoS prior sandbox	Scalar NLoS corrections, tabular deltas and simple hybrid formulas did not solve blocked-link tails. The remaining error was spatial and distributional.
54–66	PMHHNet/topology experts	Height FiLM, physical channels and six topology experts improved the easy LoS regime, but NLoS stayed dominant even after substantial backbone engineering.
67–75	Stabilisation and histogram diagnosis	Knife-edge channels, quad heads and uncertainty improved diagnosis more than final RMSE. Histogram plots made the NLoS and spread-tail mismatch explicit.
76	Distribution-first PL GMM	Predicting the target distribution before assigning pixels gave the first convincing route for hard path-loss tails.
77	Distribution-first spread GMM	Delay and angular spread behaved like spike-plus-tail targets rather than smooth images, motivating distribution-aware output heads.
78–79	Frozen calibrated priors	The strongest non-neural evidence came from height-aware LoS calibration and log-domain spread ridge calibration, both fitted only on training cities.
80	Final prior-anchored residual GMM model	The selected system is a calibrated-prior estimator plus bounded learned residual distribution head, not a black-box direct regressor.

are not valid receiver locations. This is why the final thesis does not mix early scores with the final leaderboard. They remain useful as a development trace, but the valid comparison contract starts once the ground mask is fixed.

A.1.2 Why the prior returned

The first physical-prior attempts were not successful as formulas alone. A raw prior could be far worse than a neural model, especially in NLoS. However, train-only calibration made the prior informative enough to act as an anchor. This distinction matters: the prior is not presented as a magic replacement for learning, but as a structured baseline that gives the network the correct coordinate system.

The negative evidence was equally important. PMNet-style residual models, topology experts and NLoS-focused objectives improved parts of the map, but the hard tails survived. The project therefore moved away from simply enlarging the encoder and toward separating regimes, modelling distributions and freezing strong calibrated estimators.

A.1.3 Why Try 80 is the endpoint

The last attempts are the only ones that form the final methodology. Try 78 provides the path-loss prior, with a height-aware LoS branch and calibrated NLoS residual structure. Try 79 provides delay-spread and angular-spread priors with separate LoS/NLoS, height and topology calibration. Try 80 then learns bounded residual corrections over these frozen priors with a shared multi-task GMM-head model.

In other words, the final route is evolutionary but not arbitrary: direct regression established feasibility, masked evaluation fixed the contract, PMHHNet identified the remaining NLoS bottleneck, distribution-first models showed why tails mattered, and the frozen priors made the final residual model small enough to improve rather than overwrite the physical estimator.

APPENDIX B

Representative Final Model Panels

The full generated panel set contains one candidate for each (open, mixed, dense) × (low, mid, high) audit group. The expanded reduced thesis keeps three full-size qualitative panels: open/low, mixed/mid and dense/mid. The remaining panels and their manifest are kept in `img/thesis_figures/try80_appendix_panels/` and can be regenerated with the panel-generation script. The removed open/high Nazareth panel is not used because its angular-spread map is almost flat and the robust colour scale collapses to a misleading near-zero range.

Table B.1: *Representative final-model panel manifest.*

Audit group	City	Sample	Status	Reason
open/low	Segovia	sample_13213	PDF	Open low-altitude case with visible geometry and residual correction.
mixed/mid	Nice	sample_10828	PDF	Mixed morphology and mid-height regime.
dense/mid	Vancouver	sample_15265	PDF	Dense held-out test morphology; also used as the main paper/TFG qualitative panel.

Each panel has three columns (PL, DS and AS) and six rows: ground truth, frozen prior, final prediction, support/context, prior error and model correction. The last two rows are the diagnostic rows: a useful residual model should correct the sign and location of the prior error.

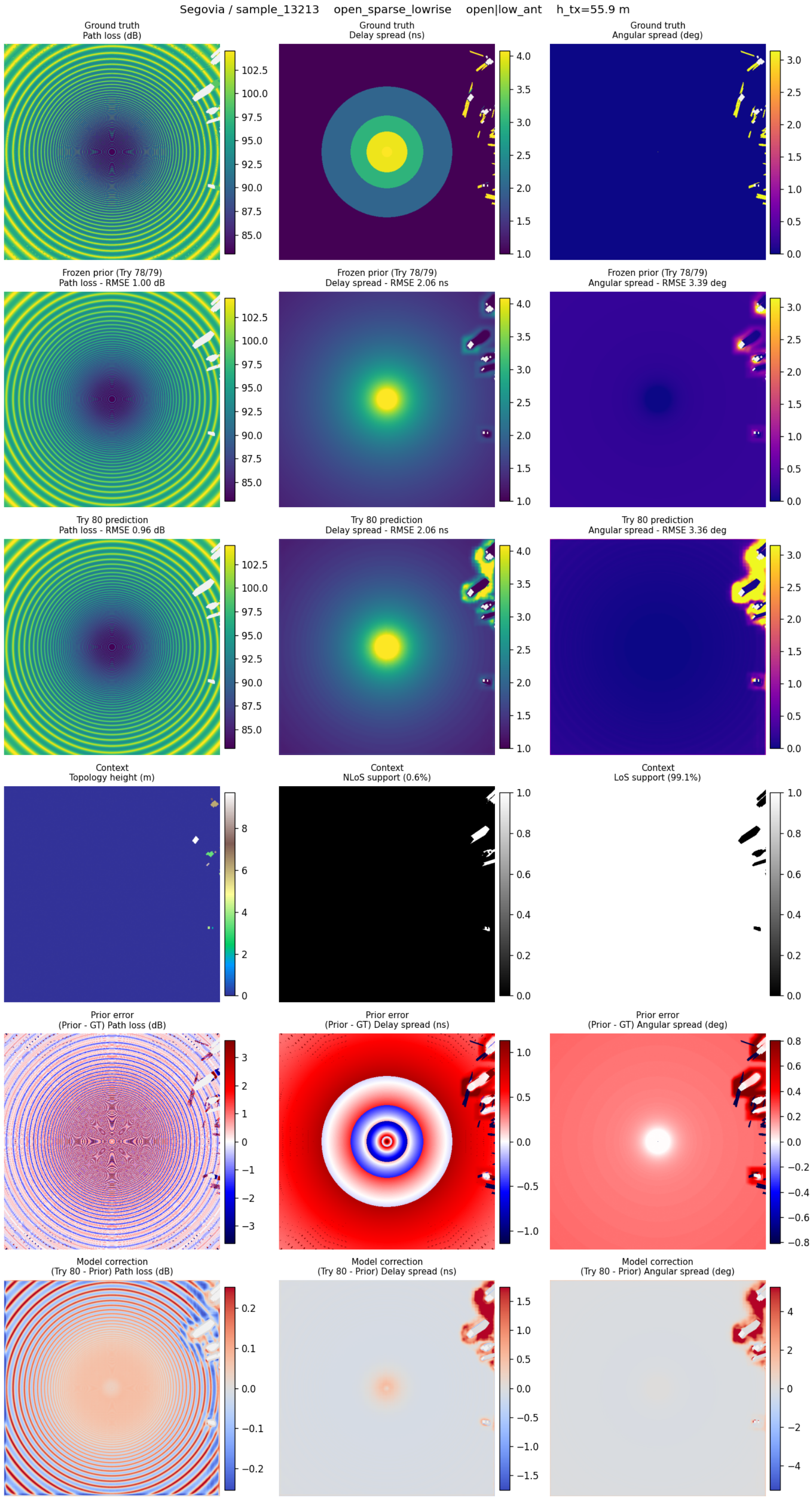


Figure B.1: *Final residual GMM-head inspection panel for an open, low-antenna validation sample in Segovia.*

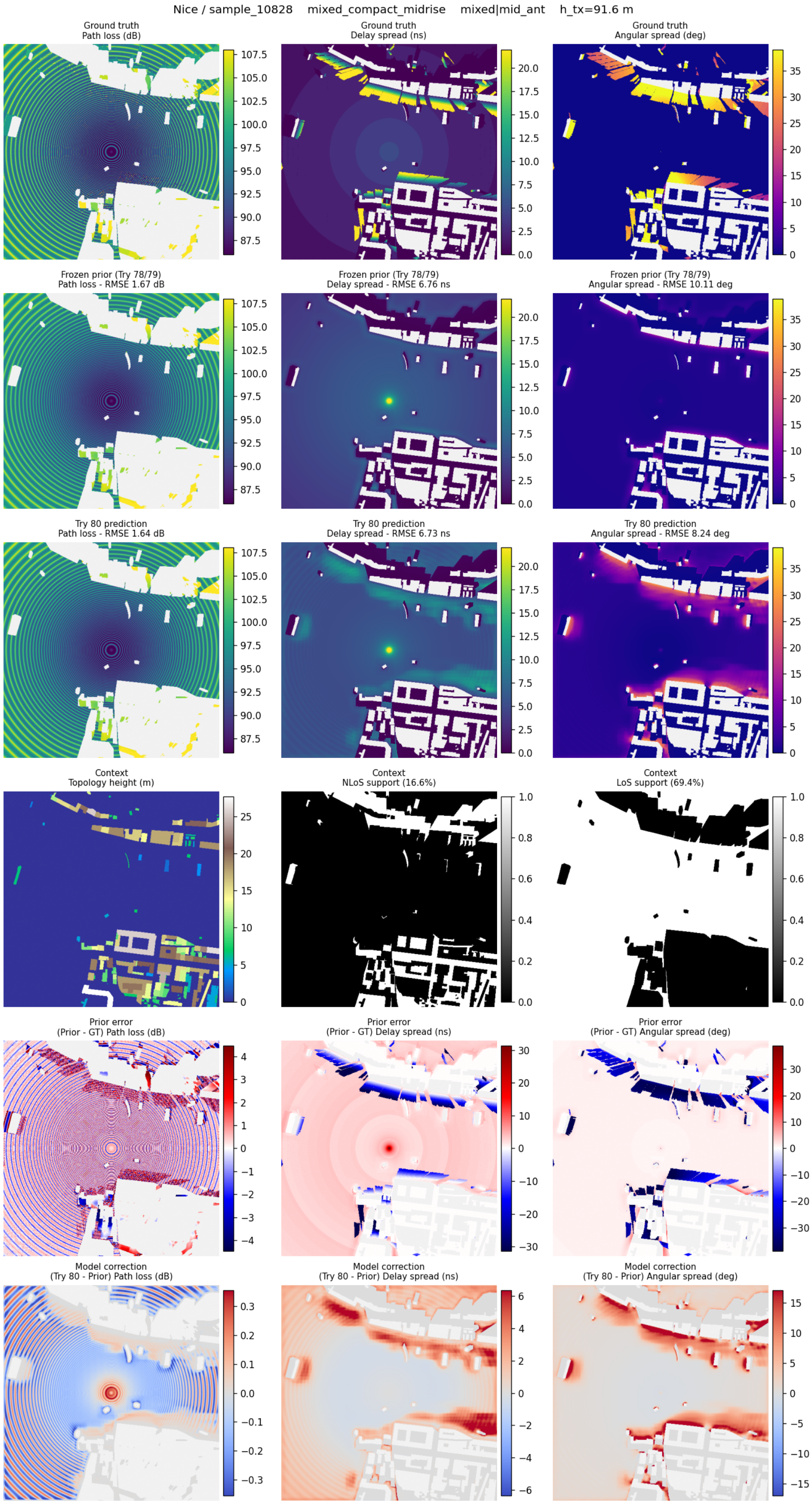


Figure B.2: *Final residual GMM-head inspection panel for a mixed, mid-antenna validation sample in Nice.*

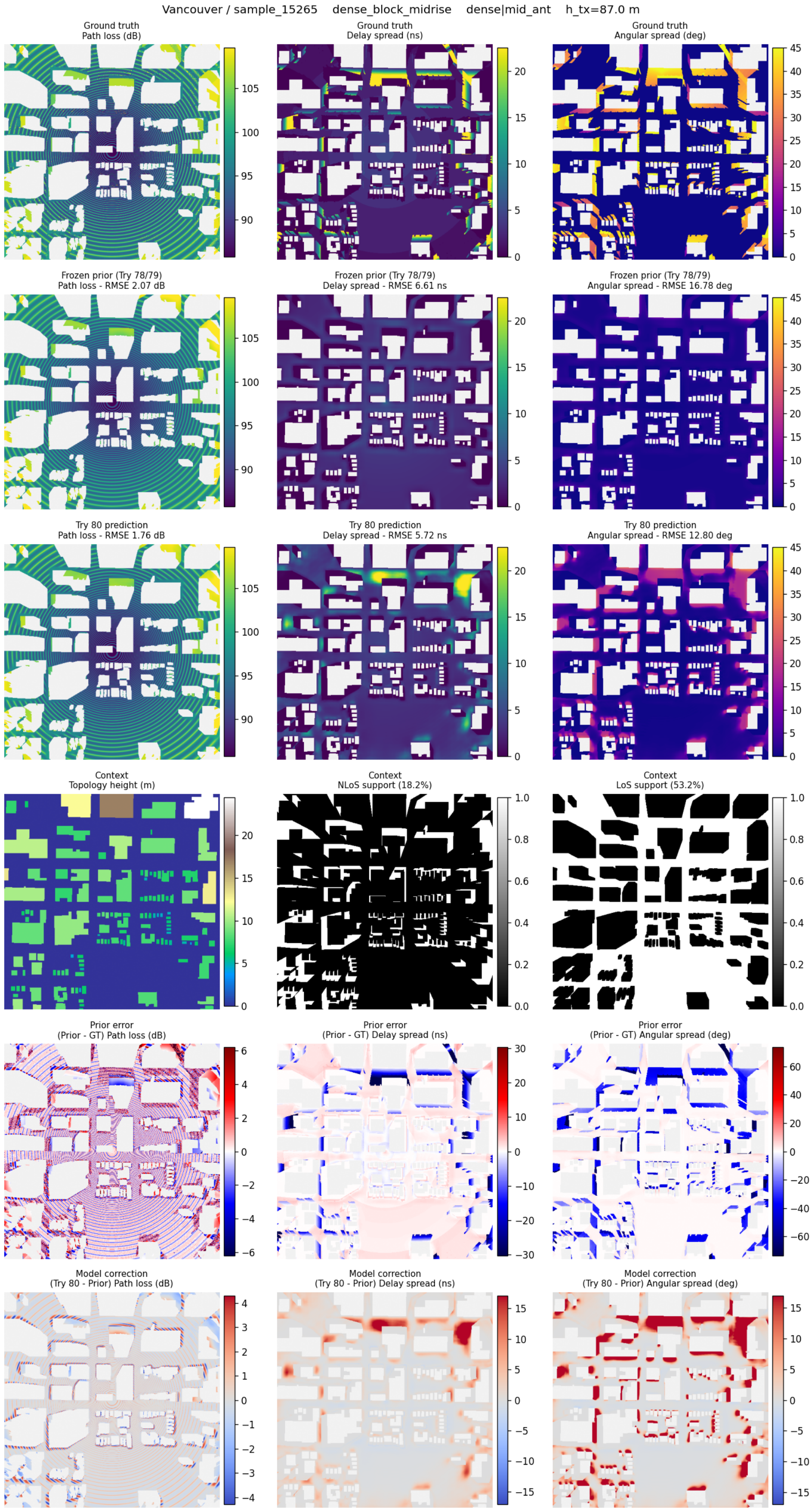


Figure B.3: *Final residual GMM-head inspection panel for a dense, mid-antenna held-out test sample in Vancouver.*

APPENDIX C

CKMGenerator Interface and Reproducibility Tool

This appendix documents the standalone CKMGenerator delivered with the final prior-anchored residual GMM-head pipeline. The complete user manual belongs in the repository README, but the thesis keeps the scientific and reproducibility contract: which inputs are accepted, how they are converted to the model grid, how receivers and LoS/NLoS masks are defined, what the tool exports, and which performance switches do not change the mathematical model.

C.1 Purpose and source record

CKMGenerator is a deployment wrapper around the final thesis pipeline. It accepts topology inputs, prepares them with the same geometric conventions used for training and evaluation, computes or imports LoS/NLoS masks, evaluates the frozen calibrated path-loss, delay-spread and angular-spread priors, optionally runs the final neural residual model, and exports the numerical arrays used for reproducible analysis. The Streamlit interface is intended for inspection and demonstrations; the command-line interface uses the same backend and is intended for scripted runs.

The reproducibility record is split across public repositories. The standalone generator source is in `CKMGenerator`; final training and evaluation code is in `Final_Code_TFG`; the broader experimental trace is preserved in `TFGAllProgress_Tries_and_Attempts`; and the thesis source is kept in `TFG_ActualText_LaTeX` [30, 31, 33, 32]. The final checkpoint is loaded from `models/best_model.pt` by default, or from an explicit `–checkpoint` path.

C.2 Interface and execution modes

Figures C.1–C.4 show the retained CKMGenerator screenshot set. The control panel is shown as a single cropped sidebar capture so the PDF documents the actual interface without wasting a full page on the blank app workspace.

The same backend can be run from the terminal. A typical HDF5 command is:

```
python ckm_generate.py ^
  --input inputs\samples\Abidjan_sample_00001.h5 ^
  --run-name abidjan_check ^
  --mixed-precision
```

For image-only topology inputs, the transmitter height must be supplied unless it is stored in a structured sidecar file:

```
python ckm_generate.py ^
  --input C:\path\topology_map.png ^
  --height 56.65 ^
  --topology-max-m 120 ^
  --meters-per-pixel 1.0 ^
  --mask-source auto ^
  --mixed-precision
```

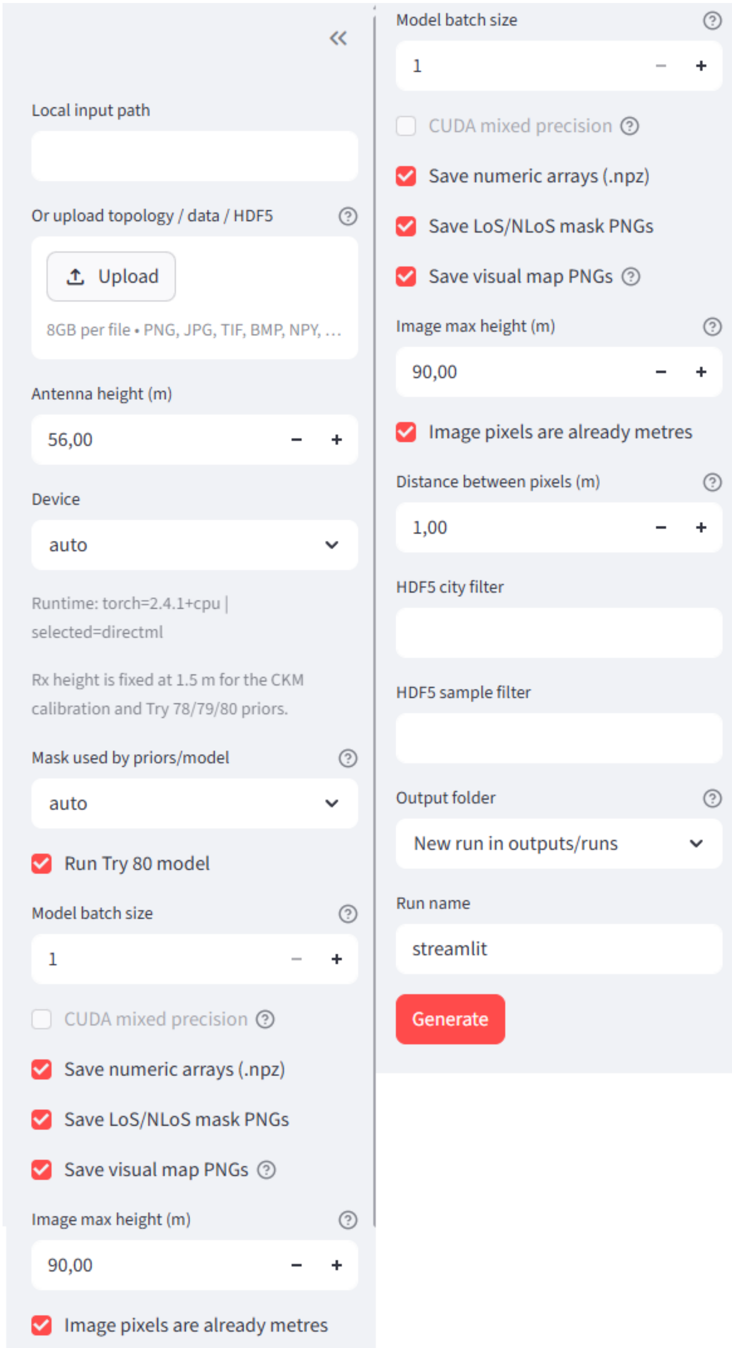
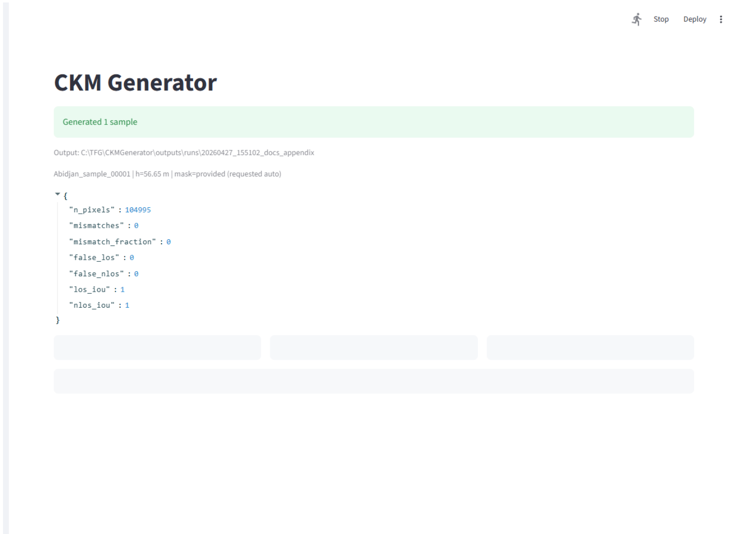
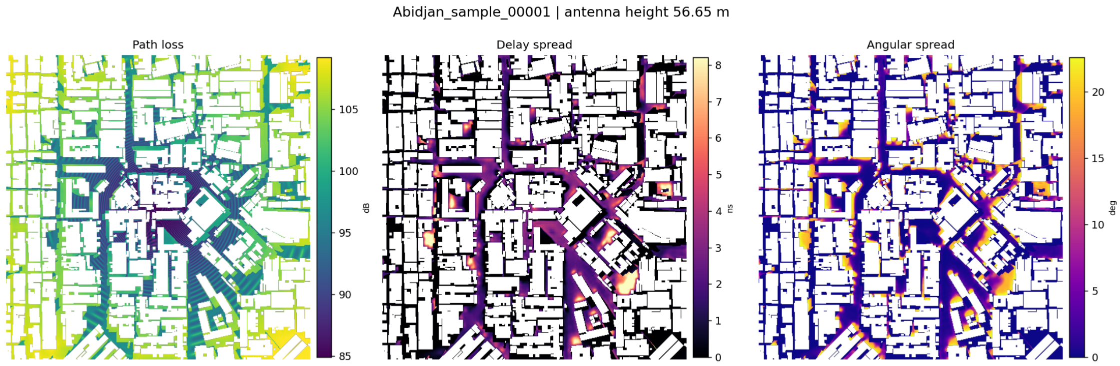


Figure C.1: *CKMGenerator Streamlit controls. The sidebar records the assumptions used by the generator: receiver height fixed at* 1.5 m*, variable antenna height, mask selection, optional Try 80 inference, mixed-precision availability, pixel scale and export format.*

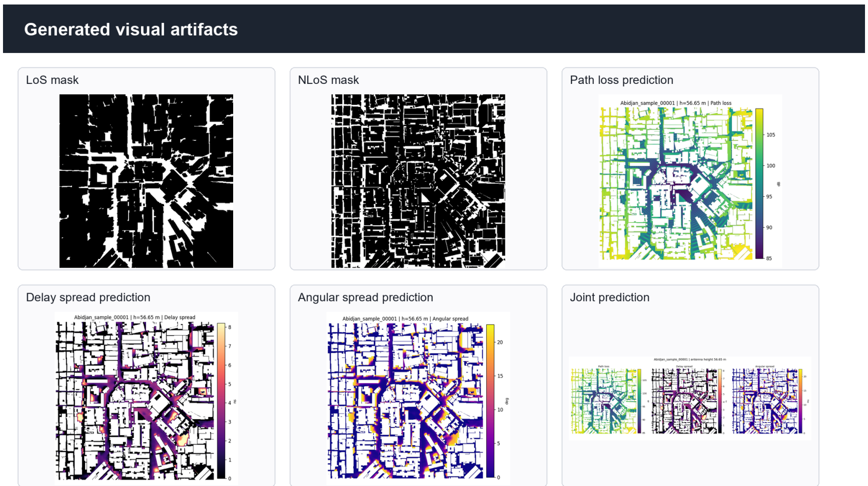


(a) *Application run summary, resolved mask policy and exported sample metadata preview.*

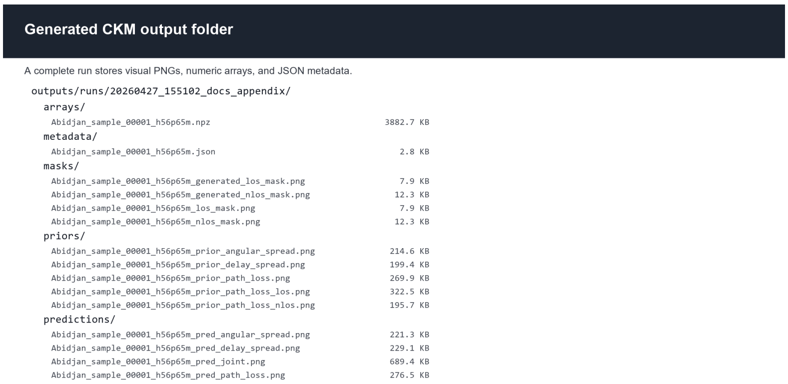


(b) *Joint path-loss, delay-spread and angular-spread prediction output.*

Figure C.2: *CKMGenerator application output after running one HDF5 sample.*



(a) *Saved visual products: masks, final prediction maps and joint panel.*



(b) *Generated output tree with masks, priors, predictions, arrays and metadata folders.*

Figure C.3: *CKMGenerator exported visual files and directory structure.*

Metadata and numeric array contents


```
{
  "sample_id": "Abidjan_sample_00001",
  "antenna_height_m": 56.647,
  "mask_source": "provided",
  "requested_mask_source": "auto",
  "topology_class_6": "dense_block_midrise",
  "antenna_bin": "low_ant",
  "npz_contains": [
    "topology",
    "ground_mask",
    "los_mask",
    "nlos_mask",
    "prior_path_loss",
    "prior_delay_spread",
    "prior_angular_spread",
    "pred_path_loss",
    "pred_delay_spread",
    "pred_angular_spread"
  ]
}
```


Figure C.4: *CKMGenerator metadata and numerical array archive. The `.npz` file is the authoritative output for reproducible analysis; PNG files are inspection artifacts.*

C.3 Input contract and grid fitting

The generator accepts raster topology images (`.png`, `.jpg`, `.tif`, `.bmp`), numeric arrays (`.npy`, `.npz`, `.csv`, `.txt`, `.json`) and HDF5 files. HDF5 and NPZ inputs may provide topology, LoS/NLoS masks and antenna height directly. Raster inputs normally store intensities rather than metre-valued building heights, so the selected maximum image height is used to convert pixel values to metres unless the option `image-values-are-metres` is enabled.

All inputs are fitted to the 513×513 grid used by the final model. The model grid represents 1 m per pixel, with the transmitter antenna located at the centre of the map and with variable UAV height. If an input is supplied with another physical spacing, `–meters-per-pixel` is used before the centred crop or padding step. Topology arrays are resampled with bilinear interpolation; masks are resampled with nearest-neighbour interpolation and then binarised again.

The receiver convention is the same one used throughout the thesis. Ground pixels are zero-height topology pixels, and every ground pixel is a receiver at $h_{rx} = 1.5$ m. Building pixels and padded outside pixels are not receiver locations. The receiver-valid mask is therefore

$$m_{\text{ground}}(x) = \mathbf{1}\{h_{\text{topology}}(x) \leq 10^{-6}\}.$$

This convention matters because the model predicts dense maps only over valid ground receivers, not over roofs or artificial padding.

C.4 Mask, prior and neural-model path

The model consumes one LoS mask and one NLoS mask as input channels, and the same masks are also used by the priors. CKMGenerator supports three policies: `provided` requires masks stored in the input or sidecar files; `generated` ray-casts masks from topology and antenna height;

and `auto` uses provided masks when present and otherwise generates them. For CKM HDF5 samples, the NLoS support is derived from the raw LoS mask and the ground mask,

$$m_{\text{LoS}} = m_{\text{LoS,raw}}\, m_{\text{ground}}, \qquad m_{\text{NLoS}} = (1 - m_{\text{LoS,raw}})\, m_{\text{ground}}.$$

The multiplication by m_{ground} is essential: otherwise building pixels would be counted as NLoS receiver pixels.

Generated masks make topology-only inference possible, but they are not claimed to be bit-identical to the CKM ray-tracing masks. They are a deployment fallback that preserves the model interface when a user does not have the original HDF5 mask fields.

After preprocessing, the tool evaluates the frozen priors from the final pipeline. These are the same height-aware, calibrated priors described in Chapter 3: a LoS/NLoS path-loss prior with centre-antenna geometry and antenna-height calibration, and spread priors with separate calibrated structure for delay spread and angular spread. When neural inference is enabled, the prepared topology, masks, priors and height context are passed to the final residual GMM-head model. The generator therefore changes neither the model definition nor the validation contract; it packages the final method for reproducible use.

C.5 Outputs and performance controls

Each run writes to a timestamped output folder, or to the explicit folder passed with `–out`. The authoritative output is the NumPy archive in `arrays/`. PNG files are optional visual inspection products.

Table C.1: *CKMGenerator artifacts and controls retained in the thesis appendix.*

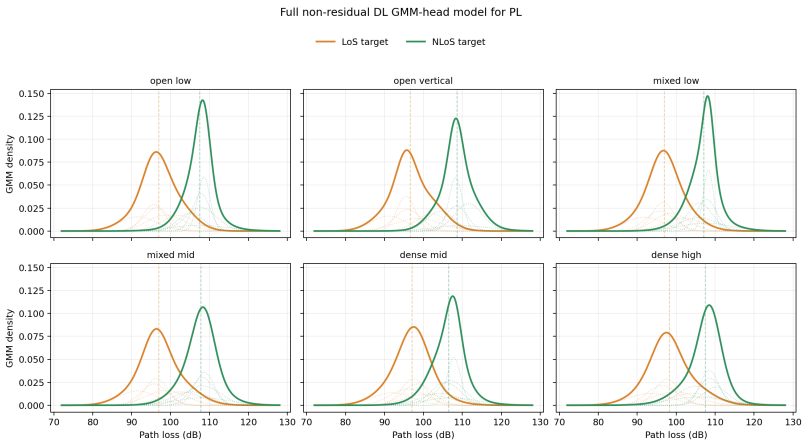
Area	Reproducibility detail
Numerical archive	Stores topology, ground mask, LoS/NLoS masks, generated or reference masks when available, PL/DS/AS priors, PL/DS/AS predictions and metadata on the final 513 × 513 grid.
Metadata	Records sample id, antenna height, source file, checkpoint path, requested and resolved mask policy, topology class, antenna bin, save options, timings and generated paths.
Visual outputs	Optional PNG exports include binary masks, prior maps, prediction maps and the joint prediction panel used for inspection.
Runtime	CUDA is preferred; CPU and DirectML are fallback paths. The `ckm_doctor.py` check verifies PyTorch, CUDA/DirectML availability, checkpoint presence and memory before inference.
Mixed precision	CUDA autocast can be enabled for speed. It was checked to change RMSE only at a negligible scale compared with validation error.
Batch speed	Disabling visual map PNGs, mask PNGs and compressed arrays avoids Matplotlib rendering and compression overhead while preserving the numerical arrays.

The benchmarked fast path reported in Chapter 4 uses vectorized LoS/NLoS mask generation, torch prior evaluation, mixed-precision neural inference and default numerical `.npz` export after the checkpoint has already been loaded. These optimizations affect runtime and storage, not the scientific definition of the generator.

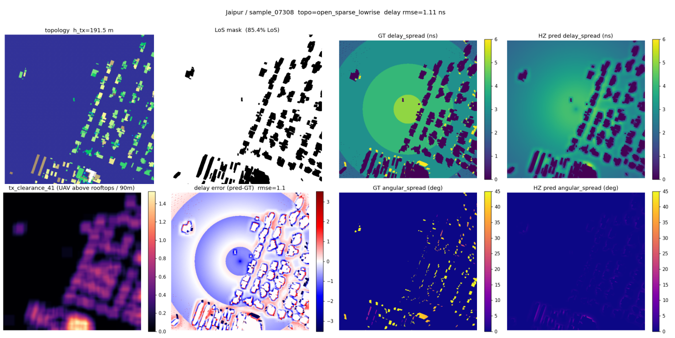
APPENDIX D

Additional Results and Qualitative Diagnostics

This one-page appendix keeps only the two diagnostics that are useful to audit the final interpretation. Full split bars, all-city subexpert tables, the full qualitative gallery and long derivations remain in the source archive and generated image folders.



(a) *Full non-residual DL GMM-head model for PL. Thick curves show summed LoS/NLoS GMM densities; faint dotted curves show the individual Gaussian components.*



(b) *Spread-target quantisation: predictions are continuous while the released targets contain staircase-like and sparse integer-valued structures.*

Figure D.1: *Compact additional diagnostics. The GMM plot exposed different LoS/NLoS marginals and motivated distribution-aware heads; the quantisation panel records the target-storage limitation discussed in Chapter 4.*

Additional source material. The nine-panel gallery, row-level 80-attempt log, CKMGenerator screenshot manual, all-city subexpert tables and split bars are omitted from this compact appendix but remain in `appendices.tex` and the generated figure folders. The Try 78/79/80 mathematical derivations are not repository-only: they are included in Chapter 3 through `prior_detail_try78.tex`, `prior_detail_try79.tex` and `prior_detail_try80.tex`, which also remain as separate source files for traceability.